AFCEC-APE-895

AIRFIELD PAVEMENT EVALUATION

Maxwell AFB, Alabama

ICAO: KMXF

January 2016

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14. ABSTRACT
An Air Force Civil Engineer Center (AFCEC) pavement evaluation team, composed of those individuals listed in Block 6, completed a comprehensive airfield pavement evaluation at Maxwell Air Force Base, Alabama from 22 June to 1 July 2015. The primary objective of this evaluation was to determine the surface condition, structural capacity, and runway friction characteristics of airfield pavement sections. For the runways at Maxwell AFB, the U.S. Air Force recommended standard Pavement Classification Numbers (PCNs) for publishing are:

Runway 15/33: 66 R/B/W/T
Assault Runway 009/189: 56 R/B/W/T

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EXECUTIVE SUMMARY

Based on the current eight-year structural evaluation cycle for most Air Force airfield assets, eight members from the Air Force Civil Engineer Center (AFCEC) conducted the field portion of an airfield pavement evaluation at Maxwell AFB, Alabama, during the period 22 June to 1 July 2015.

The objective of this evaluation was to determine the structural capacity, pavement surface condition of the airfield pavements, and friction characteristics of the runways at Maxwell AFB, Alabama. This report summarizes the analysis of the data collected in order to identify the most important observations, common themes, areas of concern, and maintenance and repair (M&R) recommendations for Maxwell AFB airfield pavements. The data in the appendices may be used by various entities, including personnel within different areas of airfield operations, base civil engineering, and higher headquarters and staff agencies in order to assist in operations, pavement management and maintenance, and project programming and funding. A separate reference file located at the AFCEC Airfield Evaluation Community of Practice describes the background information and testing procedures used to determine the airfield characteristics (link on page 17). The appendices to this report present the test results for each section of the evaluation in visual maps and various tables. The reference file also discusses the procedures used to analyze the collected data to determine the structural capacity and surface condition of the individual pavement sections. Allowable gross loads (AGL), Pavement Classification Numbers (PCN), Pavement Condition Index (PCI) data, and the Engineering Assessment (EA) are presented for each pavement section.

Conclusions

1. **Structural Capability:** Most airfield pavements are capable of supporting current mission requirements. There is only one section (T19A) on Taxiway Alpha with AGL limitations based on the C-130H primary mission aircraft. Overall, the runway and taxiway networks are in GOOD condition and suffer from few load-related distresses. The “Using the Structural Results” section of the reference file provides general examples of how to use the AGL tables to answer specific questions. These techniques will help airfield managers and base leadership make informed decisions on pavement use and overload scenarios to preserve pavement life while balancing mission requirements. The standard PCNs published in National Geospatial-Intelligence Agency (NGA) references for U.S. Air Force airfields are based on the most restrictive primary runway section (full width over 1,000 feet at the ends and the 75-foot-wide keel section for the entire length of the runway) and maximum-weight C-17 aircraft at 50,000 passes. **For the Maxwell AFB primary runway and paved assault strip, the recommended PCNs for publication are:**

Runway 15/33: 66 R/B/W/T

Assault Runway 009/189: 56 R/B/W/T

2. **Surface Condition:** The overall surface condition of the airfield pavements at Maxwell AFB ranged from GOOD to SERIOUS, with the majority of pavements by area in GOOD condition. Most pavements will only require routine M&R to sustain the condition. However, the pavements that rated FAIR or worse are due for more major M&R and, in some cases, a new design and complete reconstruction may be required. Of the 56 sections surveyed (not including shoulders), 30 are portland cement concrete (PCC), 25 are asphalt concrete (AC), and one is double bituminous surface treatment (DBST).

   a. Concrete Distresses: The primary distresses observed on the PCC pavements included joint seal damage, faulting, and small patches.

b. Asphalt Concrete Distresses: The primary distresses observed on the AC pavements included low- and medium-severity weathering, block cracking, and longitudinal and transverse cracks.

Recommendations

1. **Asphalt Concrete Sections**

   a. Reconstruct Taxiway A section T19A to provide long-term structural support for mission aircraft (C-130H).

   b. Accomplish a mill and overlay project on Runway 15/33 sections R05A1, R04C1, and R09C.

   c. Accomplish a selective crack-sealing project on Assault Runway 009/189 (R02A), including the turn-around taxiway (T06A).

   d. Program a mill and overlay project for section T06A in FY17. Particular attention should be paid to correcting slope deficiencies between the Assault Runway and Runway 15/33 to restore positive drainage.

   e. Apply an asphalt surface treatment to the shoulder sections of Runway 15/33 and shoulders of Assault Runway 009/189.

2. **Portland Cement Concrete Sections**

   a. If not already programmed, schedule a project to provide M&R of global crack and joint sealant on all PCC pavement sections identified with medium- and high-severity longitudinal/transverse cracking or joint sealant damage. This will slow deterioration of the PCC pavement and reduce spalling. See the extrapolated distress report in Appendix F to identify concrete pavements with Distress Code 63 or 65 at high or medium severity. Focus efforts on Runway 15/33, Assault Runway 009/189, the Maintenance Hangar Access Apron, the West Apron, Taxiway Alpha, and the North Apron.

   b. Continue replacing slabs as necessary and sealing cracks on various pavement sections, focusing funds on Runway 15/33 (R03A1) and the West Apron (A05B).

TABLE OF CONTENTS

EXECUTIVE SUMMARY I
TABLE OF CONTENTS III
LIST OF FIGURES V
INTRODUCTION 1
BACKGROUND INFORMATION 3
CONCLUSIONS AND RECOMMENDATIONS 15
DISTRIBUTION 17

APPENDICES

APPENDIX A—AIRFIELD MAPS A-1
APPENDIX B—REAL PROPERTY INFORMATION B-1
APPENDIX C—PCI COMPARISON AND DETERIORATION C-1
APPENDIX D—PHYSICAL PROPERTY DATA D-1
APPENDIX E—WORK HISTORY REPORT E-1
APPENDIX F—PCI RESULTS F-1
APPENDIX G—FOD POTENTIAL RATING G-1
APPENDIX H—STRUCTURAL EVALUATION RESULTS H-1
APPENDIX I—FRICTION TEST RESULTS I-1
APPENDIX J—ENGINEERING ASSESSMENT SUMMARY TABLE J-1
APPENDIX K—PROBABILITY OF FAILURE DATA K-1

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LIST OF FIGURES

FIGURE 1—ANNUAL TEMPERATURE SUMMARY, MAXWELL AFB, AL 4
FIGURE 2—AVERAGE ANNUAL CLIMATE DATA, MAXWELL AFB, AL 5
FIGURE 3—PONDING ON RUNWAY 15/33 (R04C2) 6
FIGURE 4—PONDING NEAR SECTION T06A.... 6
FIGURE 5—ANALYSIS OF TOTAL AIRFIELD SURFACE CONDITION 7
FIGURE 6—ANALYSIS OF RUNWAY AND OVERRUN SURFACE CONDITION 7
FIGURE 7—ANALYSIS OF TAXIWAY SURFACE CONDITION.... 7
FIGURE 8—ANALYSIS OF APRON SURFACE CONDITION 7
FIGURE 9—OVERRUN, ALZ RUNWAY 009/189 (O02C) 9
FIGURE 10—OVERVIEW RUNWAY 15/33 (R06A1) 9
FIGURE 11—DAMAGE JOINT SEALANT (R06A1) 10
FIGURE 12—JOINT SEAL DAMAGE (R03A).... 10
FIGURE 13— FAULTING OF PCC SLABS (R03A).... 10
FIGURE 14— BLOCK CRACKING/ WEATHERING (R09C).... 10
FIGURE 15— WEATHERING/ LONGITUDINAL CRACKING (R02A).... 10
FIGURE 16—JOINT SPALLING AND JOINT SEAL DAMAGE.... 10
FIGURE 17—OVERVIEW, TAXIWAY ALPHA (T04A).... 11
FIGURE 18—OVERVIEW RUNWAY 009/189 TURN AROUND (T11A), BLOCK CRACKING.... 11
FIGURE 19—JOINT SEAL DAMAGE (T03A) 12
FIGURE 20—JOINT SEAL DAMAGE AND FAULTING (T03A) 12
FIGURE 21—OVERVIEW NORTH APRON (A04B) 13
FIGURE 22—MEDIUM SEVERITY PATCHING ON NORTH APRON (A04B).... 13
FIGURE 23— LOW SEVERITY PATCHING ON WEST APRON (A05B).... 13
FIGURE 24—SHRINKAGE CRACKING, WEST APRON (T29B, A05B) 13
FIGURE 25— SHATTERED SLABS, AERO CLUB APRON (A10C) 13
FIGURE 26— MEDIUM SEVERITY TRANSVERSE CRACKING, AERO CLUB APRON (A10C)) 13
FIGURE 27—BRANCH-LEVEL WEIGHTED AVERAGE PCI BY AREA 14

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INTRODUCTION

Authority

Based on the eight-year structural evaluation cycle for Air Force airfields, eight members from the Asset Visibility Division of the Air Force Civil Engineer Center (AFCEC) Operations Directorate conducted the field portion of an airfield pavement evaluation at Maxwell AFB during the period of 22 June to 1 July 2015.

Description of Appendices

The detailed appendices report the information gathered during the evaluation. Table 1 provides a description of each appendix.

Table 1—Description of Appendices

Appendix	Description
A	**Airfield Maps:** Graphically depicts the different pavement facilities, branches, section designations, rank, surface condition of the airfield, the overall Engineering Assessment (EA) ratings, and core test locations.
B	**Real Property Information:** Includes facility information and branch-level designations.
C	**PCI Deterioration Rate Table:** Historical deterioration by pavement section.
D	**Summary of Physical Property Data**: Tabulates the physical properties data of each evaluated pavement section. Section area, material types, layer thickness, and engineering properties are included.
E	**Construction History:** Provides a brief summary of the construction and maintenance activities on each section.
F	**Section Condition Report and Extrapolated Distresses:** Provides the PCI rating and the distresses recorded for each section.
G	**FOD Potential Rating:** Provides the FOD Potential Rating (as related to the FOD Index or PCI of FOD-producing distresses) used as part of the EA.
H	**Pavement Classification Number (PCN) Table:** Lists PCNs, a standardized method of reporting pavement strength, for each section. **Allowable Gross Loads (AGLs):** Lists the allowable loads for every section at four pass intensity levels for each aircraft group. **Aircraft Classification Number (ACN) Charts:** Provides ACN charts for the 14 standard aircraft groups, plus some additional aircraft. **Related Data:** Includes aircraft group indices, gross weight limits for each aircraft group, and pass intensity levels. **Structural Index Report:** Lists the structural index for the EA by section.
I	**Friction Results:** Includes the texture table (surface type, rubber buildup, and outflow meter times for assessing macro-texture), average friction ratings at 40 and 60 mph for every 500-foot section of each runway, CFME readings (graphical representation) for each runway, and the friction index table as part of the EA.
J	**Engineering Assessment Summary Table**: Presents EA ratings by section for each of the four categories (PCI, Structural Index, FOD Potential Rating, and Friction Index).
K	**Probability of Failure (PoF):** Uses a method outlined in the AFCEC business rules to determine a score for project prioritization along with the Consequence of Failure (CoF). **Project Development:** Provides list of projects recommended by AFCEC engineers after assessing all of the data collected during the evaluation.

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BACKGROUND INFORMATION

Location

Maxwell AFB is located in the northwest corner of Montgomery, Alabama, in the Alabama River Valley. The base is bounded by the Alabama River along the northeast. The valley consists of broad, flat, well-drained areas that include both farmland and wooded areas. The airfield at Maxwell AFB (ICAO: KMXF) is located at an elevation of 171 feet above mean sea level.

Construction History

There have been several recent and ongoing airfield projects since the last structural evaluation in November 2006 and PCI in December 2010. A detailed list by section for all airfield pavements can be found in Appendix E, with a summary below of recent projects completed and programmed.

Recently Completed Projects:

Repair Asphalt Runway 15/33 – 2012

Repair Assault Strip 009/189 – 2012

Mill and Overlay Overrun, Runway 15/33 and 009/189 – 2013

Mill and Overlay Taxiways A, B, C, D, and E – 2014

Programmed Projects:

Replace Sections Runway 15/33 – FY16, PNQS157400, PNQS167500, PNQS172593 ($17M)

Replace Concrete Pads North End Assault Zone – FY16, PNQS137001 ($40K)

Soil Conditions

A soil map of the Montgomery area from the United States Department of Agriculture (USDA), Natural Resource Conservation Service (NRCS) website for Montgomery County, Alabama, shows the predominant subgrade surface soils in the vicinity of Maxwell AFB to be a number of clays (CL), silts (ML), and sands (SC) in the Unified Soil Classification System (USCS). Although very little published data is presented for the airfield area, the most dominant subgrade soil in the vicinity of the airfield is Roanoke Silt Loam. The airfield is constructed in an alluvial valley that contains varying layers of low-plasticity clay (CL), low-plasticity silt (ML), and clayey silts (SC). Both visual field classifications and formal laboratory classifications of subgrade soils were consistent with predominant clay soil subgrades within the pavement structures. Due to the proximity to the river and alluvial deposition of the subgrade soils, high soil moisture contents and low in-place compaction values have been reported during previous evaluations. In general, these subgrade soils are not well suited for supporting pavements as these subgrade soils are known to have moisture sensitivity issues, including strength loss. The pavement structure has performed relatively well, however, despite its age. This is likely the result of comparatively light aircraft traffic loading. Macadam PCC layers were found under many pavement sections before reaching the clay or silt subgrade. Results of laboratory testing are included in the Summary of Physical Properties in Appendix D for all soil layers encountered beneath the pavement to formally classify these soils using the USCS symbol. For more detailed soil information, the USDA's NRCS soil map can be found at the following website: http://websoilsurvey.sc.egov.usda.gov/App/WebSoilSurvey.aspx.

Climate

Maxwell AFB lies in the Alabama River Valley. The climate during most of the year is influenced by the Bermuda High which brings warm, humid maritime air masses from the Gulf of Mexico, located approximately 140 miles to the south. High temperatures in the summer months of May through September are routinely above 90 °F. Temperatures below freezing are not common throughout the year. The average annual precipitation is approximately 47 inches, with July and December being the wettest months. Figures 1 and 2 give temperature and average annual rainfall summaries; more detailed engineering weather data (EWD) can be obtained from the 14th Weather Squadron website at https://www.climate.af.mil/. The area averages 17 freeze-thaw cycles per year and the design freezing index at Maxwell AFB is 34 degree-days. The average depth of frost penetration is a maximum of 9 to 20 inches. The frost depth of 20 inches is not significant enough to provide freezing temperatures in any frost-susceptible materials due to relatively thick pavement layers. Therefore, thaw-weakened AGLs were not produced for this evaluation.

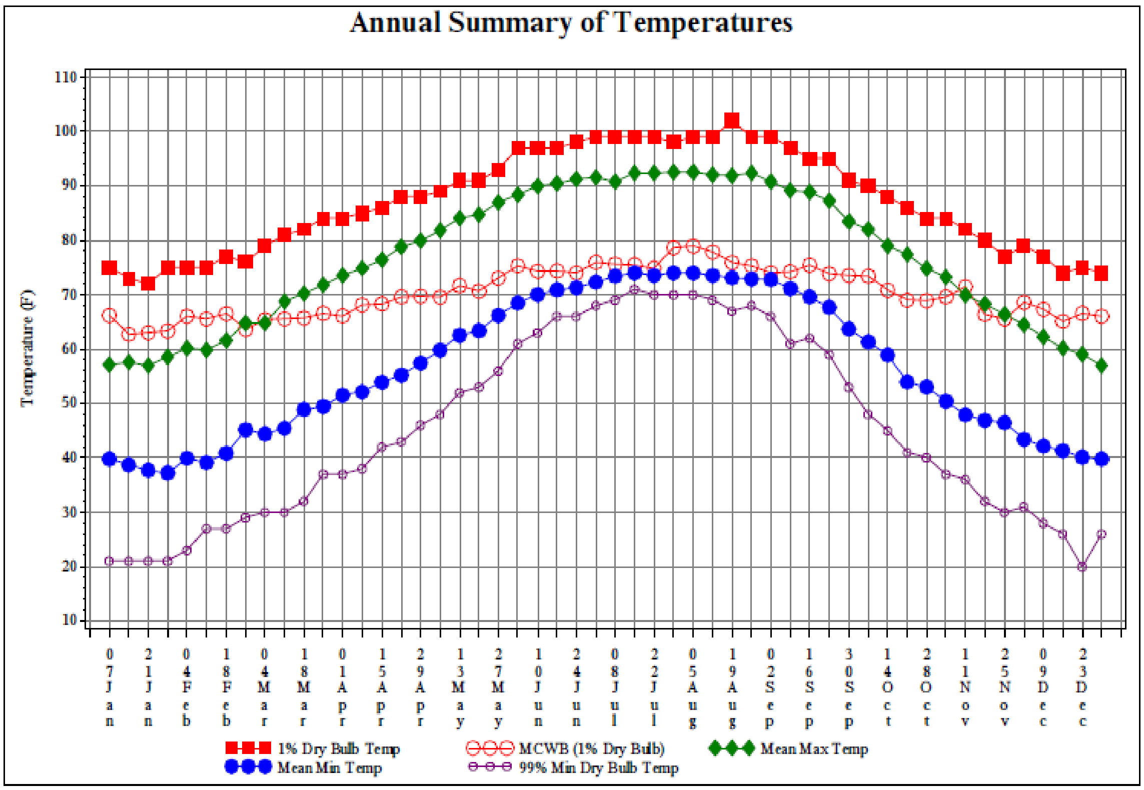

Figure 1—Annual Temperature Summary, Maxwell AFB, AL

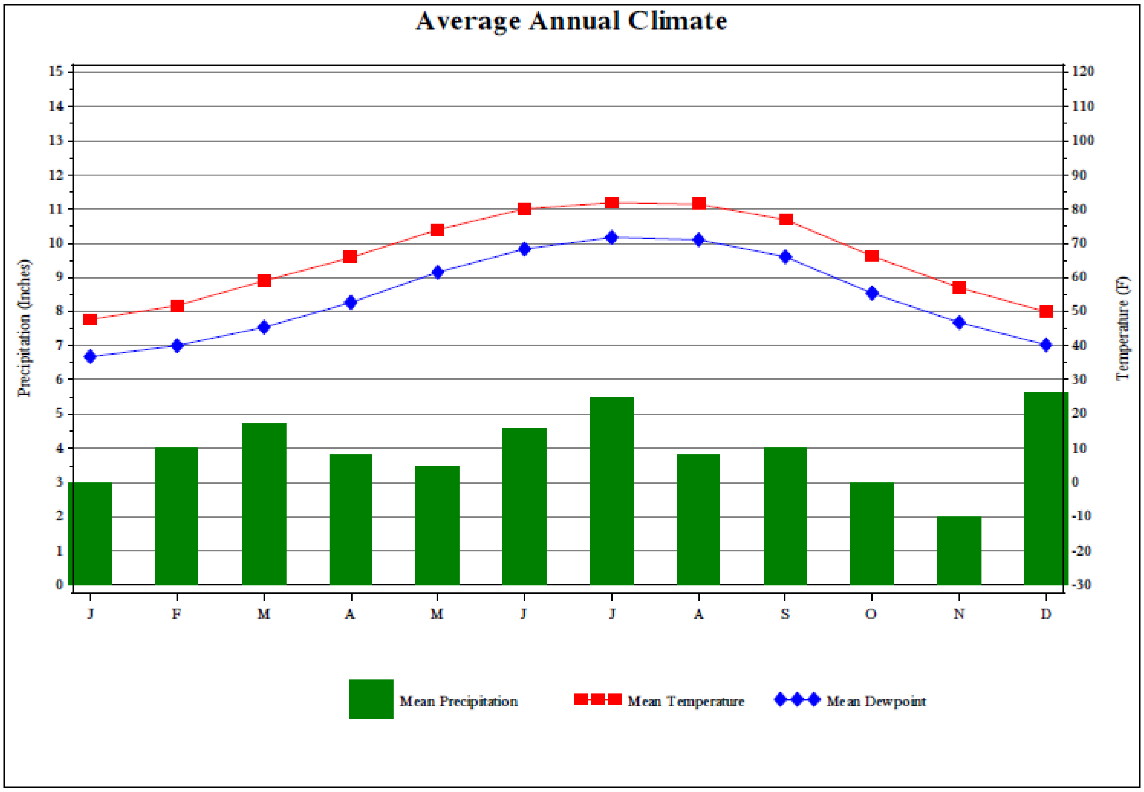

Figure 2—Average Annual Climate Data, Maxwell AFB, AL

Frost Evaluation

Local climate and conditions are such that the potential for freezing and subsequent thaw-weakening of the subgrade need not be addressed and analyzed. The climate history at Maxwell AFB indicates an average of only 17 freeze-thaw cycles annually. For frost susceptibility to be a concern, three conditions must be present: (1) the presence of frost-susceptible soils, (2) the existence of a discernable water source, and (3) freezing temperatures such that the frost line reaches saturated and frost-susceptible soils. If frost-driven distresses are encountered at an airfield, particular attention should be paid to using the thaw-weakened AGLs and PCNs during the spring thaw for the location. At Maxwell AFB, several sections exhibited signs of potentially expansive soil-related distresses, including faulting among PCC slabs, structural distresses (longitudinal/transverse cracking) in non-trafficked areas, and accelerated distress conditions. These pavement distresses are likely the result of expansive soil interactions with local variations in moisture conditions versus the effects of frost.

Depth of Freeze: The ModBerg program developed by the U.S. Army Corps of Engineers Cold Regions Research and Engineering Laboratory indicates that the design air freezing index for Maxwell AFB is 34 °F-Days. Using a Modified Berggren analysis, the design depth of frost penetration under pavements is approximately 9 to 20 inches, depending on pavement thickness and soil type.

Frost Susceptibility: Soil samples collected during this evaluation and from previous evaluations indicated that the base course materials were primarily poorly graded gravels with silt mixtures (GP-GM) or clayey sands (SC). These types of soil generally have a low susceptibility to freeze-thaw action. Subgrade soils were primarily low-plasticity clays (CL) or low-plastic silts (ML). These types of soils generally have a moderate susceptibility to freeze-thaw action.

Water: Soils under the pavements were saturated in some areas. The water table at Maxwell AFB is relatively shallow and has been reported at depths between 20 and 23 feet below the ground surface. Subsurface drainage problems have been previously reported, but are believed to have been corrected by earlier maintenance and construction efforts. Previous evaluations have determined soil moisture contents well above the optimum. As seen in Figure 3 and Figure 4, some ponding of water was noticed during the evaluation on sections of Runway 15/33 and Assault Landing Zone 009/189 (including shoulders), the North Apron, and Taxiway Alpha. However, the presence of shallow groundwater does provide some limited potential for a sufficient amount of water to be available for the formation of frost lenses in the pavement system, leading to vertical displacements at the surface.

Figure 3—Ponding on Runway 15/33 (R04C2)

Figure 4—Ponding Near Section T06A

The need to conduct a frost evaluation has been consistently ruled out during previous structural evaluations conducted at Maxwell AFB dating back to 1980. As such, the effects of freezing, and subsequent thaw weakening of the subgrade, were not considered in the production of PCNs and AGLs as part of this evaluation.

Aircraft Traffic

Aircraft stationed at Maxwell AFB include C-130H aircraft assigned to the 908th Airlift Wing. Various transient aircraft use the airfield, including fighter and cargo aircraft.

Previous Evaluations

Several structural and surface condition pavement evaluations have previously been performed at Maxwell AFB, dating back to 1980 for the first structural evaluation and the first airfield PCI in 2000. The field work for the last PCI evaluation was accomplished in December of 2010.

The previous reports listed below provided reference data for various parts of the evaluation, mainly in terms of past construction history, soil classifications, and previous conditions that may have changed due to new construction. All referenced reports can be found at the AFCEC SharePoint site link: Pavement Evaluation Reports (https://tyndall.eim.acc.af.mil/apps/afcec/Pavement%20Reports/default.aspx)

- "Airfield Pavement Evaluation, Maxwell Air Force Base, Alabama," HQ Air Force Engineering and Services Center (AFESC), Tyndall AFB, Florida, September 1980
- "Airfield Pavement Evaluation, Maxwell Air Force Base, Alabama," Air Force Civil Engineering Center Support Agency (AFCESA), Tyndall AFB, Florida, October 1998
- "Airfield Pavement Evaluation, Maxwell Air Force Base, Alabama," Air Force Civil Engineer Support Agency (AFCESA), Tyndall AFB, Florida, March 2008
- "Airfield Pavement Condition Index Survey Report, Maxwell Air Force Base, Alabama," Applied Research Associates, Inc., December 2010

Pavement Condition – Maxwell AFB Pavements

A pavement condition survey was conducted as part of this airfield pavement evaluation to establish the condition ratings for airfield pavement sections and to identify the types of distresses observed on the airfield. The following pages discuss the findings of the visual survey, along with other aspects of the pavement such as structural and friction observations. All references to pavement condition in the following text use the standard (not simplified) pavement condition ratings defined in ASTM D5340, *Standard Test Method for Airport Pavement Condition Index Surveys*, and are described further in the reference file.

Overall Airfield

The surface condition observed on the Maxwell AFB airfield pavements is generally GOOD, as shown in Figure 5. The condition has improved since the December 2010 PCI due to an aggressive mill and overlay maintenance program on the asphalt sections of the airfield network, including projects on both runways (including their overruns) and Taxiways Alpha, Bravo, Charlie, Delta, and Echo. Approximately 91 percent of the 5.67 million total square feet of pavement sections by area are in GOOD or SATISFACTORY condition, and 6 percent are in FAIR condition. Only two sections were rated as POOR (O01C) or SERIOUS (A10C). Perhaps more importantly, 81 percent of the runway and overrun sections by weighted area are in GOOD or SATISFACTORY condition.

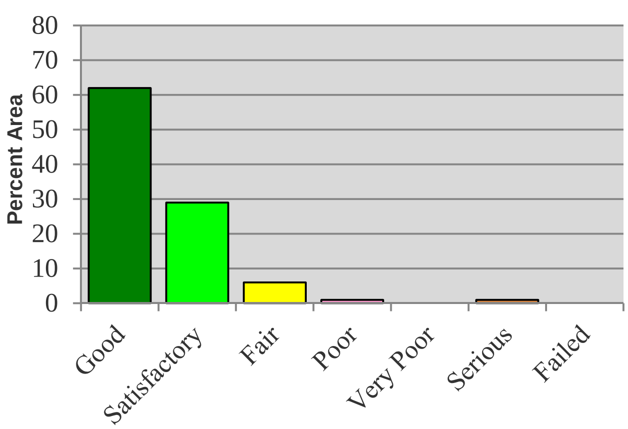

Figure 5—Analysis of Total Airfield Surface Condition

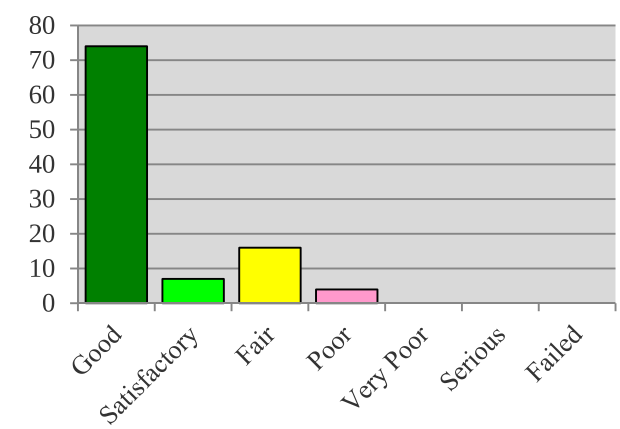

Figure 6—Analysis of Runway and Overrun Surface Condition

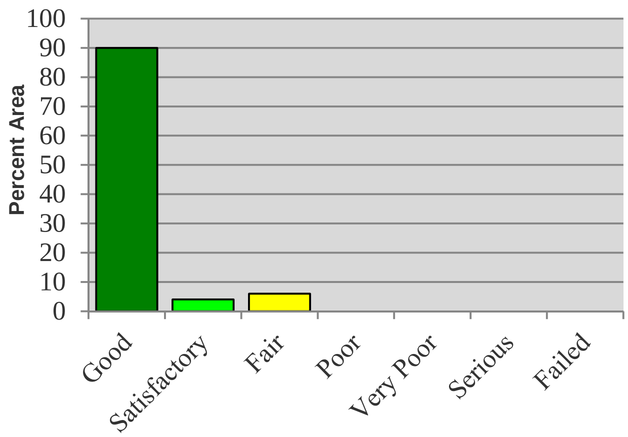

Figure 7—Analysis of Taxiway Surface Condition

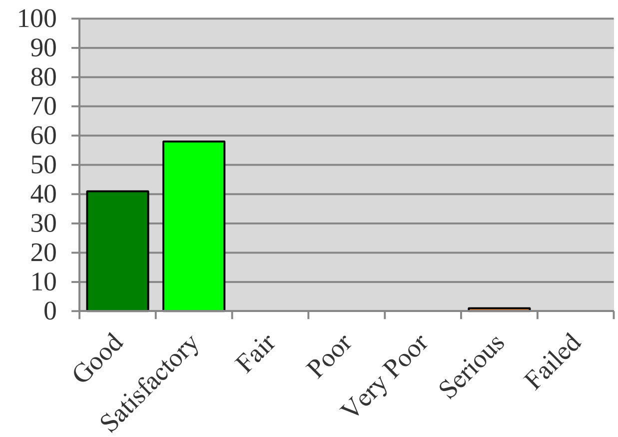

Figure 8—Analysis of Apron Surface Condition

As the overwhelming majority of pavements at Maxwell AFB are in GOOD condition, the only routine maintenance actions that should be required in the near term are to correct low- and medium-severity distresses. Maintenance practices as simple as re-sealing joints and cracks and patching concrete pavements or applying an asphalt surface seal to AC pavements can increase some section PCIs to the next higher category. Preserving pavements in this condition and averting future major M&R projects minimizes operational downtime, extends the overall pavement service life, and requires much less capital expense.

A rating of FAIR generally indicates that pavements are in transition from a state of high-quality surface condition to low-quality condition. FAIR pavements require timely M&R funds to improve the surface condition at a less-expensive long-term cost versus complete reconstruction. Pavements rated POOR or below on the PCI scale generally require major M&R or possibly reconstruction, depending on several factors such as pavement purpose, use, traffic frequency and loads, and the integrity of underlying base and subbase foundation layers of the pavement structure. Base personnel have already programmed projects to complete repairs on all sections rated as FAIR. Section O01C is currently rated as POOR; however, no corrective action is recommended. Only one section on the entire Maxwell AFB airfield rated SERIOUS (Section A10C), the Aero Club Apron. Load reductions were made on the AGL tables in Appendix H for this section.

Appendix F includes a summary of the findings of the pavement surface condition survey, including the condition rating and the types and severity levels of observed pavement distresses. The full seven-category rating scale and visual map of conditions is provided in Appendix A. Refer to UFC 3-260-16FA, *Airfield Pavement Condition Survey Procedures*, and ASTM D5340 for descriptions and definitions of the types and severity levels of the distresses listed in this section of the report or the extrapolated distresses for each section found in Appendix F.

The following paragraphs describe in more detail the observed surface condition of the airfield pavements, broken down at the branch level of runways, taxiways, and aprons.

Runways

The main runway at Maxwell AFB is 8,016 feet long by 150 feet wide; the assault landing zone is 3,000 feet long by 60 feet wide. The main runway, along with the ALZ surfaces and associated overruns, make up approximately 33 percent of the airfield network area. The runways and overruns consist of 23 pavement sections (including seven overrun sections), 18 of which are a part of the main runway. Overall, several project efforts on Runway 15/33 have resulted in a significant PCI increase for this branch since the December 2010 survey and approximately 81 percent of the sections by weighted area are in GOOD or SATISFACTORY condition. However, the majority of the AC keel sections on Runway 15/33 are currently rated FAIR. The current overrun of ALZ 009/189 is 1,000 feet in length, with the nearest 300 feet to the ALZ composed of newly placed AC (Figure 9); the remaining 700 feet of the overrun is constructed of DBST. This portion of the overrun is currently rated in POOR condition due to significant raveling of the aggregate surface. Despite the deterioration of the surface on this portion of the overrun, no repair is currently recommended. Per ETL 09-6, only 300 feet of the overrun is required for C-130 operations and it is highly unlikely that this portion of the overrun will ever receive aircraft traffic.

Runway 15/33 is structurally sound, with no AGL limitations for assigned mission aircraft. Figure 10 shows an overview of the north end of Runway 15/33. The PCC sections of Runway 15/33 are all rated in GOOD condition; however, the joint sealant in most of these sections is generally damaged. For example, in Section R06A1, the compression sealant does not adequately prevent incompressible material from becoming deposited in the joints, which have begun to spall and produce FOD (Figure 11). The silicone joint sealant in Section R03A1 is in considerably worse shape, and is non-existent in some areas (Figure 12). Maintaining good sealant is paramount to prevent moisture infiltration from the surface into the underlying base and subgrade materials. The addition of water into the expansive clay subgrade is likely the leading contributor to faulting of the PCC slabs in this section (Figure 13).

A recent mill and overlay project on the AC sections outside of the runway keel has increased the rating of these pavement sections from FAIR to GOOD. Unfortunately, environmental distresses such as weathering and block cracking have degraded the condition of the AC keel sections of the runway from GOOD to FAIR. In some cases, the cracks have become wide enough for vegetation to appear (Figure 18).

There are also no AGL limitations for assigned mission aircraft on the Assault Strip Runway 009/189. Similar to the main runway, the ALZ is composed of both AC and PCC sections. The primary distresses observed on the AC sections were weathering and longitudinal/transverse cracking, all of which are progressing from low to medium severity (Figure 15). Joint and corner spalling is prevalent in the PCC section (R01A) and is likely exacerbated by the relatively poor condition of the joint sealant in this section (Figure 16).

The lowest-rated runway or overrun sections were the DBST overrun section on ALZ 009/189, section O01C. Besides this overrun section, sections R04C1, R05A1, and R09C on Runway 15/33 rate FAIR. These sections are weathered, with low-severity block cracking beginning to degrade to medium severity. The taxiway lighting and markings on Runway 15/33 have been moved in to decrease the operational width of the runway from 300 to 150 feet. Joint sealant is in poor or non-existent condition in the PCC shoulder sections; the AC shoulder pavements are weathered and experiencing high-severity block cracking. A project should be programmed to replace joint sealant as well as route and seal cracks on the shoulder areas to prevent moisture from infiltrating through the pavement and becoming trapped in the base and subgrade. Maintain the AC surfaces outside of the 150 feet wide markings with sprayed asphalt surface treatments in order to seal cracks and slow the rate of FOD generation.

Figure 9—Overrun, ALZ Runway 009/189 (O02C)

Figure 10—Overview Runway 15/33 (R06A1)

Figure 11—Damaged Joint Sealant (R06A1)

Figure 12—Joint Seal Damage (R03A)

Figure 13— Faulting of PCC Slabs (R03A)

Figure 14— Block Cracking/ Weathering (R09C)

Figure 15— Weathering/ Longitudinal Cracking (R02A)

Figure 16—Joint Spalling and Joint Seal Damage

The primary 8,016-foot runway and 3,000-foot ALZ Runway both consist of non-grooved PCC and weathered AC surfaces. Despite relatively high coefficient of friction (mu) values obtained during continuous friction measurement equipment (CFME) testing, outflow meter times obtained were generally in excess of 5 seconds. This indicates relatively marginal bulk water drainage characteristics; nearly 56 percent of the measured transverse slopes measured on Runway 15/33 were equal to or greater than the 1.0 percent minimum required by UFC 3-260-01. As noted previously, ponding is a common occurrence on both runways during and after rain events. Therefore, there is a **FAIR potential (moderate risk) for the runway pavement to contribute to a hydroplaning incident on both Runway 15/33 and ALZ Runway 009/189**.

The heaviest rubber buildup observed was Medium to Light (see the scale in Appendix I, page I-9 for a description). Currently Maxwell AFB performs rubber removal annually using a detergent process, which was last accomplished in July 2014.

Taxiways

Of the entire pavement area, the open taxiways account for approximately 22 percent of the total airfield and include 17 surveyed sections. The PCI breakout by area is fairly uniform in this branch, with 94 percent of taxiway pavement sections by area rated either GOOD or SATISFACTORY and the remaining 6 percent rating FAIR. This is largely due to an extensive series of mill and overlay projects completed in December 2014 to correct pavement surface deficiencies on the AC portions of Taxiways Alpha, Bravo, Charlie, Delta, and Echo. Figure 17 depicts an example of this recently completed work.

With the exception of a single section (T19A) the taxiway network is capable of supporting mission-assigned aircraft (C-130H) with no AGL limitations. Although this section recently received a structural overlay, the new surface course overlays a relative thin layer of macadam pavement and buried AC. In order to remove AGL limitations in the future, a full-depth repair project should be programmed. The section will likely perform satisfactorily for many years and the current surface should remain until serious distresses develop.

Section T06A is the sole taxiway section rated FAIR; it provides access to the ALZ runway from the main runway. The primary distresses on this section are block cracking and weathering (Figure 18—Overview Runway 009/189 Turn Around (T11A), Block Cracking). A mill and overlay project should be programmed to repair this section. Any resurfacing project should pay particular attention to correcting slope deficiencies in this area and mitigate current ponding issues.

The majority of Taxiway Alpha is composed of PCC pavements; the predominant distresses in these sections are joint seal damage, small and large patching, and faulting. Figure 19 and Figure 20 illustrate the typical high severity of damage to the joint sealant on Taxiway Alpha. The current condition of the joint sealant does not provide adequate protection against moisture intrusion into the base material and subsequently into the highly expansive subgrade. As a result, considerable faulting is present throughout these sections (Figure 19). Correcting joint sealant deficiencies is imperative to maintaining the structural life of these pavement sections and should be conducted as soon as practical via a programmed contract effort and/or an in-house phased maintenance effort.

Figure 17—Overview, Taxiway Alpha (T04A)

Figure 18—Overview Runway 009/189 Turn Around (T11A), Block Cracking

Figure 19—Joint Seal Damage (T03A)

Figure 20—Joint Seal Damage and Faulting (T03A)

Aprons

The apron branch makes up the largest percentage of the airfield pavement area at approximately 45 percent, yet only includes seven sections. The PCI distribution is captured in Figure 8 and shows 99 percent rated in GOOD or SATISFACTORY condition. The apron sections are represented primarily by three large sections that make up the interconnected parking aprons of the North (Transient) Apron and the West (C-130) Apron. Additional smaller aprons include a hangar access apron along the east side of the airfield and an engine run-up apron off of Taxiway Alpha. As with the PCC taxiway sections, the overall apron branch pavement suffers primarily from medium- and high-severity joint seal damage and the PCI would increase with an extensive joint and crack re-seal project.

The large interconnected apron sections rate GOOD (West Apron) and SATISFACTORY (North Apron). Figure 21 shows an overview of the North Apron, which suffered from higher severity distresses including joint and corner spalling as well as linear cracking and patching (Figure 24). The predominant distresses on the West Apron consist of large and small patching, as well as low- and medium-severity corner spalling (Figure 23). A thin surface treatment has previously been applied to sections of the North and West Aprons, presumably to mitigate additional growth of shrinkage cracks present on the surface. This treatment (which gives the finished concrete an odd yellow color) is beginning to degrade, and shrinkage cracks are noticeably beginning to migrate and grow (Figure 24). Although not an immediate structural concern, a reapplication of the initial surface treatment could be performed to prevent additional pavement surface damage. It should be noted that no shattered slabs were encountered on either of these aprons, indicating that the pavement is structurally capable to support mission-assigned aircraft.

Figure 25 and Figure 26 are taken on the Aero Club Apron that rates SERIOUS overall due to low- and medium-severity linear cracking and low- and medium-severity shattered slabs. This apron is primarily used by small-frame Civil Air Patrol and Aero Club aircraft. Despite the extensive damage in this section, there are no AGL restrictions for Group 1 aircraft. No major M&R project is recommended for this section (A10C) at this time, primarily due to pavement use.

Figure 21—Overview North Apron (A04B)

Figure 22—Medium Severity Patching on North Apron (A04B)

Figure 23— Low Severity Patching on West Apron (A05B)

Figure 24—Shrinkage Cracking, West Apron (T29B, A05B)

Figure 25—Shattered Slabs, Aero Club Apron (A10C)

Figure 26—Medium Severity Transverse Cracking, Aero Club Apron (A10C)

Network Summary

Overall, the airfield pavements have been well maintained and have had several reconstruction projects completed in recent years. Since the March 2010 PCI inspection, these efforts have increased the overall area weighted PCI of the airfield network from an 82 to a 91. From these results, it is evident that airfield management and civil engineers are appropriately managing the pavement. An accelerated joint and crack sealant maintenance program on the PCC runway, taxiway, and apron sections would further enhance the overall PCI rating of the airfield and preserve good condition ratings. Additionally, a strategy for routine, long-term maintenance of joint sealant is critical to maintaining the structural integrity of these pavement sections.

From a network perspective, the airfield is structurally capable of carrying primary mission aircraft and some heavier load group transient aircraft. A general lack of load-related distresses was observed during the PCI survey; weight limitations for the various sections can be found in the AGL tables in Appendix H. The average area-weighted PCI by branch level shown in Figure 27 clearly indicates that the airfield is in excellent condition overall. The Apron sections are in the most need of M&R; however, a large percentage of the available parking area is currently not required for mission operations.

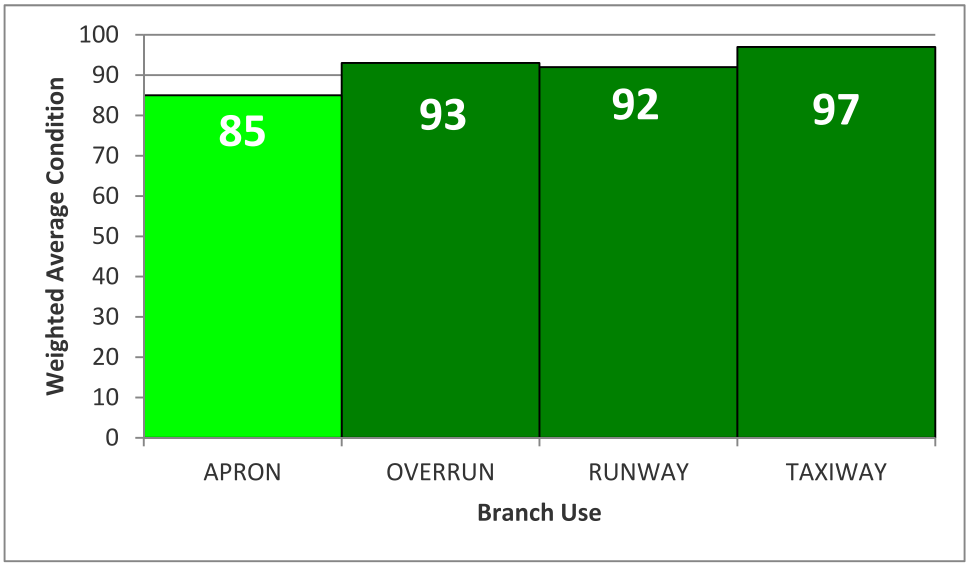

Figure 27—Branch-level Weighted Average PCI by Area

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

1. **Structural Capability:** Most airfield pavements are capable of supporting current mission requirements. There is only one section (T19A) on Taxiway Alpha with AGL limitations based on the C-130H primary mission aircraft. However, the AGL tables in Appendix H should be used to check structural adequacy of more damaging aircraft than the C-130H found in Load Group 4. Based on current use, T19A is recommended for reconstruction based on its limited structural capacity. Overall, the runway and taxiway networks are in satisfactory condition and suffer from few load-related distresses. The “Using the Structural Results” section of the reference file provides general examples of how to use the AGL tables to answer specific questions. These techniques will help airfield managers and base leadership make informed decisions on pavement use and overload scenarios to preserve pavement life while balancing mission requirements. The standard PCNs published in National Geospatial-Intelligence Agency (NGA) references for U.S. Air Force airfields are based on the most restrictive primary runway section (full width over 1,000 feet at the ends and the 75-foot-wide keel section for the entire length of the runway) and maximum-weight C-17 aircraft at 50,000 passes. **For the Maxwell AFB primary runway and paved assault strip, the recommended PCNs for publication are:**

Runway 15/33: 66 R/B/W/T

Assault Runway 009/189: 56 R/B/W/T

2. **Surface Condition:** The overall surface condition of the airfield pavements at Maxwell AFB ranged from SERIOUS to GOOD, with the majority of pavements by area in GOOD condition. Most pavements will only require routine M&R to sustain the condition. However, the pavements that rated FAIR or worse are due for more major M&R; however, complete reconstruction is not anticipated for these sections. Of the 56 sections surveyed (not including shoulders and closed pavements), 30 are PCC, 25 are AC, and one is DBST.

   a. Concrete Distresses: The primary distresses observed on the PCC pavements included joint seal damage, faulting, and small patches.

   b. Asphalt Concrete Distresses: The primary distresses observed on the AC pavements included low- and medium-severity weathering, block cracking, and longitudinal and transverse cracks.

Recommendations

1. **Asphalt Concrete Sections**

   a. Reconstruct Taxiway A section T19A to provide long-term structural support for mission aircraft (C-130H).

   b. Accomplish a mill and overlay project on Runway 15/33 sections R05A1, R04C1, and R09C.

   c. Accomplish a selective crack-sealing project on Assault Runway 009/189 (R02A), including the turn-around taxiway (T06A).

   d. Program a mill and overlay project for section T06A in FY17. Particular attention should be paid to correcting slope deficiencies between the Assault Runway and Runway 15/33 to restore positive drainage.

   e. Apply an asphalt surface treatment to the shoulder sections of Runway 15/33 and shoulders of Assault Runway 009/189.

f. Program a demolition project to selectively remove portions of closed asphalt pavements adjacent to operational surfaces. Removing these pavement sections will prevent excess moisture from infiltrating into the subgrade and becoming trapped in the pavement structure. Additionally, removing these non-operational pavements will allow for re-grading efforts along Runway 15/33 and Assault Runway 009/189 that should mitigate current ponding issues on shoulder pavements.

2. **Portland Cement Concrete Sections**

a. If not already programmed, schedule a project to provide M&R of global crack and joint sealant on all PCC pavement sections identified with medium- and high-severity longitudinal/transverse cracking or joint sealant damage. This will slow deterioration of the PCC pavement and reduce spalling. See the extrapolated distress report in Appendix F to identify concrete pavements with Distress Code 63 or 65 at high or medium severity. Focus efforts on Runway 15/33, Assault Runway 009/189, the Maintenance Hangar Access Apron, the West Apron, Taxiway Alpha, and the North Apron.

b. Continue replacing slabs as necessary and sealing cracks on various pavement sections, focusing funds on the Runway 15/33 (R03A1) and the West Apron (A05B).

c. Continue to perform maintenance repairs in accordance with UFC 3-270-03 and UFC 3-270-04. Discontinue use of silicone joint sealant products to mitigate faulting patches. Faulting, or otherwise failing, patches should be removed and replaced with an approved pavement repair material. A list of current packaged, cementitious and polymeric material for concrete airfield repair is maintained at: https://transportation.erdc.dren.mil/cacsites/TriService/pavement_repair.aspx.

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https://cs3.eis.af.mil/sites/OO-EN-CE-A6/24048/OO-EN-CE-55/APE%20Team/Forms/AllItems.aspx

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Appendix A

Airfield Maps for Maxwell AFB, AL

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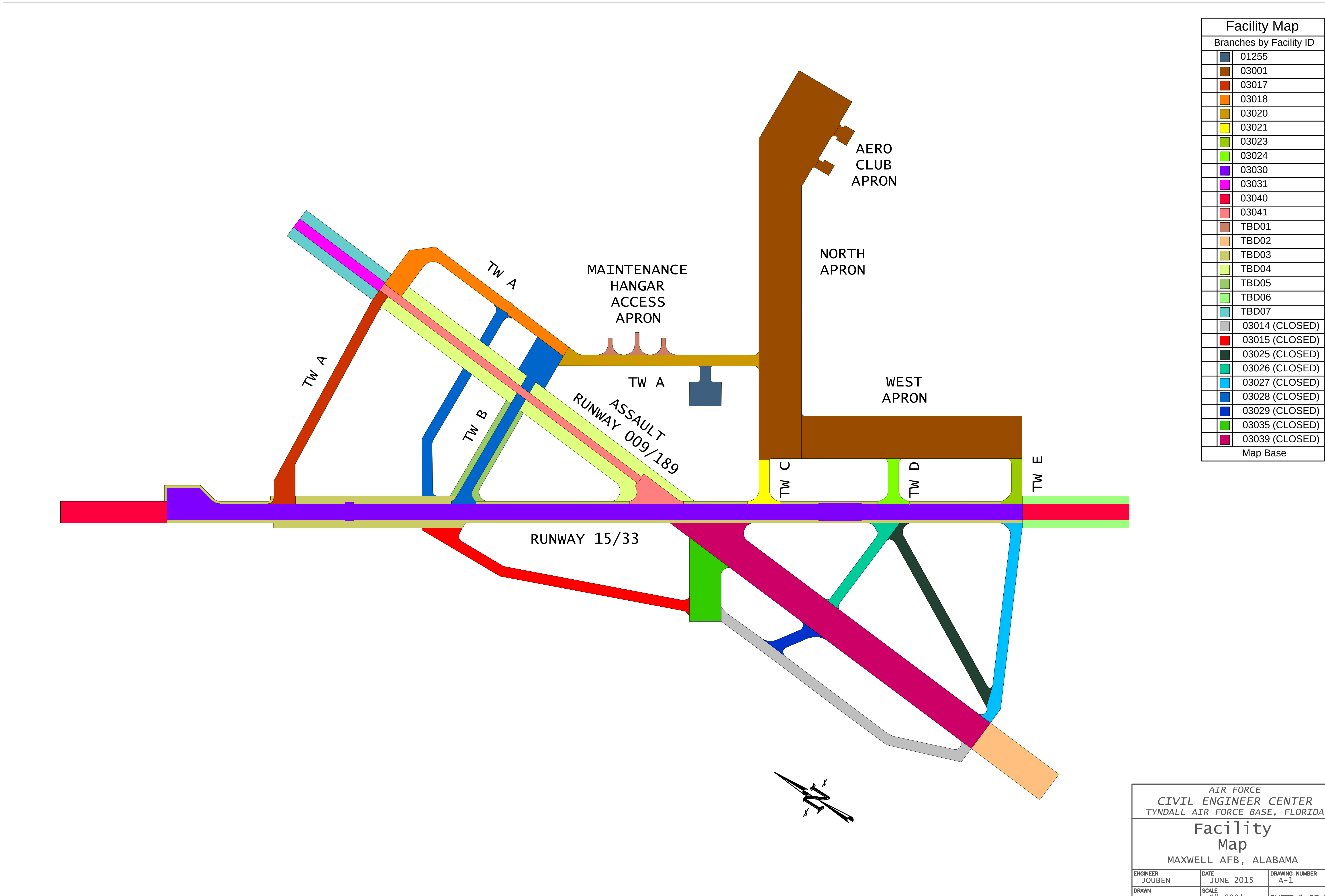

AERO
CLUB
APRON
NORTH
APRON
MAINTENANCE
HANGAR
ACCESS
APRON
TW A
TW A
TW A
TW B
ASSAULT
RUNWAY 009/189
WEST
APRON
TW C
TW D
TW E
RUNWAY 15/33
Facility Map
Branches by Facility ID
01255
03001
03017
03018
03020
03021
03023
03024
03030
03031
03040
03041
TBD01
TBD02
TBD03
TBD04
TBD05
TBD06
TBD07
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03028 (CLOSED)
03029 (CLOSED)
03035 (CLOSED)
03039 (CLOSED)
Map Base
AIR FORCE
CIVIL ENGINEER CENTER
TYNDALL AIR FORCE BASE, FLORIDA
Facility
Map
MAXWELL AFB, ALABAMA
ENGINEER
JOUBEN
DATE
JUNE 2015
DRAWING NUMBER
A-1
DRAWN
KEY
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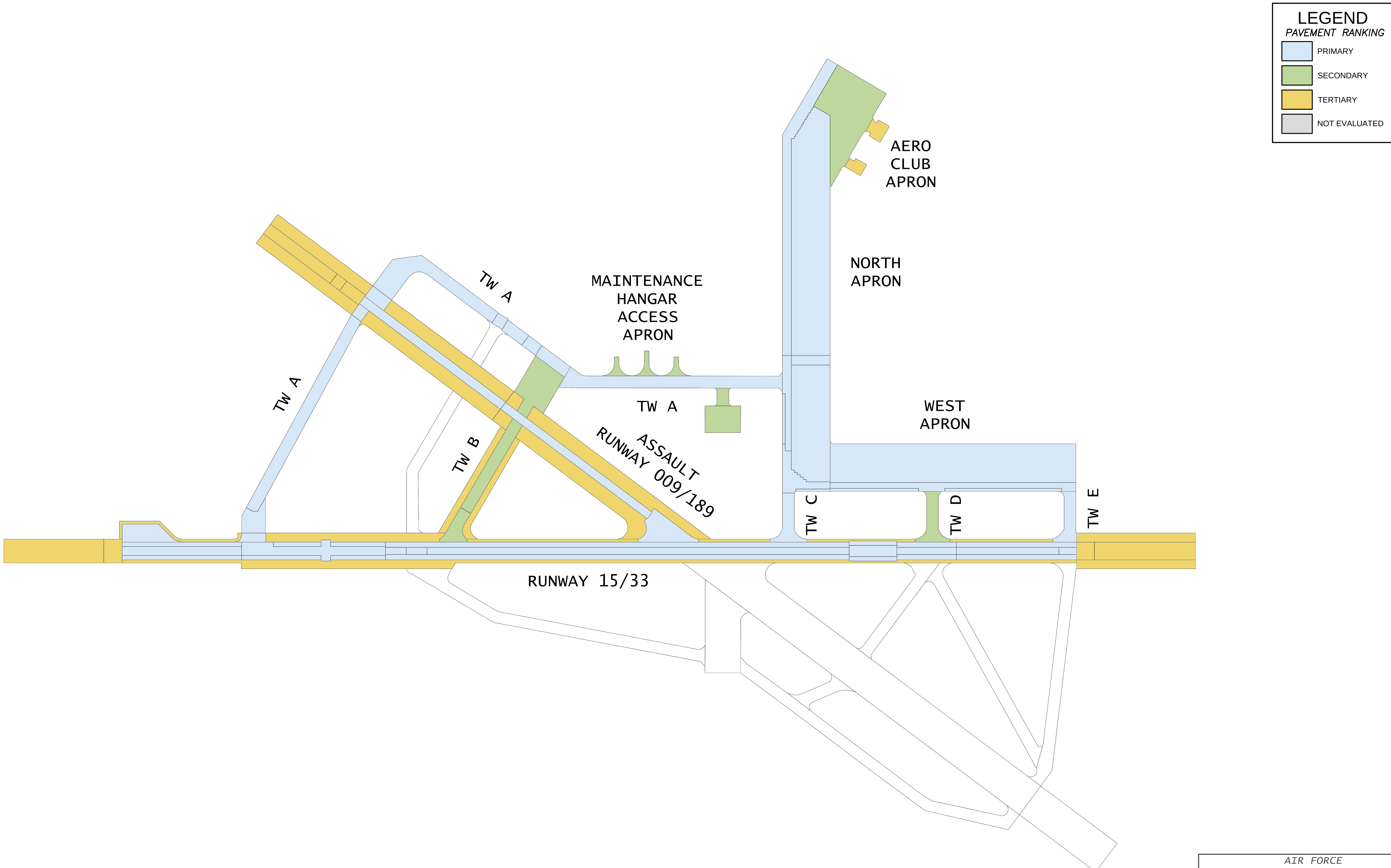
SHEET 1 OF 5



AIR FORCE
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TYNDALL AIR FORCE BASE, FLORIDA

Pavement Ranking Map

MAXWELL AFB, ALABAMA

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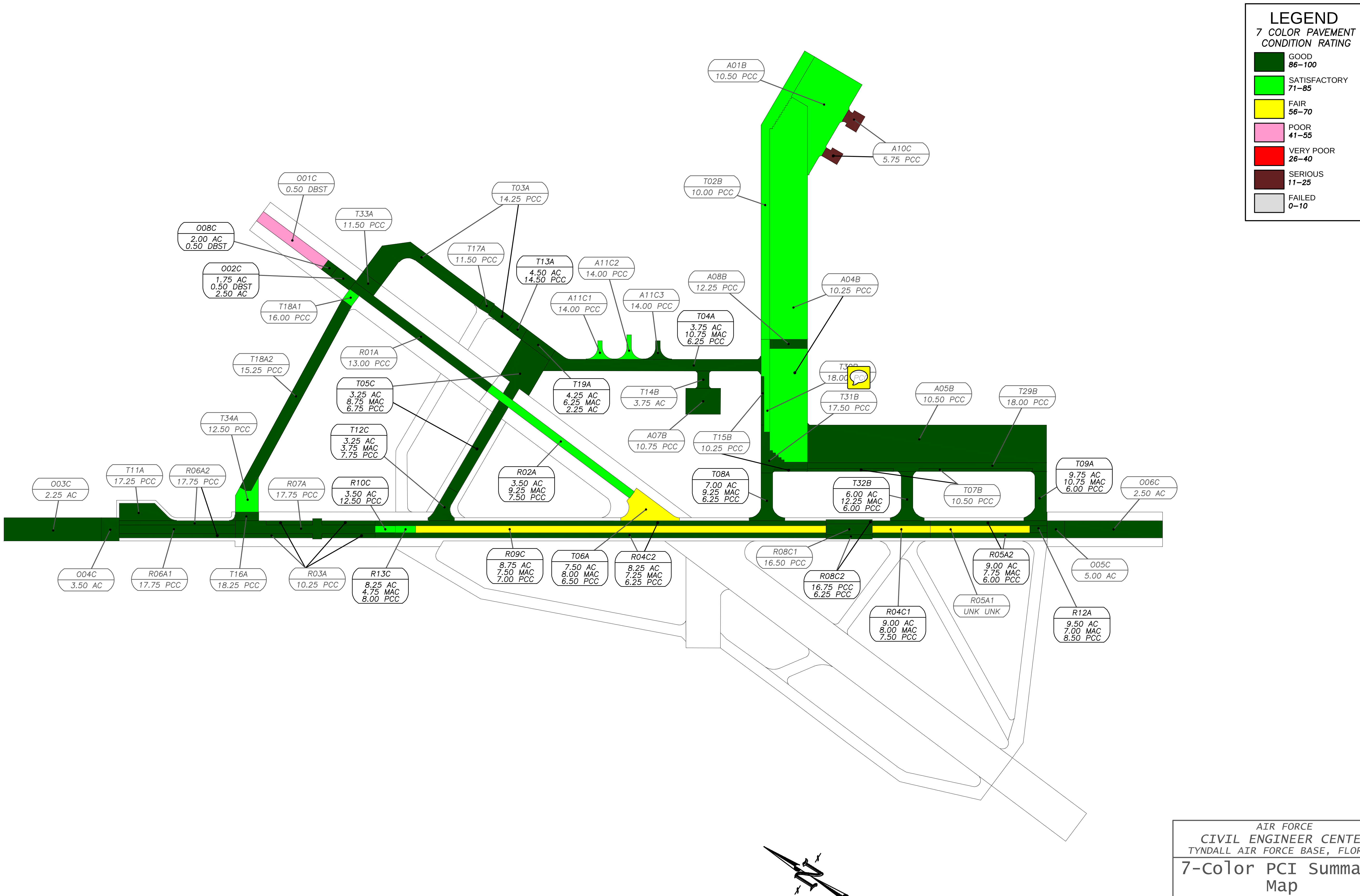
DRAWN	SCALE	
KEY	1"=800'	SHEET 2 OF 5



AIR FORCE
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TYNDALL AIR FORCE BASE, FLORIDA

7-Color PCI Summary Map

MAXWELL AFB, ALABAMA

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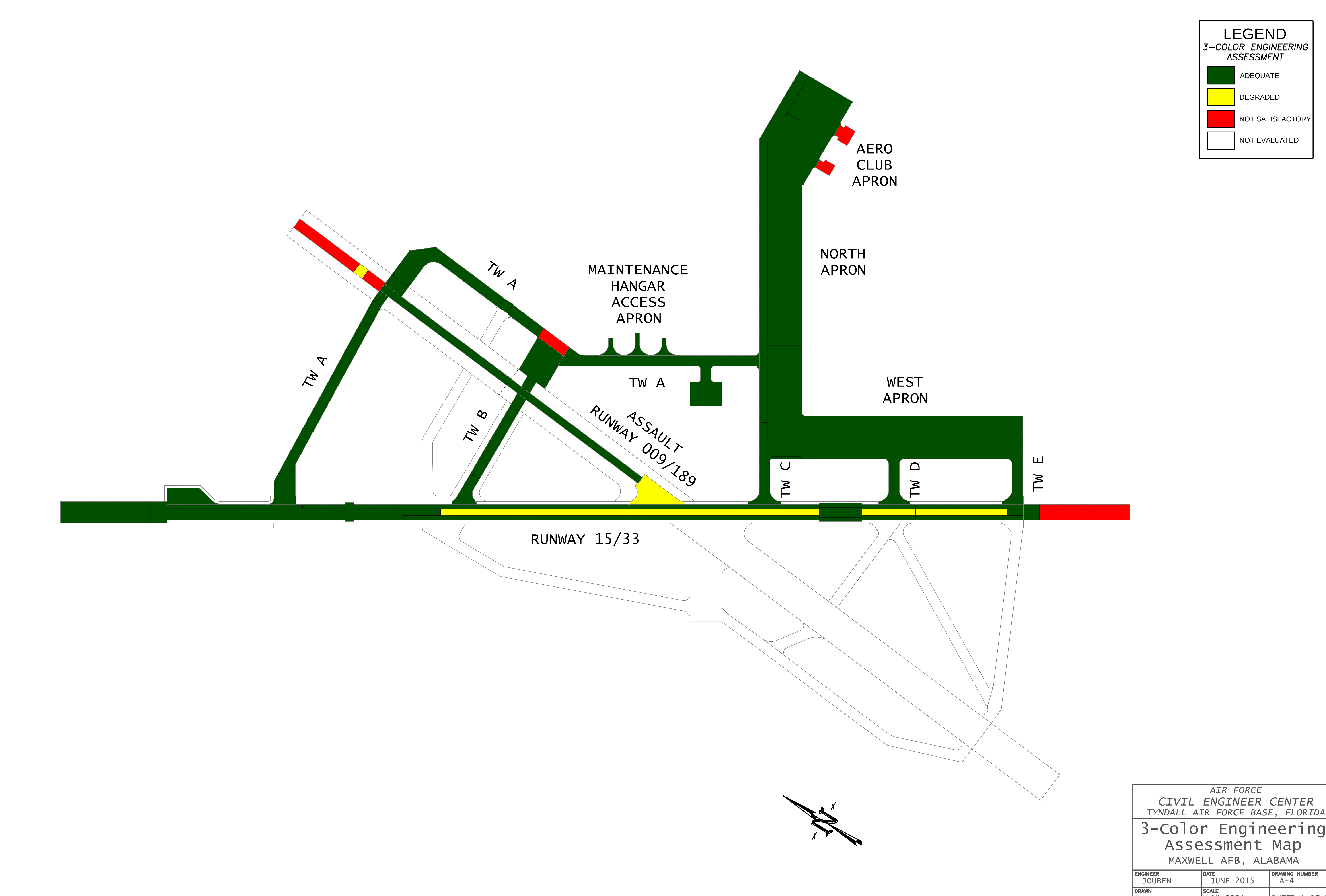
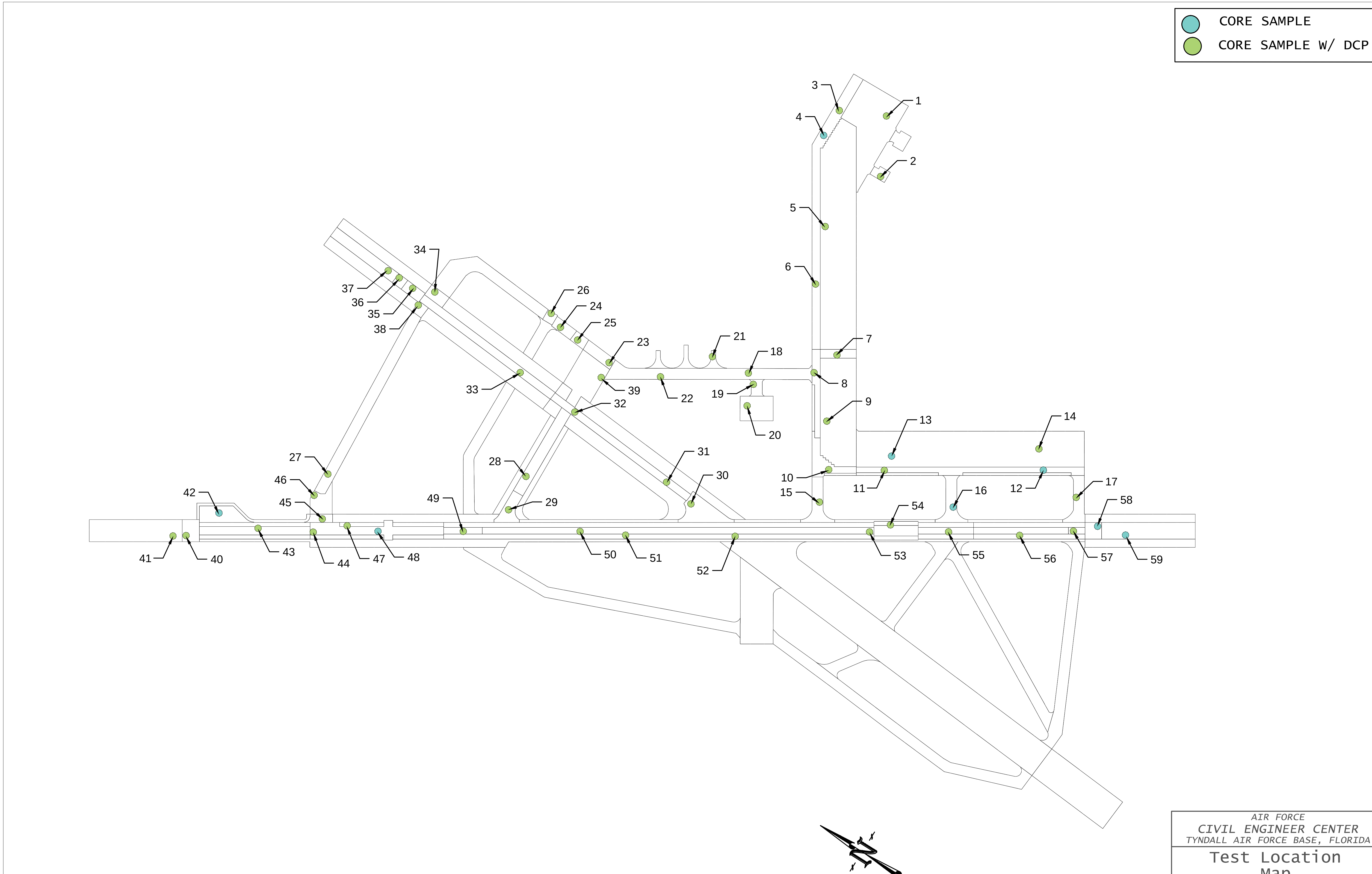

LEGEND
3-COLOR ENGINEERING ASSESSMENT
ADEQUATE
DEGRADED
NOT SATISFACTORY
NOT EVALUATED
AERO CLUB APRON
NORTH APRON
MAINTENANCE HANGAR ACCESS APRON
TW A
TW A
TW A
TW B
ASSAULT RUNWAY 009/189
WEST APRON
TW C
TW D
TW E
RUNWAY 15/33
AIR FORCE
CIVIL ENGINEER CENTER
TYNDALL AIR FORCE BASE, FLORIDA
3-Color Engineering Assessment Map
MAXWELL AFB, ALABAMA
ENGINEER JOUBEN
DATE JUNE 2015
DRAWING NUMBER A-4
DRAWN KEY
SCALE 1"=800'
SHEET 4 OF 5



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Test Location Map

MAXWELL AFB, ALABAMA

ENGINEER JOUBEN	DATE JUNE 2015	DRAWING NUMBER A-5
DRAWN KEY	SCALE 1"=800'	SHEET 5 OF 5

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Appendix B

Real Property Information

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Real Property Table - Maxwell AFB

Facility Number	RPUID	Facility Name	FAC	CATCODE	Branch ID	Branch Name	Rank	Area (SY)	PCI
ACTIVE AIRFIELD									
1255	474394	Engine Run-up Apron	1131	116664	OARUNUP	Engine Run-up Apron	S	9,175.56	94
03001	474380	Main Apron	1131	113321	APAERO	Aero Club Apron	T	4,019.78	14
					APNORTH	North Apron	P	176,603.22	85
					APWEST	West Apron	P	91,719.72	88
03017	474389	Taxiway A North	1121	112211	TWANORTH	Taxiway A (North)	P	27,960.93	89
03018	474390	Taxiway A Northwest	1121	112211	TWANW	Taxiway A (Northwest)	P	28,137.98	91
03020	474392	Taxiway A South	1121	112211	TWA	Taxiway A	P	21,283.33	100
03021	474393	Taxiway C	1121	112211	TWC	Taxiway C	P	5,688.89	100
03023	474395	Taxiway E	1121	112211	TWE	Taxiway E	P	5,400.00	100
03024	474396	Taxiway D	1121	112211	TWD	Taxiway D	S	5,888.89	100
03030	474403	Runway 15/33	1111	111111	RW15/33	Runway 15/33	P	134,551.62	85
					TW15/33 T	Runway 15/33 Turn Around	P	6,380.00	95
03031	474404	Overrun RW 009/189	1113	111115	OR19/01	Overrun Runway 009/189	T	11,111.11	60
03038	474408	Taxiway B	1121	112211	TWB	Taxiway B	S	26,754.44	100
03040	476424	Overrun RW 15/33	1113	111115	OR15/33	Overrun Runway 15/33	T	38,720.56	97
03041	476425	Runway 009/189	1111	111111	RW19/01	Runway 009/189 (Assault Strip)	P	24,270.33	86
03041	476425	Runway 009/189	1111	111111	TW19/01T	Runway 009/189 Turn Around	P	8,042.22	60
TBD01	TBD01	Hangar Access Apron	1131	113321	OAHANG	Maintenance Hangar Access Apron	S	4,009.33	82
SHOULDER PAVEMENTS									
TBD03	TBD03	Runway 15/33 Shoulder	1165	116642	SHRW15/33	Runway 15/33 Shoulders	T	57,294.44	Fair
TBD04	TBD04	Runway 009/189 Shoulder	1165	116642	SHRW186	Runway 009/189 Shoulders	T	76,194.28	Fair
TBD05	TBD05	Taxiway B Shoulder	1165	116642	SHTWB	Taxiway B Shoulders	T	11,414.33	Poor
TBD06	TBD06	Overrun RW 15/33 Shoulders	1165	116642	SHOR15/33	Runway Overrun 15/33 Shoulders	T	16,583.33	Fair
TBD07	TBD07	Overrun RW 009/189 Shoulder	1165	116642	SHOR19/01	Runway Overrun 009/189 Shoulders	T	22,177.78	Poor
INACTIVE PAVEMENTS									
03014	474387	Taxiway 4 (Closed)	1121	112211	TW4	Taxiway 4 (Closed)	U	29,337.78	Poor
03015	474388	Taxiway 7 (Closed)	1121	112211	TW7	Taxiway 7 (Closed)	U	28,954.33	Poor
03025	474397	Taxiway 2 (Closed)	1121	112211	TW2	Taxiway 2 (Closed)	U	20,311.78	Poor
03026	474398	Taxiway 3 (Closed)	1121	112211	TW3	Taxiway 3 (Closed)	U	11,237.78	Poor
03027	474399	Taxiway 1 (Closed)	1121	112211	TW1	Taxiway 1 (Closed)	U	21,636.78	Poor
03028	474400	Taxiway 8 (Closed)	1121	112211	TW8	Taxiway 8 (Closed)	U	18,015.67	Poor
03029	474401	Taxiway 5 (Closed)	1121	112211	TW5	Taxiway 5 (Closed)	U	5,163.00	Poor
03035	474405	Taxiway 6 (Closed)	1121	112211	TW6	Taxiway 6 (Closed)	U	22,671.33	Poor
03039	474409	Runway 009/189 (Closed)	1111	111111	RW19/01c	Runway 009/189 (Closed)	U	112,564.00	Poor
TBD02	TBD02	Overrun 009/189 (Closed)	1113	111115	OR19/01c	Overrun Runway 009/189 (Closed)	U	26,666.67	Poor

Appendix C

PCI Comparison and Deterioration

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PCI Deterioration and Comparison - Maxwell AFB

Branch				Section										
Name	PCI 2009	PCI 2015	PCI Change	ID	Surface Type	Section Rank	Const Date	Age	PCI 2009	PCI 2015	PCI Change	Det Rate 1*	Det Rate 2*	Comments
APRONS														
Aero Club Apron	39	14	-25	A10C	PCC	T	1945	70	39	14	-25	1.23	4.17	Severity of cracks and spalls have increased.
North Apron	89	85	-4	A01B	PCC	S	1996	19	88	85	-3	0.79	0.50	
				A04B	PCC	P	1996	19	89	84	-5	0.84	0.83	
				A08B	PCC	P	1996	19	100	99	-1	0.05	0.17	
				T02B	PCC	P	1985	30	87	84	-3	0.53	0.50	
				T15B	PCC	P	1996	19	79	81	2	1.00	-0.33	Localized patching has occurred since last inspection.
				T30B	PCC	P	1996	19	98	97	-1	0.16	0.17	
				T31B	PCC	P	1996	19	98	96	-2	0.21	0.33	
West Apron	92	88	-4	A05B	PCC	P	1996	19	91	88	-3	0.63	0.50	
				T07B	PCC	P	1985	30	98	89	-9	0.37	1.50	
				T29B	PCC	P	1985	30	88	90	2	0.33	-0.33	
Maintenance Hangar Access Apron	90	82	-8	A11C1	PCC	S	2004	11	88	81	-7	1.73	1.17	Several corner breaks and linear cracks recorded in section.
				A11C2	PCC	S	2005	10	86	77	-9	2.30	1.50	Several corner breaks and linear cracks recorded in section.
				A11C3	PCC	S	2005	10	97	90	-7	1.00	1.17	Linear cracks have developed.
Engine Run-up Apron	95	94	-1	A07B	PCC	S	1985	30	96	92	-4	0.27	0.67	
				T14B	AC	S	2014	1	92	100	8	0.00	-1.33	Major M&R in 2014.
OVERRUNS														
Overrun Runway 15/33	80	97	17	O03C	AC	T	2013	2	80	100	20	0.00	-3.33	Major M&R in 2013.
				O04C	AC	T	2013	2	78	100	22	0.00	-3.67	Major M&R in 2013.
				O05C	AC	T	2013	2	80	94	14	3.00	-2.33	Major M&R in 2013.
				O06C	AC	T	2013	2	80	94	14	3.00	-2.33	Major M&R in 2013.
Overrun Runway 009/189	80	60	-20	O01C	DBST	T	2002	13	80	43	-37	4.38	6.17	The DBST is raveling away significantly.
				O02C	AC	T	2013	2	80	98	18	1.00	-3.00	Major M&R in 2013.
				O08C	AC	T	2013	2	N/A	100	N/A	0.00	N/A	Major M&R in 2013.
RUNWAYS														
Runway 15/33	74	85	11	R03A	PCC	P	1996	19	90	88	-2	0.63	0.33	
				R04C1	AAC	P	1999	16	81	58	-23	2.63	3.83	L&T cracking developed into block cracking; North end of section experiencing swelling.
				R04C2	AAC	P	2013	2	59	93	34	3.50	-5.67	Major M&R in 2013.
				R05A1	AAC	P	1999	16	78	61	-17	2.44	2.83	L&T cracking developed into block cracking.
				R05A2	AAC	P	2013	2	65	94	29	3.00	-4.83	Major M&R in 2013.
				R06A1	PCC	P	1996	19	99	96	-3	0.21	0.50	
				R06A2	PCC	P	1996	19	98	96	-2	0.21	0.33	
				R07A	PCC	P	1996	19	100	97	-3	0.16	0.50	
				R08C1	PCC	P	1996	19	100	93	-7	0.37	1.17	Joint seal damage has increased in severity.
				R08C2	PCC	P	1996	19	100	98	-2	0.11	0.33	
				R09C	AAC	P	1999	16	76	60	-16	2.50	2.67	L&T cracking developed into block cracking.
				R10C	AAC	P	2013	2	72	80	8	10.00	-1.33	Major M&R in 2013.
				R12A	AAC	P	1996	19	N/A	100	N/A	0.00	N/A	
				R13C	AC	P	2013	2	N/A	83	N/A	8.50	N/A	Major M&R in 2013.
Runway 009/189 (Assault Strip)	93	86	-7	R01A	PCC	P	1945	70	89	88	-1	0.17	0.17	
				R02A	AAC	P	1996	19	97	83	-14	0.89	2.33	Increase in quantity of L&T cracking.
TAXIWAYS														
Runway 15/33 Turn Around	98	95	-3	T11A	PCC	P	1996	19	98	95	-3	0.26	0.50	
Runway 009/189 Turn Around	76	60	-16	T06A	AC	P	1946	69	76	60	-16	0.58	2.67	L&T cracking developed into block cracking.
Taxiway A	62	100	38	T04A	AAC	P	2014	1	62	100	38	0.00	-6.33	Major M&R in 2014.
Taxiway A (North)	91	89	-2	T16A	PCC	P	1996	19	93	93	0	0.37	0.00	
				T18A1	PCC	P	1986	29	83	80	-3	0.69	0.50	
				T18A2	PCC	P	1986	29	93	91	-2	0.31	0.33	
				T34A	PCC	P	1986	29	85	81	-4	0.66	0.67	

PCI Deterioration and Comparison - Maxwell AFB

Branch				Section										
Name	PCI 2009	PCI 2015	PCI Change	ID	Surface Type	Section Rank	Const Date	Age	PCI 2009	PCI 2015	PCI Change	Det Rate 1*	Det Rate 2*	Comments
Taxiway A (Northwest)	87	91	4	T03A	PCC	P	1986	29	91	89	-2	0.38	0.33	
				T13A	AAC	P	2014	1	71	100	29	0.00	-4.83	Major M&R in 2014.
				T17A	PCC	P	1986	29	86	86	0	0.48	0.00	
				T19A	AC	P	2014	1	65	100	35	0.00	-5.83	Major M&R in 2014.
				T33A	PCC	P	1986	29	89	92	3	0.28	-0.50	
Taxiway B	60	100	40	T05C	AC	S	2014	1	60	100	40	0.00	-6.67	Major M&R in 2014.
				T12C	AC	S	2014	1	76	100	24	0.00	-4.00	Major M&R in 2014.
Taxiway C	72	100	28	T08A	AC	P	2014	1	72	100	28	0.00	-4.67	Major M&R in 2014.
Taxiway D	71	100	29	T32B	AC	S	2014	1	71	100	29	0.00	-4.83	Major M&R in 2014.
Taxiway E	68	100	32	T09A	AC	P	2014	1	68	100	32	0.00	-5.33	Major M&R in 2014.

* Deterioration Rate 1 = PCI reduction per year since initial construction date.
* Deterioration Rate 2 = PCI reduction per year since last PCI inspection date.

Appendix D

Physical Property Data

PHYSICAL PROPERTY DATA MAXWELL AFB, AL															
SECT	IDENT	OVERLAY PAVEMENT			PAVEMENT			BASE			SUBBASE			SUBGRADE	
		THICK (in)		FLEX (psi)	THICK (in)	DESCRP	FLEX (psi)	THICK (in)	DESCRP	K/CBR	THICK (in)	DESCRP	K/CBR	DESCRP	K/CBR
A01B	NORTH APRON	-	-	-	10.50	PCC	800	-	-	-	-	-	-	FAT CLAY W/SAND (CH)[6]	175 -
A04B	NORTH APRON	-	-	-	10.25	PCC	800	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	150 -
A05B	WEST APRON	-	-	-	10.50	PCC	800	8.00	FAT CLAY (CH)[6]	100 -	-	-	-	SILTY SAND (SM)	225 -
A07B	ENGINE RUN-UP APRON	-	-	-	10.75	PCC	785	6.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	250 -	-	-	-	CLAYEY SAND (SC)[6]	150 -
A08B	NORTH APRON	-	-	-	12.25	PCC	800	5.00	POORLY GRADED GRAVEL W/SAND (GP)[6]	275 -	-	-	-	SANDY LEAN CLAY (CL)[6]	170 -
A10C	AERO-CLUB APRON	-	-	-	5.75	PCC	800	10.00	SILTY SAND W/GRAVEL (SC)[6]	110 -	12.00	LEAN CLAY W/SAND (CL)[6]	150 -	FAT CLAY W/SAND (CH)[6]	225 -
A11C1[4]	MAINTENANCE HANGAR ACCESS APRON	-	-	-	14.00	PCC	665	7.00	WELL GRADED GRAVEL	460 -	-	-	-	CLAYEY SAND (SC)[6]	225 -
A11C2[4]	MAINTENANCE HANGAR ACCESS APRON	-	-	-	14.00	PCC	665	7.00	WELL GRADED GRAVEL	460 -	-	-	-	CLAYEY SAND (SC)[6]	225 -
A11C3	MAINTENANCE HANGAR ACCESS APRON	-	-	-	14.00	PCC	665	7.00	POORLY GRADED GRAVEL W/ SILT & SAND (GP-GM)	460 -	-	-	-	CLAYEY SAND (SC)[6]	225 -
O01C	OVERRUN RUNWAY 009/189	-	-	-	0.50	DBST	-	10.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- 30	-	-	-	FAT CLAY W/SAND (CH)[6]	- 7
O02C	OVERRUN RUNWAY 009/189	1.75	AC	-	.50 2.50	DBST AC	-	12.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- 30	-	-	-	CLAYEY SAND W/GRAVEL (SC)	- 7
O03C	OVERRUN RUNWAY 15/33	-	-	-	2.25	AC	-	29.00	POORLY GRADED GRAVEL W/ SILT & SAND (GP-GM)	- 100	-	-	-	FAT CLAY W/SAND (CH)[6]	- 10
O04C	OVERRUN RUNWAY 15/33	-	-	-	3.50	AC	-	29.00	POORLY GRADED GRAVEL W/ SILT & SAND (GP-GM)	- 100	-	-	-	FAT CLAY W/SAND (CH)[6]	- 10
O05C	OVERRUN RUNWAY 15/33	0.50	DBST	-	5.00	AC	-	13.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- 40	-	-	-	CLAYEY SAND W/GRAVEL (SC)[6]	- 20
O06C	OVERRUN RUNWAY 15/33	-	-	-	2.50	AC	-	8.50	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- 20	-	-	-	LEAN CLAY W/SAND (CL)[6]	- 12
O08C	OVERRUN RUNWAY 009/189	2.00	AC	-	0.50	DBST	-	10.50	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- 30	-	-	-	FAT CLAY W/SAND (CH)[6]	- 7
R01A	RUNWAY 009/189 (ASSULT STRIP)	-	-	-	13.00	PCC	800	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	150 -
R02A	RUNWAY 009/189 (ASSULT STRIP)	-	-	-	3.50	AC	-	9.25 7.50	MACADAM PCC	- 100	10.00	CLAYEY SAND W/GRAVEL (SC)	- 7	SANDY LEAN CLAY (CL)[6]	- 10
R03A	RUNWAY 15/33	-	-	-	10.25	PCC	800	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	200 -
R04C1	RUNWAY 15/33	-	-	-	8.25	AC	-	7.25 6.25	MACADAM PCC	- 100	12.50	CLAYEY SAND W/ GRAVEL (SC)	- 8	SANDY LEAN CLAY (CL)[6]	- 11

PHYSICAL PROPERTY DATA MAXWELL AFB, AL																
SECT	IDENT	OVERLAY PAVEMENT			PAVEMENT			BASE			SUBBASE			SUBGRADE	
		THICK (in)		FLEX (psi)	THICK (in)	DESCRP	FLEX (psi)	THICK (in)	DESCRP	K/CBR	THICK (in)	DESCRP	K/CBR	DESCRP	K/CBR
R04C2	RUNWAY 15/33	-	-	-	8.25	AC	-	7.25 6.25	MACADAM PCC	- / 100	12.50	CLAYEY SAND W/ GRAVEL (SC)	- / 8	SANDY LEAN CLAY (CL)[6]	- / 11
R05A1[4]	RUNWAY 15/33	-	-	-	9.00	AC	-	7.75 6.00	MACADAM PCC	- / 100	9.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- / 30	SANDY LEAN CLAY (CL)[6]	- / 11
R05A2	RUNWAY 15/33	-	-	-	9.00	AC	-	7.75 6.00	MACADAM PCC	- / 100	9.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- / 30	SANDY LEAN CLAY (CL)[6]	- / 11
R06A1	RUNWAY 15/33	-	-	-	17.25	PCC	725	10.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	450 / -	-	-	-	SANDY LEAN CLAY (CL)[6]	200 / -
R06A2	RUNWAY 15/33	-	-	-	17.25	PCC	725	10.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	450 / -	-	-	-	SANDY LEAN CLAY (CL)[6]	200 / -
R07A	RUNWAY 15/33	-	-	-	17.75	PCC	705	6.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	375 / -	-	-	-	SANDY LEAN CLAY (CL)[6]	150 / -
R08C1[4]	RUNWAY 15/33	-	-	-	16.75	PCC	735	6.25	PCC	-	-	-	-	CLAYEY SAND W/GRAVEL (SC)[6]	200 / -
R08C2	RUNWAY 15/33	-	-	-	16.75	PCC	655	6.25	PCC	-	-	-	-	CLAYEY SAND W/GRAVEL (SC)[6]	200 / -
R09C	RUNWAY 15/33	-	-	-	8.75	AC	-	7.50 7.00	MACADAM PCC	- / 100	11.00	CLAYEY SAND W/GRAVEL (SC)[6]	- / 11	SANDY LEAN CLAY (CL)[6]	- / 21
R10C[6]	RUNWAY 15/33	3.50	AC	-	12.50	PCC	800	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	- / 7
R12A	RUNWAY 15/33	-	-	-	9.50	AC	-	7.00 8.50	MACADAM PCC	- / 100	11.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- / 20	SANDY SILT (ML)[7]	- / 17
R13C	RUNWAY 15/33	-	-	-	8.25	AC	-	4.75 8.0	MACADAM PCC	- / 100	12.00	SILTY SAND (SM)	- / 5	SANDY LEAN CLAY (CL)[6]	- / 3
T02B	NORTH APRON TAXILANE	-	-	-	10.00	PCC	800	9.00	SANDY LEAN CLAY (CL)[6]	175 / -	8.00	-	-	FAT CLAY W/SAND (CH)[6]	225 / -
T03A	TAXIWAY A (NORTHWEST)	-	-	-	15.00	PCC	800	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	125 / -
T04A	TAXIWAY A	-	-	-	3.75	AC	-	10.75 6.25	MACADAM PCC	- / 100	-	-	-	CLAYEY SAND (SC)[6]	- / 5
T05C	TAXIWAY B	-	-	-	3.25	AC	-	8.75 6.75	MACADAM PCC	- / 100	12.00	CLAYEY SAND W/GRAVEL (SC)	- / 7	FAT CLAY W/SAND (CH)[6]	- / 10
T06A	RUNWAY 009/189 TURN AROUND	-	-	-	7.50	AC	-	8.00 6.50	MACADAM PCC	- / 100	10.00	CLAYEY GRAVEL (GC)[6]	- / 6	SANDY LEAN CLAY (CL)[6]	- / 8
T07B[4]	WEST APRON TAXILANE	-	-	-	10.50	PCC	785	8.00	FAT CLAY (CH)[6]	100 / -	-	-	-	CLAYEY SAND (SC)[6]	225 / -
T08A	TAXIWAY C	-	-	-	7.00	AC	-	9.25 6.25	MACADAM PCC	- / 100	-			SAND (SP)[8]	- / 15
T09A	TAXIWAY E	-	-	-	9.75	AC	-	10.75 6.00	MACADAM PCC	- / 100	-	-	-	SAND (SP)[8]	- / 13

PHYSICAL PROPERTY DATA
MAXWELL AFB, AL

SECT	IDENT	OVERLAY PAVEMENT			PAVEMENT			BASE			SUBBASE			SUBGRADE	
		THICK (in)		FLEX (psi)	THICK (in)	DESCRP	FLEX (psi)	THICK (in)	DESCRP	K/CBR	THICK (in)	DESCRP	K/CBR	DESCRP	K/CBR
T11A	RUNWAY 15/33 TURNAROUND	-	-	-	17.25	PCC	710	10.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	450 / -	-	-	-	SANDY LEAN CLAY (CL)[6]	200 / -
T12C	TAXIWAY B	-	-	-	3.25	AC	-	3.75 7.75	MACADAM PCC	- / 100	8.00	CLAYEY GRAVEL (GC)[6]	- / 3	SANDY LEAN CLAY (CL)[8]	- / 8
T13A	TAXIWAY A (NORTHWEST)	-	-	-	4.50	AC	-	14.50	PCC	-	-	-	-	SANDY LEAN CLAY (CL)[6]	- / 4
T14B	ENGINE RUN-UP APRON ACCESS TAXIWAY	-	-	-	3.75	AC	-	22.00	POORLY GRADED GRAVEL W/SILT & SAND (GP-GM)[6]	- / 60	-	-	-	CLAYEY SAND (SC)[6]	- / 6
T15B[4]	NORTH APRON TAXILANE	-	-	-	10.25	PCC	785	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	150 / -
T16A	TAXIWAY A (NORTH)	-	-	-	18.25	PCC	685	28.00	POORLY GRADED GRAVEL W/ SILT & SAND (GP-GM)	300 / -			450 / -	SANDY LEAN CLAY (CL)6	150 / -
T17A	TAXIWAY A (NORTHWEST)	-	-	-	11.50	PCC	715	-	-	-	26.50	SAND (SP)[8]	250 / -	SANDY LEAN CLAY (CL)[6]	100 / -
T18A1	TAXIWAY A (NORTH)	-	-	-	16.00	PCC	800	-	-	-	-	-	-	FAT CLAY W/SAND (CH)[6]	150 / -
T18A2	TAXIWAY A (NORTH)	-	-	-	15.25	PCC	800	14.50	CLAYEY SAND W/GRAVEL (SC)	150 / -	-	-	-	SANDY LEAN CLAY (CL)[8]	150 / -
T19A	TAXIWAY A (NORTHWEST)	-	-	-	4.25	AC	-	6.25 2.25	MACADAM AC	- / 100	8.00	CLAYEY SAND W/GRAVEL (SC)	- / 15	SANDY LEAN CLAY (CL)[6]	- / 5
T29B	WEST APRON TAXILANE	-	-	-	18.00	PCC	675	4.00	POORLY GRADED GRAVEL W/SAND (GP)[6]	350 / -	-	-	-	SILT (ML)[8]	250 / -
T30B	NORTH APRON TAXILANE	-	-	-	18.00	PCC	770	5.00	POORLY GRADED GRAVEL W/SAND (GP)[6]	270 / -	-	-	-	SANDY LEAN CLAY (CL)[6]	225 / -
T31B	NORTH APRON TAXILANE	-	-	-	17.50	PCC	575	4.50	POORLY GRADED GRAVEL W/SAND (GP)[6]	300 / -	-	-	-	SILT (ML)[8]	200 / -
T32B	TAXIWAY D	-	-	-	6.00	AC	-	12.25 6.00	MACADAM SILT	- / 100	6.00	PCC	- / 100	SILTY SAND (SM)	- / 10[5]
T33A	TAXIWAY A (NORTHWEST)	-	-	-	11.50	PCC	789	-	-	-	-	-	-	SANDY LEAN CLAY (CL)[6]	125 / -
T34A	TAXIWAY A (NORTH)	-	-	-	12.50	PCC	800	-	-	-	-	-		SANDY CLAY	150 / -

NOTES:

1. **Flexural strengths for PCC pavements determined using split tensile method on samples taken from in-situ concrete from current and previous evaluations, capped at 800 psi.**
2. **PCC - Portland Cement Concrete.**
   RPCC - Reinforced Portland Cement Concrete
   AC - Asphaltic Concrete.
   HQ STAB - High Quality Stabilized Base
   STAB - Stabilized Base
   CTB - Cement Treated Base
   CSB - Cement Stabilized Base
   RAP - Recycled Asphalt Pavement
   NT - Not Testable
3. **A soil classification with no notation indicates laboratory USCS soil classification in 2015. All laboratory soil classifications conform to ASTM standards.**
4. **Data taken from neighboring feature.**
5. **Information taken from the 2008 AFCESA report.**
6. **Information taken from the 1998 AFCESA report.**
7. **Information taken from the 1990 AFESC report.**
8. **Information taken from the 1980 AFESC report.**

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Appendix E

Work History Report

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Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** APAERO Aero Club Apron **Section:** A10C **Surface:** PCC
L.C.D.: 1/1/1945 **Use:** APRON **Rank:** T **Length:** 137.30 (Ft) **Width:** 154.00 (Ft) **True Area:** 36,178.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1945	NU-IN	New Construction - Initial	0.00	5.75	☑	Original Construction

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** A01B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** S **Length:** 572.70 (Ft) **Width:** 475.00 (Ft) **True Area:** 274,424.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	10.00	☑	Random slab replacement, reseal joints,
1/1/1981	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Reseal joints and random slab replaceme
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☑	COE apron reconstruction and extension

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** A04B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** P **Length:** 2,945.60 (Ft) **Width:** 325.00 (Ft) **True Area:** 957,320.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	10.00	☑	Random slab replacement, patch spalls, s
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, repair cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE arpon reconstruction and extension

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** A08B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** P **Length:** 325.00 (Ft) **Width:** 80.00 (Ft) **True Area:** 26,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	12.00	☑	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, repair cracks,
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** T02B **Surface:** PCC
L.C.D.: 1/1/1985 **Use:** APRON **Rank:** P **Length:** 2,816.90 (Ft) **Width:** 75.00 (Ft) **True Area:** 211,267.50 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	SL-RP	Slab Replacement	0.00	10.00	☑	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/2/1945	NU-IN	New Construction - Initial	0.00	10.50	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** T15B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** P **Length:** 650.00 (Ft) **Width:** 25.00 (Ft) **True Area:** 16,250.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☑	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/2/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE original construction

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** T30B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** P **Length:** 640.00 (Ft) **Width:** 75.00 (Ft) **True Area:** 48,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☑	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** APNORTH North Apron **Section:** T31B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** P **Length:** 748.90 (Ft) **Width:** 75.00 (Ft) **True Area:** 56,167.50 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☑	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/2/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** APWEST West Apron **Section:** A05B **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** APRON **Rank:** P **Length:** 2,063.80 (Ft) **Width:** 325.00 (Ft) **True Area:** 670,735.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	10.00	☑	Random slab replacement, patch spalls, s
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, repair cracks,
1/1/1980	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, reseal joints,
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE original constuction

Network: Maxwell Air Force Ba **Branch:** APWEST West Apron **Section:** T07B **Surface:** PCC
L.C.D.: 1/1/1985 **Use:** APRON **Rank:** P **Length:** 1,723.70 (Ft) **Width:** 25.00 (Ft) **True Area:** 43,092.50 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	SL-RP	Slab Replacement	0.00	10.00	☑	Random slab replacement, seal cracks,
1/1/1980	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, MAX 79-009
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** APWEST West Apron **Section:** T29B **Surface:** PCC
L.C.D.: 1/1/1985 **Use:** APRON **Rank:** P **Length:** 2,233.00 (Ft) **Width:** 50.00 (Ft) **True Area:** 111,650.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	SL-RP	Slab Replacement	0.00	10.00	☑	Random slab replacement, seal cracks,
1/1/1980	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, MAX 79-009
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** OAHANG Maintenance Hang **Section:** A11C1 **Surface:** PCC
L.C.D.: 1/1/2004 **Use:** APRON **Rank:** S **Length:** 284.50 (Ft) **Width:** 40.00 (Ft) **True Area:** 11,380.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/2004	NU-IN	New Construction - Initial	0.00	14.00	☑	Original construction, PNQS02-9010

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** OAHANG Maintenance Hang **Section:** A11C2 **Surface:** PCC
L.C.D.: 1/1/2005 **Use:** APRON **Rank:** S **Length:** 333.10 (Ft) **Width:** 40.00 (Ft) **True Area:** 13,324.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/2005	NU-IN	New Construction - Initial	0.00	14.00	☑	PNQS95-9004

Network: Maxwell Air Force Ba **Branch:** OAHANG Maintenance Hang **Section:** A11C3 **Surface:** PCC
L.C.D.: 1/1/2005 **Use:** APRON **Rank:** S **Length:** 284.50 (Ft) **Width:** 40.00 (Ft) **True Area:** 11,380.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/2005	NU-IN	New Construction - Initial	0.00	14.00	☑	PNQS95-9004

Network: Maxwell Air Force Ba **Branch:** OARUNUP Engine Run-up Ap **Section:** A07B **Surface:** PCC
L.C.D.: 1/1/1985 **Use:** APRON **Rank:** S **Length:** 223.40 (Ft) **Width:** 300.00 (Ft) **True Area:** 67,020.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	NU-IN	New Construction - Initial	0.00	10.75	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** OARUNUP Engine Run-up Ap **Section:** T14B **Surface:** AC
L.C.D.: 12/12/201 **Use:** APRON **Rank:** S **Length:** 155.60 (Ft) **Width:** 100.00 (Ft) **True Area:** 15,560.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,593.00	2.00	☑	
1/1/1997	NU-IN	New Construction - Initial	0.00	3.25	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** OR15/33 Overrun Runway 1 **Section:** O03C **Surface:** AC
L.C.D.: 11/18/201 **Use:** OVERRU **Rank:** T **Length:** 840.00 (Ft) **Width:** 200.00 (Ft) **True Area:** 168,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
11/18/2013	CR-AC	Complete Reconstruction - AC	558,427.00	2.00	☑	
1/1/2000	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks
1/1/1997	NU-IN	New Construction - Initial	0.00	0.50	☑	Original construction, COE 94B0048

Network: Maxwell Air Force Ba **Branch:** OR15/33 Overrun Runway 1 **Section:** O04C **Surface:** AC
L.C.D.: 11/18/201 **Use:** OVERRU **Rank:** T **Length:** 155.50 (Ft) **Width:** 200.00 (Ft) **True Area:** 31,100.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	558,427.00	2.00	☑	
1/1/1996	NU-IN	New Construction - Initial	0.00	2.50	☑	Original construciton, COE 94B0048

Network: Maxwell Air Force Ba **Branch:** OR15/33 Overrun Runway 1 **Section:** O05C **Surface:** AC
L.C.D.: 11/18/201 **Use:** OVERRU **Rank:** T **Length:** 150.00 (Ft) **Width:** 150.00 (Ft) **True Area:** 22,500.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	558,427.00	2.00	☑	
1/1/2000	CS-AC	Crack Sealing - AC	0.00	0.00	☐	
1/1/1998	CR-AC	Complete Reconstruction - AC	0.00	0.00	☑	Complete reconstruction with DBST

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** OR15/33 Overrun Runway 1 **Section:** O05C **Surface:** AC
L.C.D.: 11/18/201 **Use:** OVERRU **Rank:** T **Length:** 150.00 (Ft) **Width:** 150.00 (Ft) **True Area:** 22,500.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1958	NU-IN	New Construction - Initial	0.00	2.50	☑	0.5" DBST over AC, Original constructi

Network: Maxwell Air Force Ba **Branch:** OR15/33 Overrun Runway 1 **Section:** O06C **Surface:** AC
L.C.D.: 11/18/201 **Use:** OVERRU **Rank:** T **Length:** 845.90 (Ft) **Width:** 150.00 (Ft) **True Area:** 126,885.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	558,427.00	2.00	☑	
1/1/2000	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks
1/1/1998	CR-AC	Complete Reconstruction - AC	0.00	0.00	☑	Complete reconstruction with DBST
1/1/1958	NU-IN	New Construction - Initial	0.00	0.50	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** OR009/189 Overrun Runway 0 **Section:** O01C **Surface:** DBST
L.C.D.: 1/1/2002 **Use:** OVERRU **Rank:** T **Length:** 700.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 70,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/2002	SU-DB	Surface Course - Double Bitum.	0.00	0.00	☑	Reconstructed with DBST
1/1/1958	NU-IN	New Construction - Initial	0.00	0.50	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** OR009/189 Overrun Runway 0 **Section:** O02C **Surface:** AC
L.C.D.: 11/18/201 **Use:** OVERRU **Rank:** T **Length:** 200.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 20,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	84,401.00	2.00	☑	
1/1/2002	CR-AC	Complete Reconstruction - AC	0.00	0.00	☑	Complete reconstruction with DBST
1/1/1958	NU-IN	New Construction - Initial	0.00	0.00	☑	

Network: Maxwell Air Force Ba **Branch:** OR009/189 Overrun Runway 0 **Section:** O08C **Surface:** AC
L.C.D.: 11/11/201 **Use:** OVERRU **Rank:** T **Length:** 100.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 10,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
11/11/2013	NC-AC	New Construction - AC	0.00	0.00	☑	

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R03A **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,961.20 (Ft) **Width:** 38.00 (Ft) **True Area:** 74,525.60 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	1.50	☑	Random slab replacement, patch spalls, s
1/1/1983	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal PCC joints, MAX 84-0014
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R04C1 **Surface:** AAC
L.C.D.: 1/1/1999 **Use:** RUNWAY **Rank:** P **Length:** 500.00 (Ft) **Width:** 60.00 (Ft) **True Area:** 30,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R04C1 **Surface:** AAC

L.C.D.: 1/1/1999 **Use:** RUNWAY **Rank:** P **Length:** 500.00 (Ft) **Width:** 60.00 (Ft) **True Area:** 30,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00		Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R04C2 **Surface:** AAC

L.C.D.: 2/11/2013 **Use:** RUNWAY **Rank:** P **Length:** 4,391.40 (Ft) **Width:** 90.00 (Ft) **True Area:** 395,226.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	1,107,582.00	2.00	☑	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00		Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R05A1 **Surface:** AAC

L.C.D.: 1/1/1999 **Use:** RUNWAY **Rank:** P **Length:** 1,008.40 (Ft) **Width:** 60.00 (Ft) **True Area:** 51,504.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1999	OL-AS	Overlay - AC Structural	0.00	0.00	☑	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00		Seal cracks, MAX 84-0049
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R05A2 **Surface:** AAC

L.C.D.: 2/11/2013 **Use:** RUNWAY **Rank:** P **Length:** 1,008.40 (Ft) **Width:** 90.00 (Ft) **True Area:** 90,756.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	1,107,582.00	2.00	☑	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	0.00	☑	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00		Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R05A2 **Surface:** AAC

L.C.D.: 2/11/2013 **Use:** RUNWAY **Rank:** P **Length:** 1,008.40 (Ft) **Width:** 90.00 (Ft) **True Area:** 90,756.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R06A1 **Surface:** PCC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,000.80 (Ft) **Width:** 75.00 (Ft) **True Area:** 75,060.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	NU-IN	New Construction - Initial	0.00	17.00	☑	Original construciton, COE 94B0048

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R06A2 **Surface:** PCC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,000.80 (Ft) **Width:** 75.00 (Ft) **True Area:** 75,060.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	NU-IN	New Construction - Initial	0.00	17.00	☑	Original construction, COE 94B0048

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R07A **Surface:** PCC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,110.50 (Ft) **Width:** 98.00 (Ft) **True Area:** 108,829.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	17.00	☑	Replace keel slabs, COE 94B0048
1/1/1983	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 84-0014
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R08C1 **Surface:** PCC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 400.00 (Ft) **Width:** 93.80 (Ft) **True Area:** 37,520.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☑	Remove AC and replace with PCC, COE
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R08C2 **Surface:** PCC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 400.00 (Ft) **Width:** 75.00 (Ft) **True Area:** 30,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☑	Remove AC and replace with PCC, COE
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R08C2 **Surface:** PCC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 400.00 (Ft) **Width:** 75.00 (Ft) **True Area:** 30,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R09C **Surface:** AAC

L.C.D.: 1/1/1999 **Use:** RUNWAY **Rank:** P **Length:** 3,541.40 (Ft) **Width:** 60.00 (Ft) **True Area:** 212,484.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R10C **Surface:** AAC

L.C.D.: 2/11/2013 **Use:** RUNWAY **Rank:** P **Length:** 175.00 (Ft) **Width:** 60.00 (Ft) **True Area:** 10,500.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	31,920.00	2.00	☑	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Mill and overlay

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R12A **Surface:** AAC

L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 150.00 (Ft) **Width:** 60.00 (Ft) **True Area:** 9,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Seal cracks, overlay 150' wide, MAX 84
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☑	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☑	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/1/1943	NC-AC	New Construction - AC	0.00	0.00	☑	

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** RW15/33 Runway 15/33 **Section:** R13C **Surface:** AC
L.C.D.: 2/11/2013 **Use:** RUNWAY **Rank:** P **Length:** 175.00 (Ft) **Width:** 60.00 (Ft) **True Area:** 10,500.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
2/11/2013	NC-AC	New Construction - AC	0.00	0.00	☑	

Network: Maxwell Air Force Ba **Branch:** RW009/189 Runway 009/189 **Section:** R01A **Surface:** PCC
L.C.D.: 1/1/1945 **Use:** RUNWAY **Rank:** P **Length:** 1,478.00 (Ft) **Width:** 75.00 (Ft) **True Area:** 110,850.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1981	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 77-0237
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☑	COE original construction

Network: Maxwell Air Force Ba **Branch:** RW009/189 Runway 009/189 **Section:** R02A **Surface:** AAC
L.C.D.: 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,536.90 (Ft) **Width:** 70.00 (Ft) **True Area:** 107,583.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☑	Mill and overlay, COE 94B0048
1/1/1985	ST-SS	Surface Treatment - Slurry Seal	0.00	0.00	☐	Slurry seal 70' along center line, MAX 8
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** TW15/33 T Runway 15/33 Tur **Section:** T11A **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** TAXIWAY **Rank:** P **Length:** 382.80 (Ft) **Width:** 150.00 (Ft) **True Area:** 57,420.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	NU-IN	New Construction - Initial	0.00	17.00	☑	Original construction, COE 94B0048

Network: Maxwell Air Force Ba **Branch:** TW009/189 Runway 009/189 Tur **Section:** T06A **Surface:** AC
L.C.D.: 1/1/1946 **Use:** TAXIWAY **Rank:** P **Length:** 517.00 (Ft) **Width:** 140.00 (Ft) **True Area:** 72,380.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	ST-SS	Surface Treatment - Slurry Seal	0.00	0.00	☐	Slurry seal, MAX 85-0022
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction project
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** TWANORTH Taxiway A North **Section:** T16A **Surface:** PCC
L.C.D.: 1/1/1996 **Use:** TAXIWAY **Rank:** P **Length:** 75.70 (Ft) **Width:** 212.00 (Ft) **True Area:** 16,048.40 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☑	Remove and replace, COE 94B0048
1/1/1986	SL-PC	Slab Replacement - PCC	0.00	11.50	☐	Random slab replacement and seal crack
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☑	COE original construction and extension

Network: Maxwell Air Force Ba **Branch:** TWANORTH Taxiway A North **Section:** T18A1 **Surface:** PCC
L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 112.10 (Ft) **Width:** 100.00 (Ft) **True Area:** 11,210.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** TWANORTH Taxiway A North **Section:** T18A1 **Surface:** PCC
L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 112.10 (Ft) **Width:** 100.00 (Ft) **True Area:** 11,210.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☑	Random slab replacement and seal crack
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	10.75	☑	COE original construction and extension

Network: Maxwell Air Force Ba **Branch:** TWANORTH Taxiway A North **Section:** T18A2 **Surface:** PCC
L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 1,834.90 (Ft) **Width:** 100.00 (Ft) **True Area:** 183,490.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☑	Random slab replacement and seal crack
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☑	COE original construction and extension

Network: Maxwell Air Force Ba **Branch:** TWANORTH Taxiway A North **Section:** T34A **Surface:** PCC
L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 204.50 (Ft) **Width:** 200.00 (Ft) **True Area:** 40,900.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☑	Random slab replacement and seal crack
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☑	COE original construction and extension

Network: Maxwell Air Force Ba **Branch:** TWANW Taxiway A Northw **Section:** T03A **Surface:** PCC
L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 1,766.70 (Ft) **Width:** 100.00 (Ft) **True Area:** 176,670.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☑	Random slab replacement, seal cracks,
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☑	COE runway extension project

Network: Maxwell Air Force Ba **Branch:** TWANW Taxiway A Northw **Section:** T13A **Surface:** AAC
L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** P **Length:** 141.26 (Ft) **Width:** 100.00 (Ft) **True Area:** 14,126.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	423,597.00	2.00	☑	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☑	Mill, MAX 82-0026, Overlay, MAX 83-
1/1/1958	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Overlay
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction, MACADAM leveli
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** TWANW Taxiway A Northw **Section:** T17A **Surface:** PCC
L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 100.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 10,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☑	Random slab replacement and seal crack
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** TWANW Taxiway A Northw **Section:** T17A **Surface:** PCC

L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 100.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 10,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☑	COE original construction and extension

Network: Maxwell Air Force Ba **Branch:** TWANW Taxiway A Northw **Section:** T19A **Surface:** AC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** P **Length:** 301.70 (Ft) **Width:** 100.00 (Ft) **True Area:** 30,170.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,593.00	2.00	☑	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☑	Mill and overlay, MAX 82-0026 and M
1/1/1946	NU-IN	New Construction - Initial	0.00	3.00	☑	Original construction, COE

Network: Maxwell Air Force Ba **Branch:** TWANW Taxiway A Northw **Section:** T33A **Surface:** PCC

L.C.D.: 1/1/1986 **Use:** TAXIWAY **Rank:** P **Length:** 175.40 (Ft) **Width:** 127.00 (Ft) **True Area:** 22,275.80 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☑	Random slab replacement, seal cracks,
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☑	COE runway extension project

Network: Maxwell Air Force Ba **Branch:** TWA Taxiway A **Section:** T04A **Surface:** AAC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** P **Length:** 1,915.50 (Ft) **Width:** 100.00 (Ft) **True Area:** 191,550.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,593.00	2.00	☑	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☑	Mill, MAX 82-0026, Overlay, MAX 83-
1/1/1958	OL-AT	Overlay - AC Thin	0.00	1.50	☑	Overlay
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction, MACADAM leveli
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** TWB Taxiway B **Section:** T05C **Surface:** AC

L.C.D.: 12/2/2014 **Use:** TAXIWAY **Rank:** S **Length:** 862.50 (Ft) **Width:** 100.00 (Ft) **True Area:** 203,490.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/2/2014	CM-OL-2	2 in Cold Mill & Overlay	450,689.00	2.00	☑	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☑	Mill and overlay, MAX 82-0026, MAX
1/1/1946	NU-IN	New Construction - Initial	0.00	3.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** TWB Taxiway B **Section:** T12C **Surface:** AC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** S **Length:** 373.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 37,300.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	450,689.00	0.00	☑	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☑	Mill and overlay, MAX 82-0026, MAX
1/1/1946	NU-IN	New Construction - Initial	0.00	3.00	☑	Original construction

Work History Report - Maxwell AFB

Network: Maxwell Air Force Ba **Branch:** TWB Taxiway B **Section:** T12C **Surface:** AC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** S **Length:** 373.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 37,300.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments

Network: Maxwell Air Force Ba **Branch:** TWC Taxiway C **Section:** T08A **Surface:** AC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** P **Length:** 512.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 51,200.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,689.00	2.00	☑	
1/1/1983	OL_4	4 in overlay	0.00	4.00	☑	Mill and overlay, MAX 82-022 and MA
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction project
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	Original construction

Network: Maxwell Air Force Ba **Branch:** TWD Taxiway D **Section:** T32B **Surface:** AC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** S **Length:** 530.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 53,000.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	205,679.00	2.00	☑	
1/1/1983	OL_4	4 in overlay	0.00	4.00	☑	Mill and overlay, MAX 82-022 and MA
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction project
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	

Network: Maxwell Air Force Ba **Branch:** TWE Taxiway E **Section:** T09A **Surface:** AC

L.C.D.: 12/12/201 **Use:** TAXIWAY **Rank:** P **Length:** 486.00 (Ft) **Width:** 100.00 (Ft) **True Area:** 48,600.00 (SqFt)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	205,679.00	2.00	☑	
1/1/1983	OL_4	4 in overlay	0.00	4.00	☑	Mill and overlay, MAX 82-022 and MA
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☑	COE reconstruction project
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☑	COE original construction

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Appendix F

PCI Results

Extrapolated Distresses by Section

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PCI Results - Maxwell AFB

Branch Name	Section ID	Area (ft²)	Surface	Section PCI		Distress Related Cause		
				Index	Rating	Climate	Load	Other
APRONS								
Aero Club Apron	A10C	36,178	PCC	14	Poor	8	82	10
North Apron	A01B	274,424	PCC	85	Good	59	25	16
	A04B	957,320	PCC	84	Good	30	20	50
	A08B	26,000	PCC	99	Good	74	0	26
	T02B	211,268	PCC	84	Good	29	39	32
	T15B	16,250	PCC	81	Good	35	35	30
	T30B	48,000	PCC	97	Good	83	0	17
	T31B	56,168	PCC	96	Good	52	0	48
West Apron	A05B	670,735	PCC	88	Good	33	24	43
	T07B	43,093	PCC	89	Good	15	9	76
	T29B	111,650	PCC	90	Good	18	0	82
Maintenance Hangar Access Apron	A11C1	11,380	PCC	81	Good	25	61	14
	A11C2	13,324	PCC	77	Good	40	54	6
	A11C3	11,380	PCC	90	Good	94	0	6
Engine Run-up Apron	A07B	67,020	PCC	92	Good	83	0	17
	T14B	15,560	AC	100	Good	0	0	0
OVERRUNS								
Overrun Runway 15/33	O03C	168,000	AC	100	Good	0	0	0
	O04C	31,100	AC	100	Good	0	0	0
	O05C	22,500	AC	94	Good	100	0	0
	O06C	126,885	AC	94	Good	100	0	0
Overrun Runway 009/189	O01C	70,000	DBST	43	Poor	100	0	0
	O02C	20,000	AC	98	Good	66	0	34
	O08C	10,000	AC	100	Good	0	0	0
RUNWAYS								
Runway 15/33	R03A	74,526	PCC	88	Good	37	41	22
	R04C1	30,000	AAC	**58**	Fair	100	0	0
	R04C2	395,226	AAC	93	Good	100	0	0
	R05A1	51,504	AAC	**61**	Fair	100	0	0
	R05A2	90,756	AAC	94	Good	100	0	0
	R06A1	75,060	PCC	96	Good	43	0	57
	R06A2	75,060	PCC	96	Good	39	0	61
	R07A	108,829	PCC	97	Good	56	0	44
	R08C1	37,520	PCC	93	Good	23	24	53
	R08C2	30,000	PCC	98	Good	100	0	0
	R09C	212,484	AAC	**60**	Fair	100	0	0
	R10C	10,500	AAC	80	Good	100	0	0
	R12A	9,000	AAC	100	Good	0	0	0
	R13C	10,500	AC	83	Good	100	0	0
Runway 009/189 (Assault Strip)	R01A	110,850	PCC	88	Good	67	4	29
	R02A	107,583	AAC	83	Good	100	0	0
TAXIWAYS								
Runway 15/33 Turn Around	T11A	57,420	PCC	95	Good	84	0	16
Runway 009/189 Turn Around	T06A	72,380	AC	**60**	Fair	100	0	0
Taxiway A	T04A	191,550	AAC	100	Good	0	0	0
Taxiway A (North)	T16A	16,048	PCC	93	Good	100	0	**0**
	T18A1	11,210	PCC	80	Good	56	11	33
	T18A2	183,490	PCC	91	Good	87	3	10
	T34A	40,900	PCC	81	Good	50	19	31
Taxiway A (Northwest)	T03A	176,670	PCC	89	Good	60	10	30
	T13A	14,126	AAC	100	Good	0	0	0
	T17A	10,000	PCC	86	Good	13	43	44
	T19A	30,170	AC	100	Good	0	0	0
	T33A	22,276	PCC	92	Good	56	27	17
Taxiway B	T05C	203,490	AC	100	Good	0	0	0
	T12C	37,300	AC	100	Good	0	0	0
Taxiway C	T08A	51,200	AC	100	Good	0	0	0
Taxiway D	T32B	53,000	AC	100	Good	0	0	0
Taxiway E	T09A	48,600	AC	100	Good	0	0	0

Extrapolated Distresses by Section - Maxwell AFB

Name	Section ID	PCI	Distress Code	Severity	Distress Quantity	Units	PCI Deduct	Distress Description
Aero Club Apron	A10C	14	63	Low	3	Slabs	11.31	LINEAR CR
			63	Medium	1	Slabs	10.93	LINEAR CR
			65	High	20	Slabs	12.00	JT SEAL DMG
			72	Low	15	Slabs	50.78	SHAT. SLAB
			72	Medium	1	Slabs	18.59	SHAT. SLAB
			73	N	1	Slabs	1.12	SHRINKAGE CR
North Apron	A01B	85	63	Low	34	Slabs	3.16	LINEAR CRACKING
			63	Medium	23	Slabs	5.05	LINEAR CRACKING
			65	High	34	Slabs	12.00	JOINT SEAL DAMAGE
			65	Medium	1,054	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	3	Slabs	0.24	SMALL PATCH
			66	Medium	9	Slabs	0.66	SMALL PATCH
			67	Low	11	Slabs	0.76	LARGE PATCH/UTILITY
			67	Medium	3	Slabs	2.47	LARGE PATCH/UTILITY
			70	Low	3	Slabs	0.33	SCALING
			73	N/A	6	Slabs	0.26	SHRINKAGE CRACKING
			75	Low	6	Slabs	0.45	CORNER SPALLING
	A04B	84	62	Low	7	Slabs	0.80	CORNER BREAK
			63	Low	106	Slabs	2.83	LINEAR CRACKING
			63	Medium	13	Slabs	2.52	LINEAR CRACKING
			65	Low	2,478	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	556	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	265	Slabs	1.31	SMALL PATCH
			66	Medium	26	Slabs	0.66	SMALL PATCH
			67	Low	265	Slabs	4.52	LARGE PATCH/UTILITY
			67	Medium	33	Slabs	2.47	LARGE PATCH/UTILITY
			70	Low	13	Slabs	0.33	SCALING
			71	Low	60	Slabs	1.53	FAULTING
			71	Medium	13	Slabs	1.77	FAULTING
			73	N/A	46	Slabs	0.31	SHRINKAGE CRACKING
			74	Low	26	Slabs	0.34	JOINT SPALLING
			74	Medium	20	Slabs	0.80	JOINT SPALLING
			75	Low	7	Slabs	0.45	CORNER SPALLING
			75	Medium	7	Slabs	0.87	CORNER SPALLING
	A08B	99	Extrapolated Distress Information not available.					
	T02B	84	62	High	4	Slabs	2.98	CORNER BREAK
			62	Medium	2	Slabs	1.89	CORNER BREAK
			63	Low	43	Slabs	4.86	LINEAR CRACKING
			63	Medium	4	Slabs	2.52	LINEAR CRACKING
			65	Low	307	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	333	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	26	Slabs	0.67	SMALL PATCH
			66	Medium	2	Slabs	0.66	SMALL PATCH
			67	Low	34	Slabs	2.78	LARGE PATCH/UTILITY
			67	Medium	4	Slabs	2.47	LARGE PATCH/UTILITY
			71	Low	21	Slabs	2.43	FAULTING
			73	N/A	2	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	2	Slabs	0.34	JOINT SPALLING
			75	Low	4	Slabs	0.45	CORNER SPALLING
	T15B	81	63	Low	7	Slabs	8.96	LINEAR CRACKING
			65	Low	22	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	43	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	1	Slabs	0.36	SMALL PATCH
			67	Low	6	Slabs	5.78	LARGE PATCH/UTILITY
			73	N/A	2	Slabs	0.74	SHRINKAGE CRACKING
			75	Low	1	Slabs	0.68	CORNER SPALLING
	T30B	97	65	Low	132	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	24	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	1	Slabs	0.24	SMALL PATCH
			74	Low	1	Slabs	0.34	JOINT SPALLING
			75	Low	1	Slabs	0.45	CORNER SPALLING
			75	Medium	1	Slabs	0.87	CORNER SPALLING

Extrapolated Distresses by Section - Maxwell AFB

Name	Section ID	PCI	Distress Code	Severity	Distress Quantity	Units	PCI Deduct	Distress Description
North Apron	T31B	96	65	Low	225	Slabs	2.00	JOINT SEAL DAMAGE
			73	N/A	10	Slabs	1.05	SHRINKAGE CRACKING
			74	Low	2	Slabs	0.35	JOINT SPALLING
			75	Low	2	Slabs	0.46	CORNER SPALLING
West Apron	A05B	88	62	Medium	5	Slabs	1.89	CORNER BREAK
			63	Low	55	Slabs	2.15	LINEAR CRACKING
			63	Medium	5	Slabs	2.52	LINEAR CRACKING
			65	Low	1,249	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	1,434	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	161	Slabs	1.18	SMALL PATCH
			66	Medium	5	Slabs	0.66	SMALL PATCH
			67	Low	92	Slabs	2.39	LARGE PATCH/UTILITY
			67	Medium	18	Slabs	2.47	LARGE PATCH/UTILITY
			69	N/A	5	Slabs	1.06	PUMPING
			70	Low	14	Slabs	0.33	SCALING
			71	Low	5	Slabs	0.99	FAULTING
			73	N/A	9	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	9	Slabs	0.34	JOINT SPALLING
			74	Medium	5	Slabs	0.80	JOINT SPALLING
			75	Low	18	Slabs	0.45	CORNER SPALLING
			75	Medium	5	Slabs	0.87	CORNER SPALLING
	T07B	89	63	Low	2	Slabs	1.24	LINEAR CRACKING
			65	Low	132	Slabs	2.00	JOINT SEAL DAMAGE
			66	Low	16	Slabs	1.64	SMALL PATCH
			67	Low	15	Slabs	5.51	LARGE PATCH/UTILITY
			67	Medium	1	Slabs	2.47	LARGE PATCH/UTILITY
			73	N/A	1	Slabs	0.26	SHRINKAGE CRACKING
			75	Low	2	Slabs	0.52	CORNER SPALLING
	T29B	90	65	Low	447	Slabs	2.00	JOINT SEAL DAMAGE
			67	Low	3	Slabs	0.73	LARGE PATCH/UTILITY
			70	Low	7	Slabs	0.48	SCALING
			71	Low	3	Slabs	0.99	FAULTING
			73	N/A	124	Slabs	3.93	SHRINKAGE CRACKING
			74	Low	3	Slabs	0.34	JOINT SPALLING
			75	Low	16	Slabs	1.56	CORNER SPALLING
			75	Medium	4	Slabs	0.87	CORNER SPALLING
Maintenance Hangar Access Apron	A11C1	81	62	Low	6	Slabs	4.32	CORNER BREAK
			62	Medium	2	Slabs	3.26	CORNER BREAK
			65	Medium	94	Slabs	7.00	JOINT SEAL DAMAGE
			72	Medium	2	Slabs	9.18	SHATTERED SLAB
			74	Low	14	Slabs	3.83	JOINT SPALLING
	A11C2	77	62	Low	6	Slabs	3.93	CORNER BREAK
			63	Low	23	Slabs	12.23	LINEAR CRACKING
			65	High	26	Slabs	12.00	JOINT SEAL DAMAGE
			65	Medium	86	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	3	Slabs	0.55	SMALL PATCH
			67	Low	3	Slabs	1.71	LARGE PATCH/UTILITY
			72	Low	2	Slabs	3.61	SHATTERED SLAB
			72	Medium	1	Slabs	5.67	SHATTERED SLAB
			74	Low	2	Slabs	0.55	JOINT SPALLING
	A11C3	90	65	High	47	Slabs	12.00	JOINT SEAL DAMAGE
			65	Medium	67	Slabs	7.00	JOINT SEAL DAMAGE
			73	N/A	3	Slabs	0.68	SHRINKAGE CRACKING
			74	Low	2	Slabs	0.63	JOINT SPALLING
Engine Run-up Apron	A07B	92	65	Medium	180	Slabs	7.00	JOINT SEAL DAMAGE
			66	Medium	1	Slabs	0.66	SMALL PATCH
			74	Medium	1	Slabs	0.80	JOINT SPALLING
	T14B	100	No Distress recorded.					
Overrun Runway 15/33	O03C	100	No Distress recorded.					
	O04C	100	No Distress recorded.					
	O05C	94	Extrapolated Distress Information not available.					
	O06C	94	57	Low	126,881	SqFt	5.96	WEATHERING

Extrapolated Distresses by Section - Maxwell AFB

Name	Section ID	PCI	Distress Code	Severity	Distress Quantity	Units	PCI Deduct	Distress Description
Overrun Runway 009/189	O01C	43	52	Medium	70,000	SqFt	56.77	RAVELING
	O02C	98	48	Low	33	Ft	2.85	L&T CRACKING
			56	Low	40	SqFt	1.48	SWELLING
	O08C	100	No Distress recorded.					
Runway 15/33	R03A	88	62	Low	1	Slabs	0.80	CORNER BREAK
			63	Low	14	Slabs	4.57	LINEAR CRACKING
			63	Medium	1	Slabs	2.52	LINEAR CRACKING
			65	Low	193	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	90	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	12	Slabs	0.86	SMALL PATCH
			67	Low	6	Slabs	1.44	LARGE PATCH/UTILITY
			71	Low	2	Slabs	0.99	FAULTING
			72	Low	1	Slabs	2.28	SHATTERED SLAB
			73	N/A	3	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	3	Slabs	0.34	JOINT SPALLING
			75	Low	4	Slabs	0.60	CORNER SPALLING
			75	Medium	1	Slabs	0.87	CORNER SPALLING
	R04C1	58	Extrapolated Distress Information not available.					
	R04C2	93	48	Low	201	Ft	2.50	L&T CRACKING
			57	Low	395,226	SqFt	5.96	WEATHERING
	R05A1	61	43	Low	51,844	SqFt	33.87	BLOCK CRACKING
			57	Low	60,504	SqFt	5.96	WEATHERING
	R05A2	94	Extrapolated Distress Information not available.					
	R06A1	96	65	Low	230	Slabs	2.00	JOINT SEAL DAMAGE
			66	Low	7	Slabs	0.69	SMALL PATCH
			67	Low	1	Slabs	0.73	LARGE PATCH/UTILITY
			73	N/A	1	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	3	Slabs	0.53	JOINT SPALLING
			75	Low	1	Slabs	0.45	CORNER SPALLING
	R06A2	96	Extrapolated Distress Information not available.					
	R07A	97	Extrapolated Distress Information not available.					
	R08C1	93	Extrapolated Distress Information not available.					
	R08C2	98	65	Low	60	Slabs	2.00	JOINT SEAL DAMAGE
	R09C	60	43	Low	183,303	SqFt	33.94	BLOCK CRACKING
		60	43	Medium	1,133	SqFt	10.11	BLOCK CRACKING
		60	48	Low	176	Ft	2.50	L&T CRACKING
		60	48	Medium	167	Ft	4.00	L&T CRACKING
		60	57	Low	212,484	SqFt	5.96	WEATHERING
	R10C	80	48	Low	17	Ft	2.86	L&T CRACKING
		80	52	Low	1,517	SqFt	11.77	RAVELING
		80	57	Low	8,983	SqFt	5.80	WEATHERING
	R12A	100	No Distress recorded.					
	R13C	83	48	Low	234	Ft	8.02	L&T CRACKING
		83	52	Low	400	SqFt	5.83	RAVELING
		83	57	Low	10,500	SqFt	5.96	WEATHERING
Runway 009/189 (Assault Strip)	R01A	88	63	Low	3	Slabs	1.07	LINEAR CRACKING
			65	High	69	Slabs	12.00	JOINT SEAL DAMAGE
			65	Medium	313	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	7	Slabs	0.39	SMALL PATCH
			66	Medium	1	Slabs	0.66	SMALL PATCH
			67	Low	17	Slabs	2.70	LARGE PATCH/UTILITY
			71	Low	3	Slabs	0.99	FAULTING
			73	N/A	3	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	9	Slabs	0.66	JOINT SPALLING
			74	Medium	3	Slabs	0.80	JOINT SPALLING
			75	Low	9	Slabs	0.87	CORNER SPALLING
			75	Medium	1	Slabs	0.87	CORNER SPALLING
	R02A	83	48	Low	3,472	Ft	10.62	L&T CRACKING
			48	Medium	20	Ft	4.00	L&T CRACKING
			57	Low	106,946	SqFt	5.96	WEATHERING
			57	Medium	102	SqFt	1.20	WEATHERING

Extrapolated Distresses by Section - Maxwell AFB

Name	Section ID	PCI	Distress Code	Severity	Distress Quantity	Units	PCI Deduct	Distress Description
Runway 15/33 Turn Around	T11A	95	65	Low	124	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	40	Slabs	7.00	JOINT SEAL DAMAGE
			73	N/A	14	Slabs	1.71	SHRINKAGE CRACKING
Runway 009/189 Turn Around	T06A	60	43	Low	59,763	SqFt	35.60	BLOCK CRACKING
			57	Low	54,110	SqFt	5.87	WEATHERING
Taxiway A	T04A	100	No Distress recorded.					
Taxiway A (North)	T16A	93	65	Medium	61	Slabs	7.00	JOINT SEAL DAMAGE
	T18A1	80	63	Low	2	Slabs	4.34	LINEAR CRACKING
			65	High	20	Slabs	12.00	JOINT SEAL DAMAGE
			65	Low	10	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	15	Slabs	7.00	JOINT SEAL DAMAGE
			67	Medium	2	Slabs	9.44	LARGE PATCH/UTILITY
			73	N/A	1	Slabs	0.55	SHRINKAGE CRACKING
			74	Medium	1	Slabs	1.76	JOINT SPALLING
			75	Low	1	Slabs	0.98	CORNER SPALLING
Taxiway A (North)	T18A2	91	62	Low	2	Slabs	0.80	CORNER BREAK
			65	High	107	Slabs	12.00	JOINT SEAL DAMAGE
			65	Low	21	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	606	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	2	Slabs	0.24	SMALL PATCH
			71	Low	2	Slabs	0.99	FAULTING
			73	N/A	5	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	2	Slabs	0.34	JOINT SPALLING
			75	Low	9	Slabs	0.54	CORNER SPALLING
	T34A	81	62	Low	2	Slabs	1.02	CORNER BREAK
			62	Medium	1	Slabs	1.89	CORNER BREAK
			63	Low	4	Slabs	2.62	LINEAR CRACKING
			63	Medium	1	Slabs	2.52	LINEAR CRACKING
			64	Low	1	Slabs	0.74	DURABILITY CRACKING
			65	High	19	Slabs	12.00	JOINT SEAL DAMAGE
			65	Low	50	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	74	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	9	Slabs	1.14	SMALL PATCH
			67	Low	15	Slabs	5.62	LARGE PATCH/UTILITY
			67	Medium	2	Slabs	3.11	LARGE PATCH/UTILITY
			71	Low	1	Slabs	0.99	FAULTING
			73	N/A	1	Slabs	0.26	SHRINKAGE CRACKING
			74	Low	8	Slabs	1.67	JOINT SPALLING
			75	Low	1	Slabs	0.45	CORNER SPALLING
Taxiway A (Northwest)	T03A	89	63	Low	2	Slabs	1.07	LINEAR CRACKING
			63	Medium	4	Slabs	2.52	LINEAR CRACKING
			65	High	77	Slabs	12.00	JOINT SEAL DAMAGE
			65	Low	208	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	278	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	5	Slabs	0.24	SMALL PATCH
			66	Medium	2	Slabs	0.66	SMALL PATCH
			67	Low	2	Slabs	0.73	LARGE PATCH/UTILITY
			67	Medium	5	Slabs	2.47	LARGE PATCH/UTILITY
			70	Low	23	Slabs	0.96	SCALING
			71	Low	22	Slabs	2.91	FAULTING
			71	Medium	2	Slabs	1.77	FAULTING
			74	Low	2	Slabs	0.34	JOINT SPALLING
			75	Low	4	Slabs	0.45	CORNER SPALLING
	T13A	100	No Distress recorded.					
	T17A	86	62	Low	2	Slabs	4.07	CORNER BREAK
			63	Low	1	Slabs	2.57	LINEAR CRACKING
			65	Low	20	Slabs	2.00	JOINT SEAL DAMAGE
			70	Low	1	Slabs	0.76	SCALING
			73	N/A	4	Slabs	1.91	SHRINKAGE CRACKING
			74	Low	1	Slabs	0.84	JOINT SPALLING
			75	Low	1	Slabs	1.10	CORNER SPALLING
			75	Medium	1	Slabs	2.10	CORNER SPALLING
	T19A	100	No Distress recorded.					

Extrapolated Distresses by Section - Maxwell AFB

Name	Section ID	PCI	Distress Code	Severity	Distress Quantity	Units	PCI Deduct	Distress Description
Taxiway A (Northwest)	T33A	92	63	Low	4	Slabs	4.34	LINEAR CRACKING
			65	Low	60	Slabs	2.00	JOINT SEAL DAMAGE
			65	Medium	20	Slabs	7.00	JOINT SEAL DAMAGE
			66	Low	2	Slabs	0.51	SMALL PATCH
			73	N/A	1	Slabs	0.29	SHRINKAGE CRACKING
			75	Medium	2	Slabs	1.88	CORNER SPALLING
Taxiway B	T05C	100	No Distress recorded.					
Taxiway B	T12C	100	No Distress recorded.					
Taxiway C	T08A	100	No Distress recorded.					
Taxiway D	T32B	100	No Distress recorded.					
Taxiway E	T09A	100	No Distress recorded.					

Appendix G

FOD Potential Rating

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FOD Potential Rating - Maxwell AFB

Branch Name	Section ID	Area (ft²)	Surface	Aircraft Category	Section FOD Index	Section FOD Potential	Section FOD Rating
APRONS							
Aero Club Apron	A10C	36,178	PCC	C-17	93	67	Poor
North Apron	A01B	274,424	PCC	C-17	35	18	Good
	A04B	957,320	PCC	C-17	20	0	Good
	A08B	26,000	PCC	C-17	2	0	Good
	T02B	211,268	PCC	C-17	23	3	Good
	T15B	16,250	PCC	C-17	33	16	Good
	T30B	48,000	PCC	C-17	10	0	Good
	T31B	56,168	PCC	C-17	9	0	Good
West Apron	A05B	670,735	PCC	C-17	25	6	Good
	T07B	43,093	PCC	C-17	15	0	Good
	T29B	111,650	PCC	C-17	11	0	Good
Maintenance Hangar Access Apron	A11C1	11,380	PCC	C-17	34	17	Good
	A11C2	13,324	PCC	C-17	40	23	Good
	A11C3	11,380	PCC	C-17	37	20	Good
Engine Run-up Apron	A07B	67,020	PCC	C-17	29	11	Good
	T14B	15,560	AC	C-17	0	0	Good
OVERRUNS							
Overrun Runway 15/33	O03C	168,000	AC	C-17	0	0	Good
	O04C	31,100	AC	C-17	0	0	Good
	O05C	22,500	AC	C-17	6	4	Good
	O06C	126,885	AC	C-17	6	4	Good
Overrun Runway 009/189	O01C	70,000	DBST	C-17	57	N/A	Poor
	O02C	20,000	AC	C-17	1	0	Good
	O08C	10,000	AC	C-17	0	0	Good
RUNWAYS							
Runway 15/33	R03A	74,526	PCC	C-17	20	0	Good
	R04C1	30,000	AAC	C-17	42	31	Good
	R04C2	395,226	AAC	C-17	7	5	Good
	R05A1	51,504	AAC	C-17	39	28	Good
	R05A2	90,756	AAC	C-17	6	4	Good
	R06A1	75,060	PCC	C-17	9	0	Good
	R06A2	75,060	PCC	C-17	10	0	Good
	R07A	108,829	PCC	C-17	9	0	Good
	R08C1	37,520	PCC	C-17	8	0	Good
	R08C2	30,000	PCC	C-17	6	0	Good
	R09C	212,484	AAC	C-17	40	29	Good
	R10C	10,500	AAC	C-17	18	13	Good
	R12A	9,000	AAC	C-17	6	4	Good
	R13C	10,500	AC	C-17	7	5	Good
Runway 009/189 (Assault Strip)	R01A	110,850	PCC	C-17	32	15	Good
	R02A	107,583	AAC	C-17	17	12	Good

FOD Potential Rating - Maxwell AFB

Branch Name	Section ID	Area (ft²)	Surface	Aircraft Category	Section FOD		
					Index	Potential	Rating
TAXIWAYS							
Runway 15/33 Turn Around	T11A	57,420	PCC	C-17	13	0	Good
Runway 009/189 Turn Around	T06A	72,380	AC	C-17	40	29	Good
Taxiway A	T04A	191,550	AAC	C-17	0	0	Good
Taxiway A (North)	T16A	16,048	PCC	C-17	28	10	Good
	T18A1	11,210	PCC	C-17	42	24	Good
	T18A2	183,490	PCC	C-17	31	14	Good
	T34A	40,900	PCC	C-17	31	14	Good
Taxiway A (Northwest)	T03A	176,670	PCC	C-17	22	1	Good
	T13A	14,126	AAC	C-17	0	0	Good
	T17A	10,000	PCC	C-17	15	0	Good
	T19A	30,170	AC	C-17	0	0	Good
	T33A	22,276	PCC	C-17	16	0	Good
Taxiway B	T05C	203,490	AC	C-17	0	0	Good
	T12C	37,300	AC	C-17	0	0	Good
Taxiway C	T08A	51,200	AC	C-17	0	0	Good
Taxiway D	T32B	53,000	AC	C-17	0	0	Good
Taxiway E	T09A	48,600	AC	C-17	0	0	Good

Appendix H

Sturctural Evaluation Results

(Primary Aircraft, ACN, PCN, and Structural Index)

Pavement Classification Numbers (PCNs)

Summary of Allowable Gross Loads (AGLs)

Aircraft Classification Number (ACN) Charts PCN

Reference Data

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Structural Data
MAXWELL AFB
Non-Frost Period

SECTION	PRIMARY AIRCRAFT	ACN	PCN	STRUCTURAL INDEX	SECTION	PRIMARY AIRCRAFT	ACN	PCN	STRUCTURAL INDEX
A01B	C-130H	37/R/C	35/R/C/W/T	1.06	T12C	C-130H	32/F/B	105/F/B/W/T	0.30
A04B	C-130H	37/R/C	36/R/C/W/T	1.03	T13A	C-130H	37/R/C	101/R/C/W/T	0.37
A05B	C-130H	37/R/C	40/R/C/W/T	0.93	T14B	C-130H	27/F/A	40/F/A/W/T	0.68
A07B	C-130H	34/R/B	41/R/B/W/T	0.83	T15B	C-130H	34/R/B	76/R/B/W/T	0.45
A08B	C-130H	34/R/B	51/R/B/W/T	0.67	T16A	C-130H	31/R/A	84/R/A/W/T	0.37
A10C	T-1A	6/R/C	15/R/C/W/T	0.40	T17A	C-130H	37/R/C	37/R/C/W/T	1.00
A11C1	C-130H	34/R/B	69/R/B/W/T	0.49	T18A1	C-130H	37/R/C	58/R/C/W/T	0.64
A11C2	C-130H	34/R/B	69/R/B/W/T	0.49	T18A2	C-130H	37/R/C	60/R/C/W/T	0.62
A11C3	C-130H	34/R/B	69/R/B/W/T	0.49	T19A	C-130H	32/F/B	22/F/B/W/T	1.45
O01C	C-130H	27/F/A	1/F/A/W/T	27.00	T29B	C-130H	34/R/B	71/R/B/W/T	0.48
O02C	C-130H	27/F/A	18/F/A/W/T	1.50	T30B	C-130H	34/R/B	79/R/B/W/T	0.43
O03C	C-130H	27/F/A	150/F/A/W/T	0.18	T31B	C-130H	34/R/B	53/R/B/W/T	0.64
O04C	C-130H	27/F/A	87/F/A/W/T	0.31	T32B	C-130H	27/F/A	118/F/A/W/T	0.23
O05C	C-130H	27/F/A	52/F/A/W/T	0.52	T33A	C-130H	37/R/C	39/R/C/W/T	0.95
O06C	C-130H	27/F/A	15/F/A/W/T	1.80	T34A	C-130H	37/R/C	49/R/C/W/T	0.76
O08C	C-130H	32/F/B	24/F/B/W/T	1.33					
R01A	C-130H	34/R/B	56/R/B/W/T	0.61					
R02A	C-130H	27/F/A	125/F/A/W/T	0.22					
R03A	C-130H	37/R/C	36/R/C/W/T	1.03					
R04C1	C-130H	32/F/B	97/F/B/W/T	0.33					
R04C2	C-130H	32/F/B	88/F/B/W/T	0.36					
R05A1	C-130H	27/F/A	150/F/A/W/T	0.18					
R05A2	C-130H	27/F/A	149/F/A/W/T	0.18					
R06A1	C-130H	34/R/B	66/R/B/W/T	0.52					
R06A2	C-130H	34/R/B	66/R/B/W/T	0.52					
R07A	C-130H	34/R/B	71/R/B/W/T	0.48					
R08C1	C-130H	34/R/B	87/R/B/W/T	0.39					
R08C2	C-130H	34/R/B	80/R/B/W/T	0.43					
R09C	C-130H	32/F/B	150/F/B/W/T	0.21					
R10C	C-130H	27/F/A	150/F/A/W/T	0.18					
R12A	C-130H	27/F/A	150/F/A/W/T	0.18					
R13C	C-130H	32/F/B	150/F/B/W/T	0.21					
T02B	C-130H	37/R/C	37/R/C/W/T	1.00					
T03A	C-130H	37/R/C	62/R/C/W/T	0.60					
T04A	C-130H	32/F/B	80/F/B/W/T	0.40					
T05C	C-130H	27/F/A	118/F/A/W/T	0.23					
T06A	C-130H	27/F/A	150/F/A/W/T	0.18					
T07B	C-130H	37/R/C	38/R/C/W/T	0.97					
T08A	C-130H	32/F/B	97/F/B/W/T	0.33					
T09A	C-130H	27/F/A	141/F/A/W/T	0.19					
T11A	C-130H	34/R/B	71/R/B/W/T	0.48					

***** PCN BASED ON C-17 AIRCRAFT AT 50,000 PASSES

PAVEMENT CLASSIFICATION NUMBERS, STRUCTURAL INDEX, AND SUMMARY OF ALLOWABLE GROSS LOADS
MAXWELL AFB
Non-Frost Period

PAVEMENT CAPACITY IN KIPS FOR AIRCRAFT GROUP INDEX NUMBERS

SECTION	PCN	PASS INTENSITY LEVEL	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A01B	35/R/C/W/T	I	72	76	102	175	102	120	126	277	322	411	691	457	656	229
		II	85	90	119	197	116	135	141	311	362	457	767	513	736	270
		III	97	102	134	240	140	163	170	374	435	537	901	614	879	336
		IV	119	125	162	314	183	211	218	482	561	669	1117	787	1122	431
A04B	36/R/C/W/T	I	71	76	102	177	102	120	127	287	334	419	720	471	684	235
		II	84	89	120	200	115	136	143	323	376	466	799	529	767	277
		III	96	101	135	243	140	164	172	388	452	549	938	634	916	345
		IV	117	123	162	319	182	212	221	501	582	683	1163	812	1169	443
A05B	40/R/C/W/T	I	76	81	110	192	110	130	137	315	365	455	788	513	747	256
		II	90	95	129	217	124	146	154	354	411	506	875	576	838	302
		III	102	108	145	263	150	176	185	426	494	595	1027	690	1001	376
		IV	125	132	175	345	196	228	238	549	637	740	1274	885	1277	482
A07B	41/R/B/W/T	I	80	85	116	202	115	137	144	335	387	480	838	543	793	271
		II	94	100	136	229	130	154	162	377	436	534	931	610	889	319
		III	107	113	153	277	157	186	195	453	524	628	1093	731	1062	398
		IV	131	138	184	363	205	240	251	584	675	781	1355	937	1355	511
A08B	51/R/B/W/T	I	107	113	149	254	149	176	183	410	475	601	1020	670	968	336
		II	126	133	175	288	168	198	205	461	535	669	1133	753	1086	396
		III	144	151	197	349	204	239	246	554	642	787	1330	901	1297	494
		IV	176	184	237	457	266	309	317	714	828	979	1648	1155	1655	633
A10C	15/R/C/W/T	I	29	31	49	90	47	58	64	163	178	215	408	244	367	129
		II	34	37	57	102	53	65	72	184	201	239	454	274	411	151
		III	39	42	64	124	64	79	86	221	241	281	532	328	491	189
		IV	47	51	78	162	83	102	110	284	311	350	660	421	627	242
A11C1	69/R/B/W/T	I	151	158	205	344	205	241	247	548	634	810	1356	895	1287	452
		II	178	186	240	389	231	272	278	617	714	901	1506	1005	1444	533
		III	202	211	270	472	280	327	334	741	858	1060	1767	1203	1724	664
		IV	247	257	325	619	365	424	429	955	1106	1320	2191	1542	2200	852
A11C2	69/R/B/W/T	I	151	158	205	344	205	241	247	548	634	810	1356	895	1287	452
		II	178	186	240	389	231	272	278	617	714	901	1506	1005	1444	533
		III	202	211	270	472	280	327	334	741	858	1060	1767	1203	1724	664
		IV	247	257	325	619	365	424	429	955	1106	1320	2191	1542	2200	852
A11C3	69/R/B/W/T	I	151	158	205	344	205	241	247	548	634	810	1356	895	1287	452
		II	178	186	240	389	231	272	278	617	714	901	1506	1005	1444	533
		III	202	211	270	472	280	327	334	741	858	1060	1767	1203	1724	664
		IV	247	257	325	619	365	424	429	955	1106	1320	2191	1542	2200	852
O01C	1/F/A/W/T	I	0	0	2	1	1	1	0	1	1	2	4	1	2	1
		II	0	0	49	1	1	1	1	1	1	2	5	1	2	1
		III	1	0	55	4	1	1	1	1	1	3	9	1	2	1
		IV	11	0	69	78	2	2	1	2	2	6	380	2	3	2
O02C	18/F/A/W/T	I	64	36	114	141	95	104	90	178	157	312	704	192	298	100
		II	84	43	132	158	106	128	108	210	182	379	777	223	340	124
		III	96	50	146	191	125	164	147	278	232	487	904	285	426	173
		IV	120	64	174	251	160	212	202	399	343	609	1117	424	601	272

PAVEMENT CLASSIFICATION NUMBERS, STRUCTURAL INDEX, AND SUMMARY OF ALLOWABLE GROSS LOADS
MAXWELL AFB
Non-Frost Period

PAVEMENT CAPACITY IN KIPS FOR AIRCRAFT GROUP INDEX NUMBERS

SECTION	PCN	PASS INTENSITY LEVEL	1	2	3	4	5	6	7	8	9	10	11	12	13	14
O03C	150/F/A/W/T	I	57	75	216	780	169	316	550	929	1209	2175	3393	1974	2990	1105
		II	82	108	309	822	215	402	579	1182	1538	2292	4260	2080	3149	1433
		III	104	137	393	881	296	555	621	1472	1652	2457	4567	2230	3377	1723
		IV	144	189	542	1059	424	734	747	1770	1986	2955	5492	2681	4061	2151
O04C	87/F/A/W/T	I	60	72	166	505	157	241	358	601	703	1166	1910	1211	1675	719
		II	86	104	237	643	199	306	455	765	895	1483	2429	1541	2130	991
		III	109	132	302	887	275	422	628	1055	1234	2046	3352	2125	2939	1419
		IV	150	182	416	1269	394	604	899	1510	1766	2928	4797	3042	4206	1957
O05C	52/F/A/W/T	I	55	65	143	239	139	193	208	455	467	737	1412	625	981	437
		II	79	94	191	254	149	205	220	483	495	783	1499	663	1042	474
		III	100	112	203	275	161	222	239	523	537	848	1624	719	1129	577
		IV	115	122	220	334	196	270	291	637	653	1032	1977	875	1373	720
O06C	15/F/A/W/T	I	14	17	43	97	34	48	71	132	152	266	490	253	367	141
		II	15	19	48	103	36	51	76	141	162	284	524	270	392	154
		III	16	20	51	112	39	56	83	154	177	309	571	294	428	188
		IV	18	22	55	138	48	68	102	189	216	379	700	361	525	235
008C	24/F/B/W/T	I	33	41	83	120	74	93	105	219	226	364	696	313	486	216
		II	45	48	93	129	80	100	113	236	243	392	749	337	523	238
		III	48	52	100	142	88	111	124	260	268	432	826	372	577	294
		IV	53	57	110	176	109	137	154	321	332	534	1021	460	714	367
R01A	56/R/B/W/T	I	125	131	166	272	165	193	198	429	501	659	1101	706	1028	370
		II	147	154	192	306	185	216	221	480	561	730	1221	789	1149	435
		III	166	174	215	366	221	257	262	571	668	855	1429	935	1364	539
		IV	201	210	256	467	282	326	330	721	848	1055	1765	1179	1721	688
R02A	125/F/A/W/T	I	265	279	415	617	363	456	477	967	1099	1646	3053	1534	2276	1020
		II	288	303	451	653	384	482	504	1023	1161	1740	3228	1621	2406	1099
		III	304	320	476	703	414	519	543	1101	1251	1874	3476	1746	2592	1329
		IV	328	345	513	850	500	627	656	1332	1512	2266	4203	2111	3133	1659
R03A	36/R/C/W/T	I	69	74	98	169	97	115	122	277	325	414	722	456	675	235
		II	81	86	114	189	109	129	136	310	364	459	801	510	754	275
		III	92	97	128	226	130	153	161	368	433	537	937	604	895	342
		IV	111	118	152	289	166	194	203	466	549	662	1157	762	1129	436
R04C1	97/F/B/W/T	I	233	244	320	497	309	371	375	690	788	1154	2064	1110	1621	780
		II	264	276	363	541	336	404	408	750	857	1256	2247	1209	1764	874
		III	287	301	395	605	377	453	457	840	960	1406	2516	1353	1976	1091
		IV	322	337	442	756	471	565	571	1049	1199	1756	3142	1690	2467	1362
R04C2	88/F/B/W/T	I	202	212	280	441	270	326	330	624	716	1054	1897	1010	1474	704
		II	229	240	318	480	294	355	359	679	779	1147	2065	1099	1604	788
		III	250	261	346	537	329	398	402	760	873	1285	2313	1231	1797	984
		IV	280	293	388	671	411	497	502	949	1090	1604	2888	1537	2243	1229
R05A1	150/F/A/W/T	I	413	432	566	859	529	636	643	1209	1383	2031	3646	1947	2845	1335
		II	454	475	621	915	563	678	685	1287	1472	2162	3881	2073	3028	1451
		III	483	505	661	995	613	737	744	1399	1600	2351	4220	2254	3293	1772
		IV	525	549	719	1215	748	900	909	1709	1954	2871	5153	2753	4021	2212

PAVEMENT CLASSIFICATION NUMBERS, STRUCTURAL INDEX, AND SUMMARY OF ALLOWABLE GROSS LOADS
MAXWELL AFB
Non-Frost Period

			PAVEMENT CAPACITY IN KIPS FOR AIRCRAFT GROUP INDEX NUMBERS													
SECTION	**PCN**	**PASS INTENSITY LEVEL**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
R05A2	149/F/A/W/T	I	405	423	542	835	516	610	616	1151	1328	1945	3481	1881	2731	1286
		II	446	466	597	892	551	652	658	1228	1418	2076	3715	2008	2915	1403
		III	476	498	638	973	601	711	718	1340	1547	2265	4054	2191	3181	1719
		IV	520	543	696	1193	737	871	880	1642	1896	2776	4968	2685	3898	2147
R06A1	66/R/B/W/T	I	169	177	208	337	214	245	246	510	590	782	1275	823	1199	447
		II	198	207	241	378	240	275	275	570	661	867	1414	919	1340	524
		III	224	234	270	452	286	327	326	677	786	1015	1655	1090	1590	650
		IV	272	282	321	578	365	414	411	857	998	1253	2043	1374	2007	829
R06A2	66/R/B/W/T	I	169	177	208	338	214	245	246	511	592	784	1279	826	1203	447
		II	199	207	241	379	240	275	275	572	663	870	1419	922	1344	525
		III	225	234	270	453	286	327	326	679	789	1018	1661	1093	1595	651
		IV	272	282	321	579	365	414	411	859	1001	1257	2050	1378	2013	831
R07A	71/R/B/W/T	I	179	186	219	359	225	259	260	550	639	844	1383	892	1299	479
		II	209	218	254	402	252	290	290	616	715	936	1534	996	1452	562
		III	237	246	285	481	301	345	344	731	851	1096	1795	1181	1722	697
		IV	287	298	339	615	384	437	434	925	1080	1353	2216	1489	2174	889
R08C1	87/R/B/W/T	I	233	242	274	445	286	322	323	669	773	1023	1657	1074	1565	586
		II	273	283	317	500	321	360	361	748	866	1135	1838	1200	1749	687
		III	308	320	355	597	383	428	428	888	1030	1328	2151	1422	2075	852
		IV	374	386	422	763	489	543	539	1123	1307	1639	2656	1793	2619	1087
R08C2	80/R/B/W/T	I	210	219	248	405	259	292	293	614	711	940	1526	989	1440	536
		II	246	256	288	455	291	327	328	687	796	1042	1693	1104	1609	629
		III	279	289	322	544	347	389	388	816	947	1220	1982	1309	1910	780
		IV	338	349	383	695	443	493	489	1032	1202	1506	2447	1650	2410	995
R09C	150/F/B/W/T	I	346	362	479	740	451	548	554	1065	1214	1790	3228	1703	2498	1180
		II	386	404	534	797	486	590	597	1147	1306	1927	3473	1832	2688	1302
		III	415	434	574	879	536	651	658	1265	1441	2125	3831	2021	2965	1609
		IV	458	479	634	1087	662	805	814	1564	1782	2628	4737	2499	3666	2009
R10C	150/F/A/W/T	I	724	758	956	1416	941	1071	1090	1803	2051	2931	5157	2904	4218	2048
		II	807	845	1065	1523	1012	1152	1173	1939	2207	3153	5548	3124	4538	2258
		III	868	909	1146	1680	1116	1270	1293	2138	2433	3476	6117	3445	5004	2791
		IV	957	1003	1264	2076	1379	1569	1598	2643	3007	4296	7560	4257	6184	3484
R12A	150/F/A/W/T	I	436	456	590	898	556	664	670	1251	1432	2098	3758	2019	2945	1385
		II	478	500	648	956	592	707	714	1332	1525	2234	4002	2149	3136	1506
		III	509	533	690	1040	643	768	776	1448	1658	2430	4351	2337	3410	1840
		IV	554	579	750	1270	786	938	948	1769	2025	2968	5315	2855	4165	2297
R13C	150/F/B/W/T	I	406	426	579	883	534	654	663	1276	1464	2165	3923	2064	3018	1403
		II	448	470	639	943	570	699	708	1362	1564	2313	4190	2204	3224	1532
		III	479	502	682	1029	623	763	773	1488	1708	2525	4575	2407	3520	1879
		IV	523	548	745	1262	764	936	948	1824	2094	3096	5609	2951	4316	2346
T02B	37/R/C/W/T	I	70	75	103	181	102	122	129	301	348	430	755	488	713	244
		II	83	88	121	205	116	137	145	339	392	478	838	549	799	287
		III	95	100	136	248	140	165	174	407	471	562	984	657	955	357
		IV	116	122	164	326	183	214	224	525	607	700	1220	842	1218	459

PAVEMENT CLASSIFICATION NUMBERS, STRUCTURAL INDEX, AND SUMMARY OF ALLOWABLE GROSS LOADS
MAXWELL AFB
Non-Frost Period

			PAVEMENT CAPACITY IN KIPS FOR AIRCRAFT GROUP INDEX NUMBERS													
SECTION	PCN	PASS INTENSITY LEVEL	1	2	3	4	5	6	7	8	9	10	11	12	13	14
T03A	62/R/C/W/T	I	136	143	174	281	176	204	206	427	496	659	1078	695	1011	374
		II	160	167	201	316	197	228	230	477	555	731	1196	777	1130	439
		III	181	189	225	377	236	271	273	567	661	856	1400	921	1341	544
		IV	219	228	268	482	300	344	344	717	838	1056	1728	1161	1692	694
T04A	80/F/B/W/T	I	162	170	245	374	222	280	288	575	651	974	1790	908	1345	624
		II	181	190	274	403	239	302	310	620	702	1050	1930	979	1450	690
		III	196	205	296	446	264	334	343	685	777	1161	2134	1083	1604	854
		IV	216	227	327	552	327	413	425	849	962	1438	2643	1341	1987	1066
T05C	118/F/A/W/T	I	222	234	373	547	318	411	434	927	1024	1551	2933	1405	2130	943
		II	242	255	406	578	336	435	460	981	1083	1641	3104	1487	2254	1017
		III	256	270	430	624	363	469	496	1058	1169	1770	3348	1604	2432	1231
		IV	276	291	464	756	439	569	601	1281	1415	2144	4055	1942	2945	1537
T06A	150/F/A/W/T	I	154	175	322	783	340	468	588	1147	1297	2005	3419	1952	2923	1307
		II	220	250	461	821	433	596	616	1323	1487	2228	4133	2046	3063	1391
		III	280	318	586	874	506	647	656	1408	1583	2371	4400	2178	3261	1657
		IV	387	439	624	1040	603	770	781	1677	1886	2824	5240	2593	3884	2069
T07B	38/R/C/W/T	I	70	74	103	181	102	122	129	304	350	431	761	491	717	245
		II	83	88	121	205	115	137	145	342	394	479	845	551	804	288
		III	94	99	136	249	139	165	174	411	473	564	992	660	960	359
		IV	115	121	164	326	182	214	224	530	610	701	1230	846	1225	461
T08A	97/F/B/W/T	I	224	234	306	482	293	353	357	687	788	1161	2092	1110	1621	770
		II	252	263	344	522	318	383	387	744	854	1257	2266	1202	1756	856
		III	273	285	373	581	353	426	430	827	950	1398	2520	1337	1953	1065
		IV	303	317	415	722	440	529	535	1029	1181	1739	3135	1663	2429	1330
T09A	141/F/A/W/T	I	386	403	505	787	485	570	577	1101	1258	1840	3293	1766	2582	1222
		II	426	445	557	841	518	609	616	1176	1345	1966	3520	1888	2760	1336
		III	455	476	596	920	566	666	674	1286	1470	2150	3848	2064	3018	1639
		IV	498	520	651	1129	695	817	827	1579	1805	2639	4723	2534	3704	2047
T11A	71/R/B/W/T	I	172	179	214	350	218	252	254	544	632	833	1373	884	1288	471
		II	201	210	248	393	244	282	283	608	708	924	1523	987	1439	553
		III	228	237	277	470	292	336	336	722	842	1082	1783	1171	1707	686
		IV	276	286	330	601	372	426	423	914	1069	1335	2201	1475	2155	875
T12C	105/F/B/W/T	I	188	199	302	467	278	340	360	726	826	1241	2302	1160	1719	780
		II	209	222	337	503	299	366	387	781	888	1336	2477	1249	1850	861
		III	225	239	362	555	330	404	427	862	980	1474	2733	1377	2041	1064
		IV	249	263	400	686	408	499	528	1065	1211	1822	3378	1703	2523	1329
T13A	101/R/C/W/T	I	197	205	245	417	249	290	292	697	820	1027	1779	1137	1677	582
		II	231	240	284	468	279	325	326	779	918	1139	1973	1270	1874	683
		III	261	271	318	560	333	386	387	926	1093	1333	2310	1506	2224	847
		IV	316	328	378	716	425	489	487	1171	1387	1645	2852	1898	2807	1079
T14B	40/F/A/W/T	I	36	43	91	251	90	132	183	319	363	579	950	591	832	372
		II	52	61	130	319	115	168	232	405	462	736	1208	752	1059	513
		III	66	78	166	440	158	232	320	559	637	1016	1667	1037	1461	734
		IV	91	107	229	554	227	332	412	800	912	1454	2386	1431	2091	1012

PAVEMENT CLASSIFICATION NUMBERS, STRUCTURAL INDEX, AND SUMMARY OF ALLOWABLE GROSS LOADS
MAXWELL AFB
Non-Frost Period

			PAVEMENT CAPACITY IN KIPS FOR AIRCRAFT GROUP INDEX NUMBERS													
SECTION	PCN	PASS INTENSITY LEVEL	1	2	3	4	5	6	7	8	9	10	11	12	13	14
T15B	76/R/B/W/T	I	202	210	247	405	258	293	294	600	688	899	1438	962	1378	509
		II	238	247	289	458	292	330	331	675	775	1000	1597	1080	1546	600
		III	271	280	326	556	353	398	397	811	931	1176	1874	1294	1846	747
		IV	332	342	392	728	461	515	510	1046	1201	1463	2324	1658	2355	959
T16A	84/R/A/W/T	I	192	200	237	388	242	280	281	603	697	912	1506	969	1418	521
		II	225	235	275	436	272	313	314	675	780	1011	1671	1082	1585	612
		III	255	265	307	521	324	372	372	802	929	1183	1956	1283	1880	759
		IV	309	320	366	666	414	472	469	1014	1178	1461	2414	1617	2373	968
T17A	37/R/C/W/T	I	76	80	104	176	104	122	127	282	330	428	730	466	682	241
		II	89	94	121	197	116	136	142	315	370	475	810	521	762	283
		III	100	106	135	236	139	162	169	374	440	556	948	618	904	351
		IV	122	128	161	301	177	206	212	473	558	686	1170	778	1141	448
T18A1	58/R/C/W/T	I	145	152	178	282	184	209	209	409	471	627	1010	655	955	364
		II	171	178	206	316	206	234	234	458	527	696	1121	732	1067	428
		III	193	201	230	378	246	278	277	544	628	814	1312	868	1266	530
		IV	234	243	274	483	313	353	350	688	796	1005	1619	1093	1598	676
T18A2	60/R/C/W/T	I	138	145	174	280	177	204	206	419	485	646	1051	679	988	369
		II	162	169	201	314	199	228	230	468	543	717	1166	759	1104	433
		III	183	191	225	375	237	272	272	556	647	839	1365	899	1310	537
		IV	222	231	268	480	302	344	343	703	820	1036	1685	1134	1653	684
T19A	22/F/B/W/T	I	41	44	76	118	70	87	95	204	221	342	650	306	468	210
		II	47	50	86	129	77	95	104	223	242	374	711	334	511	237
		III	51	55	94	145	86	108	117	251	273	422	802	377	576	297
		IV	58	62	107	182	108	135	147	315	342	529	1005	472	722	371
T29B	71/R/B/W/T	I	181	188	224	370	232	266	267	561	646	840	1356	904	1296	472
		II	213	221	261	419	262	299	300	632	727	934	1506	1016	1453	556
		III	242	251	295	508	317	360	360	759	874	1099	1767	1216	1735	692
		IV	297	306	355	665	414	467	462	978	1127	1368	2191	1559	2214	889
T30B	79/R/B/W/T	I	203	211	251	414	261	298	299	622	715	932	1499	1001	1434	525
		II	240	249	293	468	295	335	336	700	805	1037	1665	1124	1608	618
		III	273	282	330	567	357	404	403	841	968	1219	1954	1346	1921	770
		IV	334	344	398	743	465	523	518	1084	1248	1518	2423	1726	2451	988
T31B	53/R/B/W/T	I	140	145	172	282	179	204	205	419	481	628	1007	673	964	356
		II	165	171	201	319	203	230	230	472	541	699	1118	756	1081	419
		III	187	194	227	387	245	277	277	567	651	822	1312	905	1291	522
		IV	229	237	273	507	320	359	356	731	839	1023	1627	1160	1647	669
T32B	118/F/A/W/T	I	83	99	223	650	214	324	476	829	951	1554	2661	1604	2249	956
		II	119	141	319	827	273	413	606	1054	1210	1977	3386	2041	2862	1319
		III	152	180	406	1141	376	569	836	1454	1670	2728	4671	2815	3949	1887
		IV	210	248	561	1633	539	815	1197	2081	2389	3904	6685	4029	5650	2604
T33A	39/R/C/W/T	I	81	85	110	182	109	128	133	286	335	441	739	475	690	248
		II	95	100	127	205	123	143	149	320	375	489	819	530	771	291
		III	107	113	142	245	147	170	176	380	447	573	959	629	915	361
		IV	130	136	169	313	187	216	222	481	567	707	1184	792	1155	460
T34A	49/R/C/W/T	I	99	104	134	222	133	156	162	353	412	540	909	583	849	304
		II	116	122	155	250	149	175	181	394	462	599	1008	651	948	356
		III	131	138	173	299	178	208	214	468	550	701	1180	772	1125	442
		IV	159	167	206	382	227	264	270	592	697	866	1457	973	1420	563

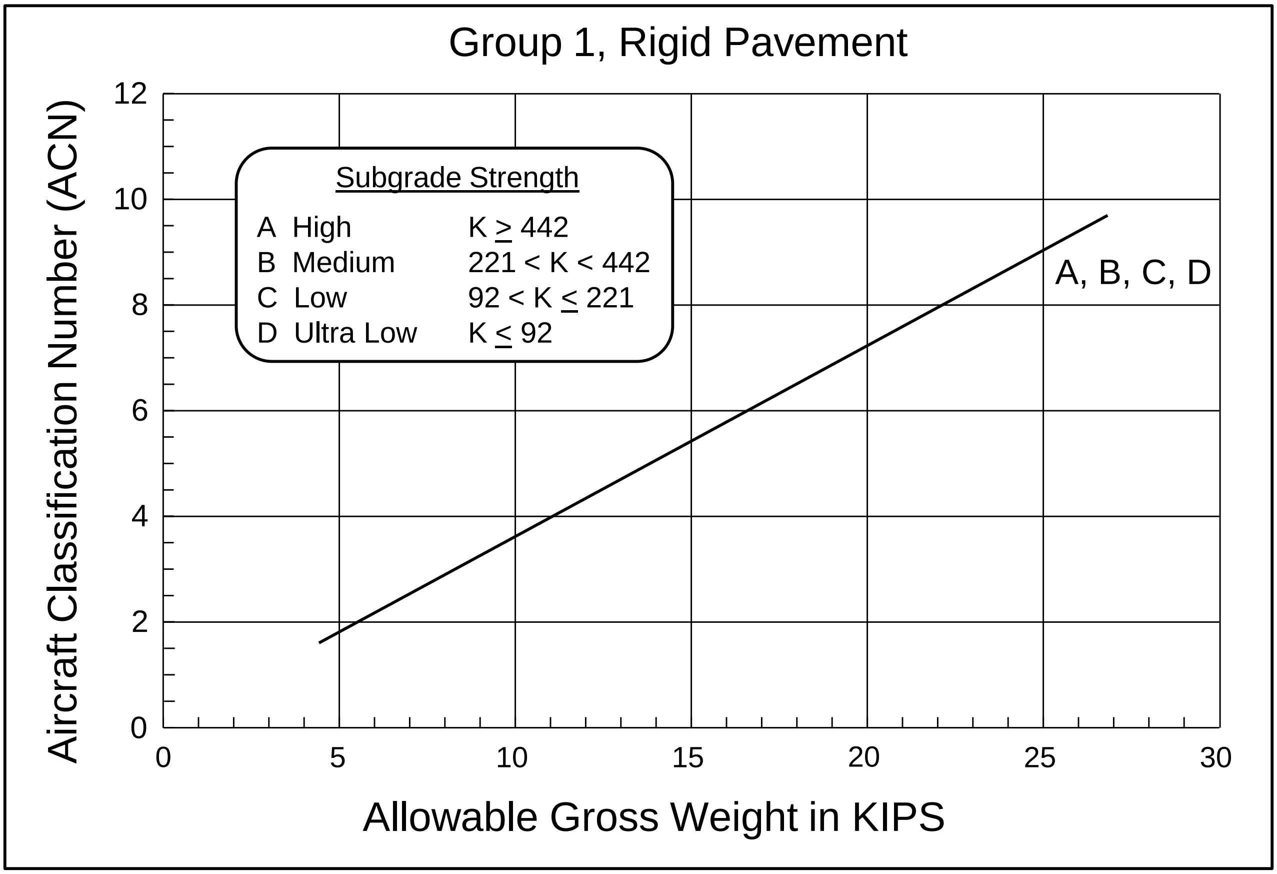
UNDERSTANDING THE PCN CODE EXAMPLE: 31/F/C/W/T				
PCN NUMERIC VALUE	**PAVEMENT TYPE**	**SUBGRADE STRENGTH**	**ALLOWABLE TIRE PRESSURE**	**METHOD OF PCN DETERMINATION**
31	**F - FLEXIBLE**	A	**W**	**T - TECHNICAL EVALUATION**
	R - RIGID	B	X	U - USING AIRCRAFT
		C	Y	
		D	Z	

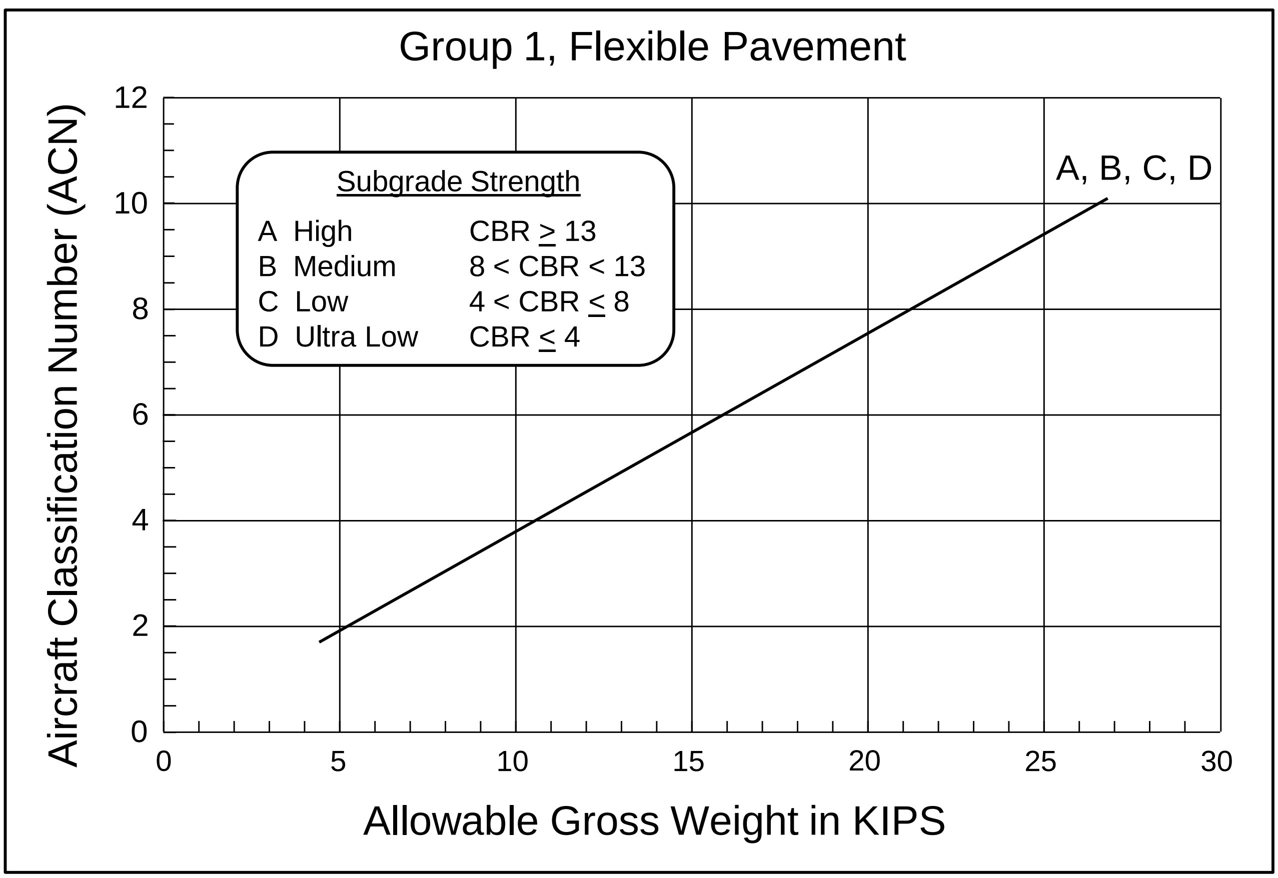
SUBGRADE STRENGTH CODES

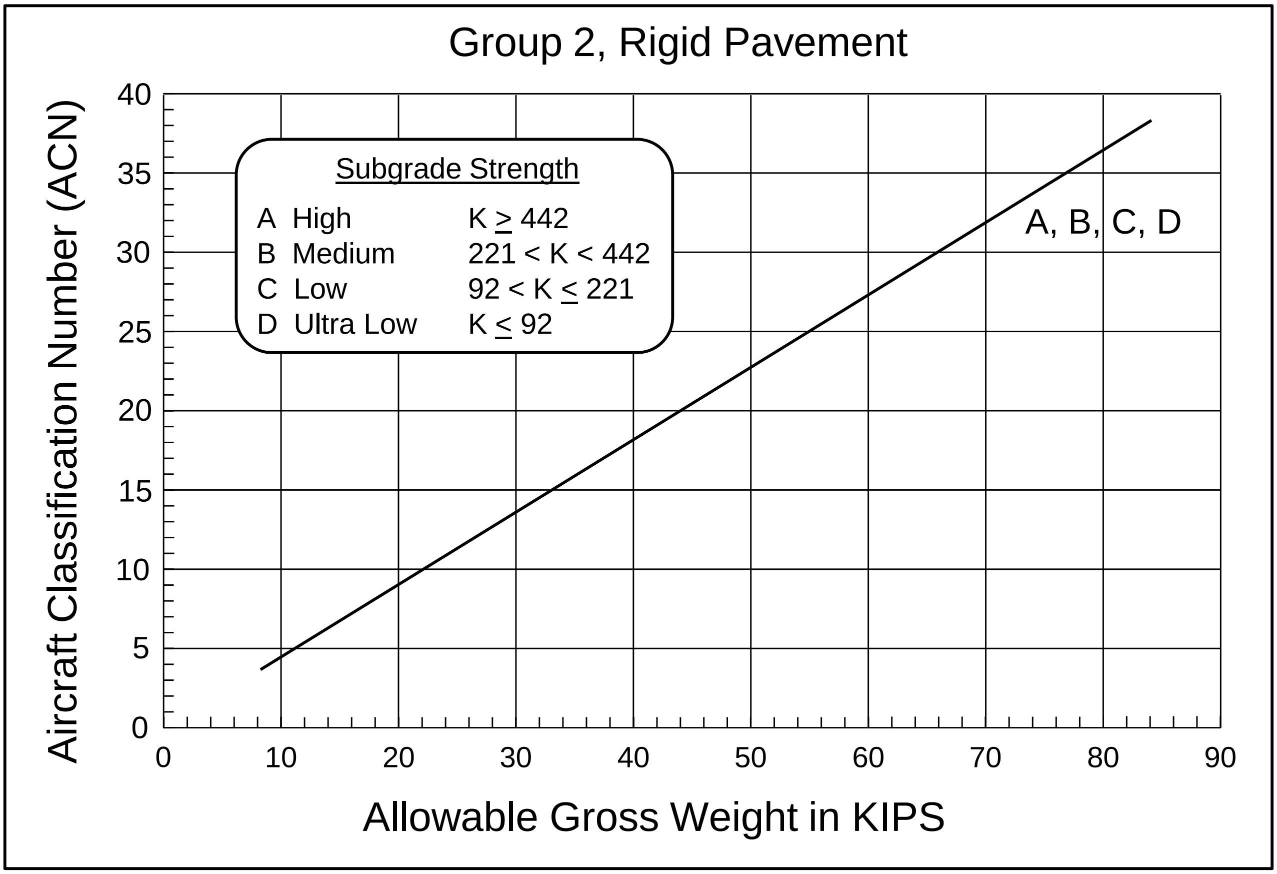
CODE	**CATEGORY**	**FLEXIBLE PAVEMENT CBR, %**	**RIGID PAVEMENT k, pci**
A	HIGH	OVER 13	OVER 442
B	MEDIUM	8-13	221 - 442
C	LOW	4-8	92 - 221
D	ULTRA LOW	< 4	< 92

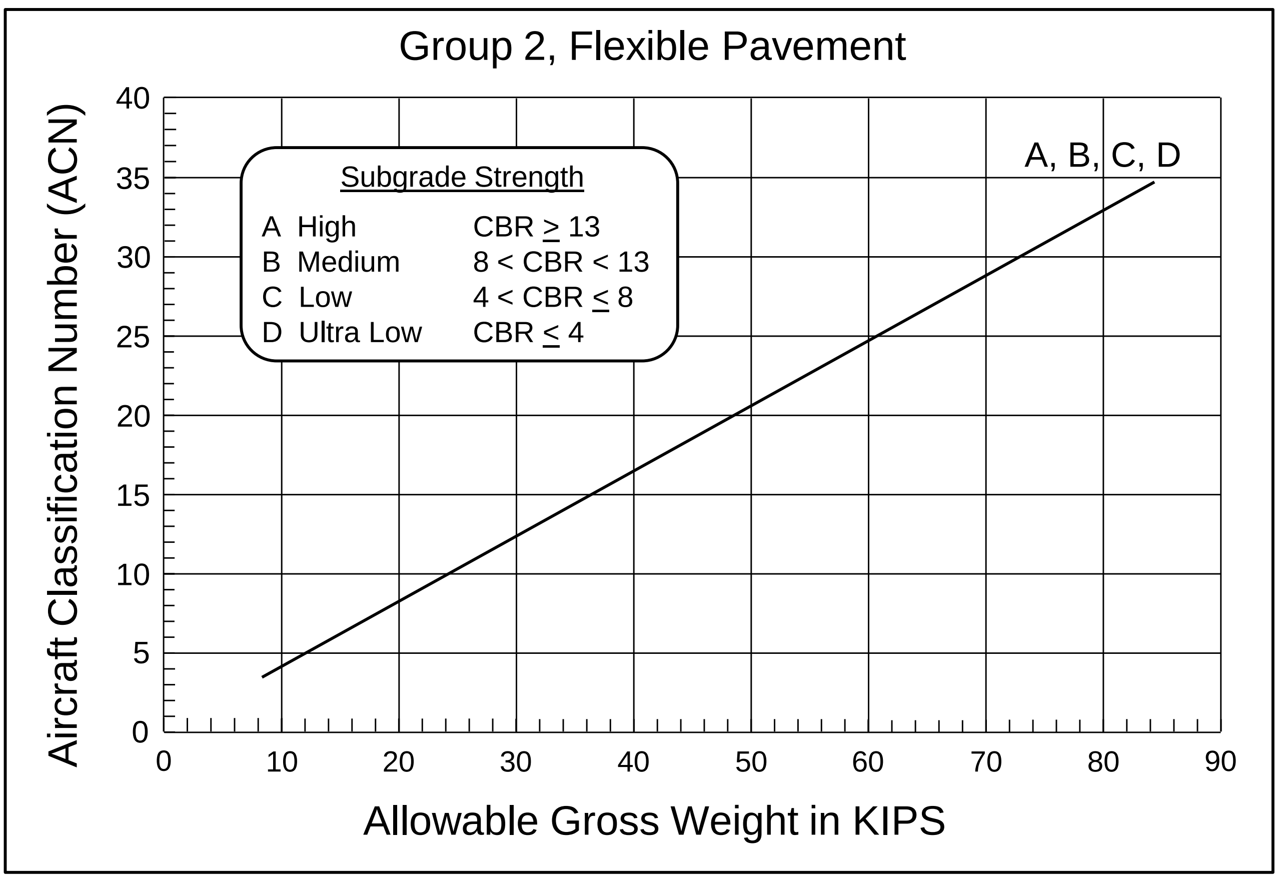
TIRE PRESSURE CODES

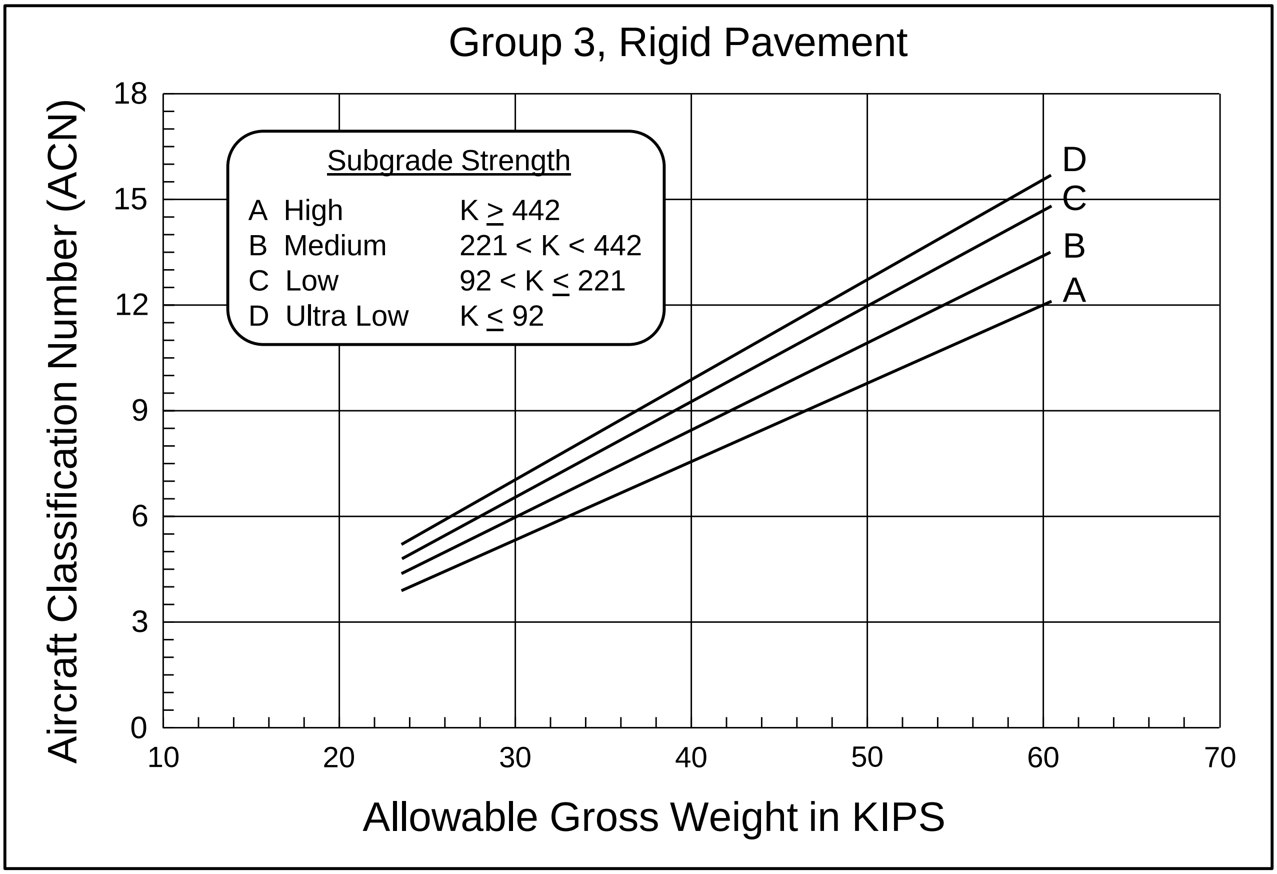
CODE	**CATEGORY**	**ALLOWABLE TIRE PRESSURE, psi**
W	HIGH	NO LIMIT
X	MEDIUM	182 - 254
Y	LOW	74 - 181
Z	ULTRA LOW	0 - 73

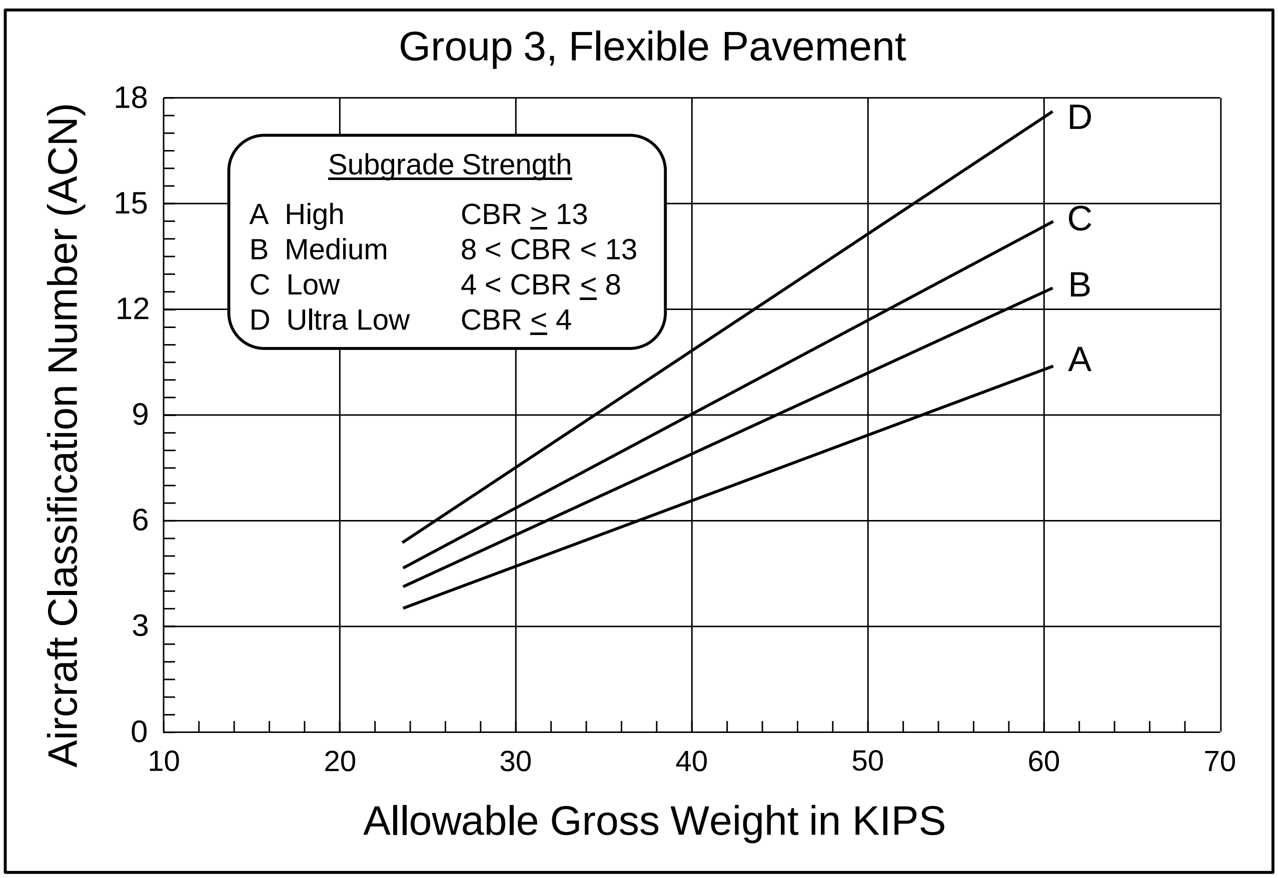

Group 1, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
A, B, C, D
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
0
2
4
6
8
10
12
0
5
10
15
20
25
30

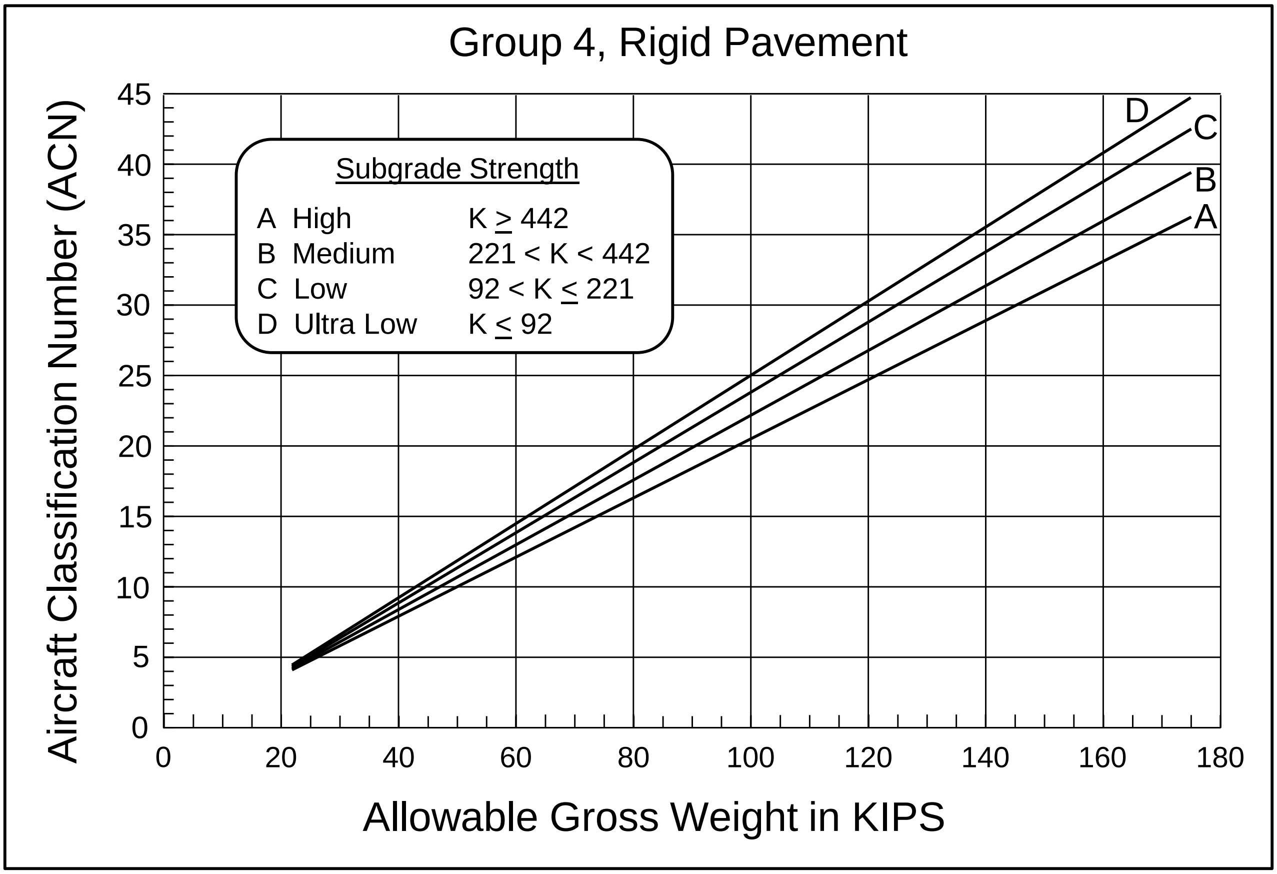

Group 1, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
A, B, C, D
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
0
2
4
6
8
10
12
0
5
10
15
20
25
30


Group 2, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
A, B, C, D
Aircraft Classification Number (ACN)
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50
60
70
80
90
Allowable Gross Weight in KIPS

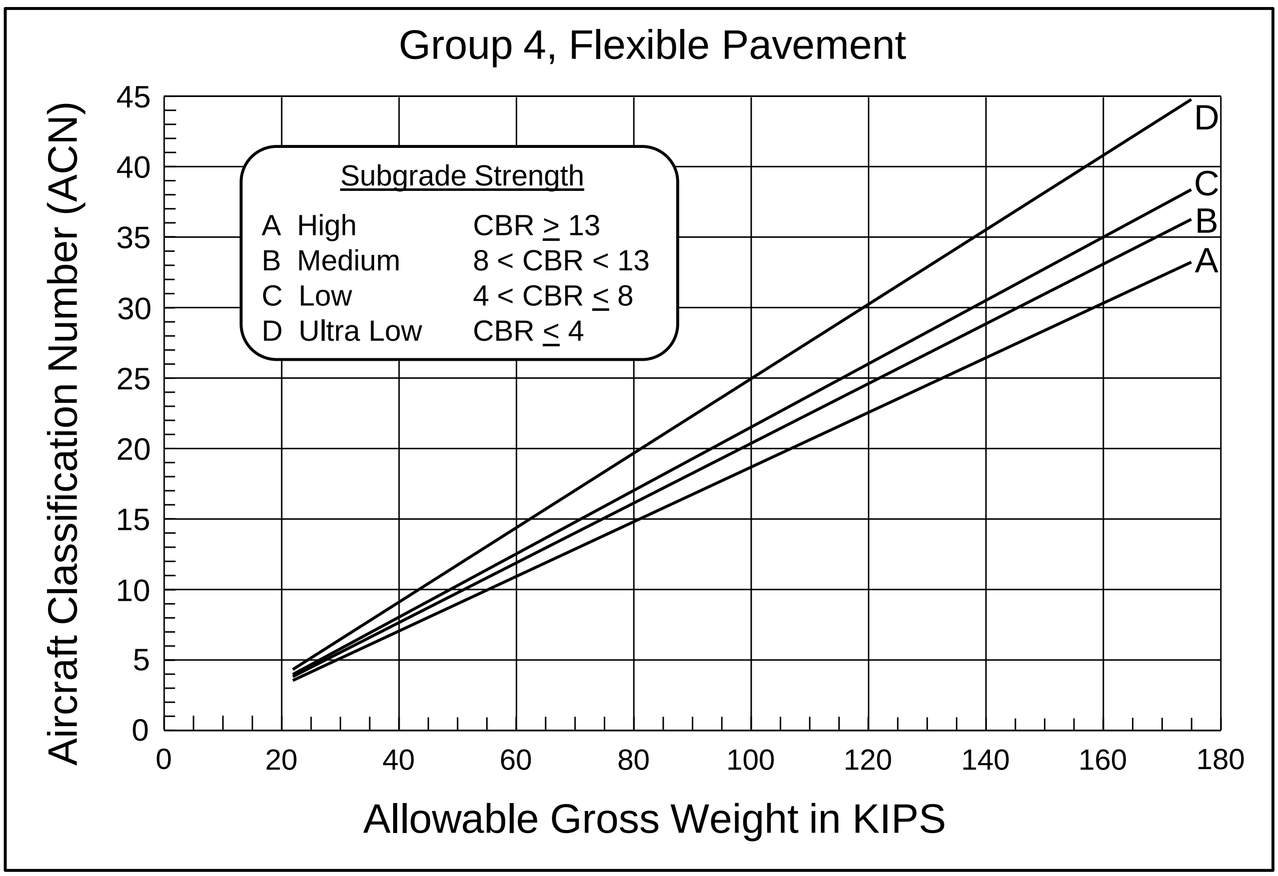

Group 2, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
A, B, C, D
Aircraft Classification Number (ACN)
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50
60
70
80
90
Allowable Gross Weight in KIPS


Group 3, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
A
B
C
D
Aircraft Classification Number (ACN)
0
3
6
9
12
15
18
10
20
30
40
50
60
70
Allowable Gross Weight in KIPS

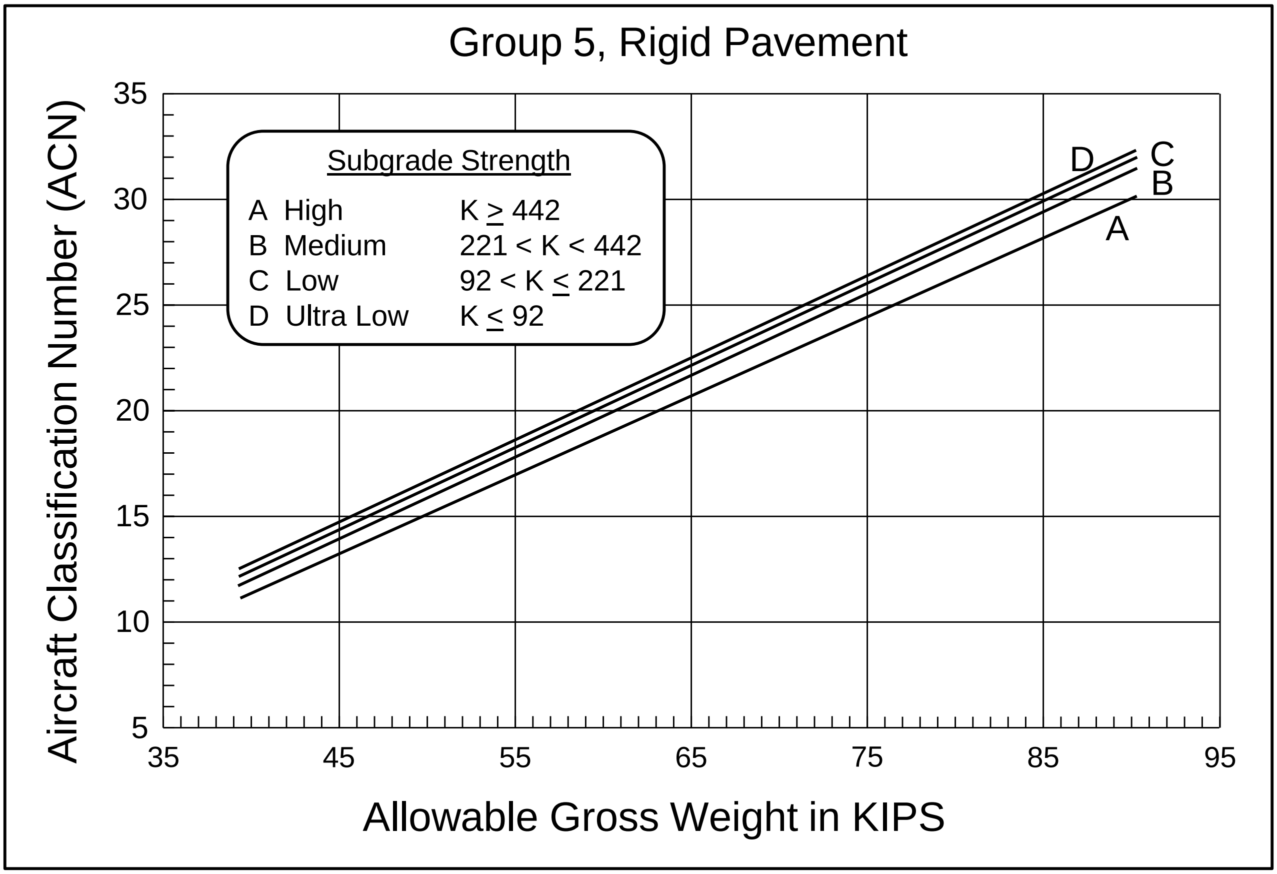

Group 3, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
A
B
C
D
Aircraft Classification Number (ACN)
0
3
6
9
12
15
18
10
20
30
40
50
60
70
Allowable Gross Weight in KIPS



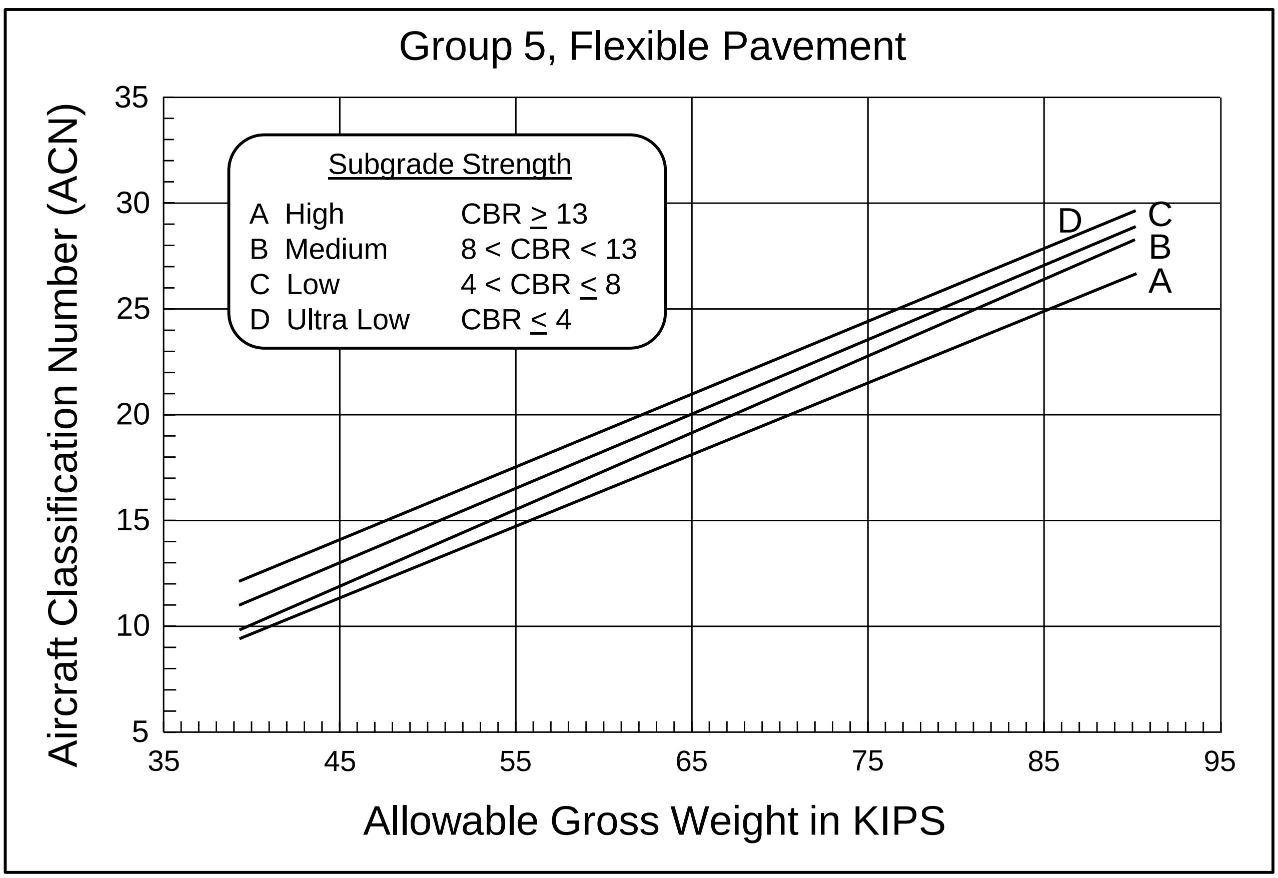
Group 4, Rigid Pavement
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
A
B
C
D
0
5
10
15
20
25
30
35
40
45
0
20
40
60
80
100
120
140
160
180



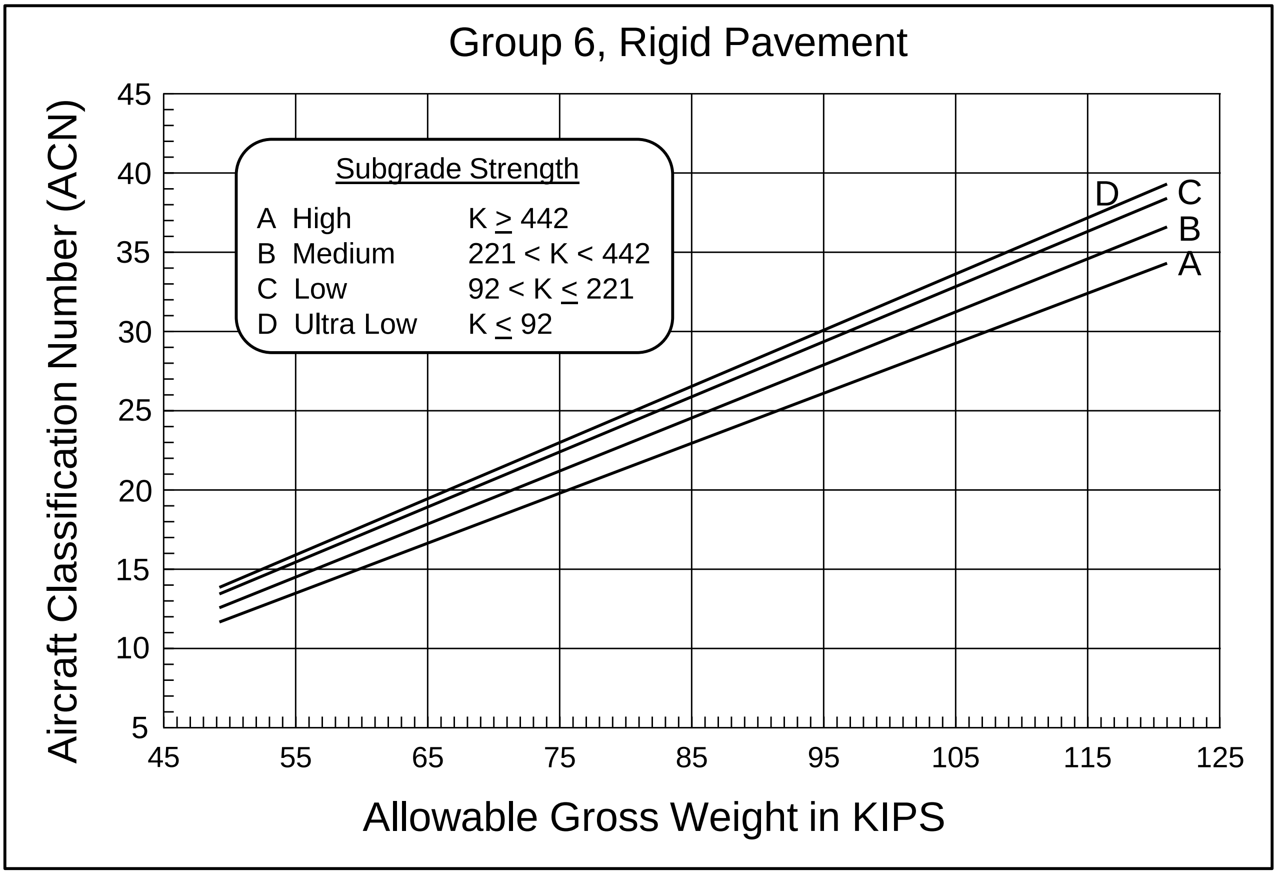
Group 4, Flexible Pavement
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
A
B
C
D
0
5
10
15
20
25
30
35
40
45
0
20
40
60
80
100
120
140
160
180

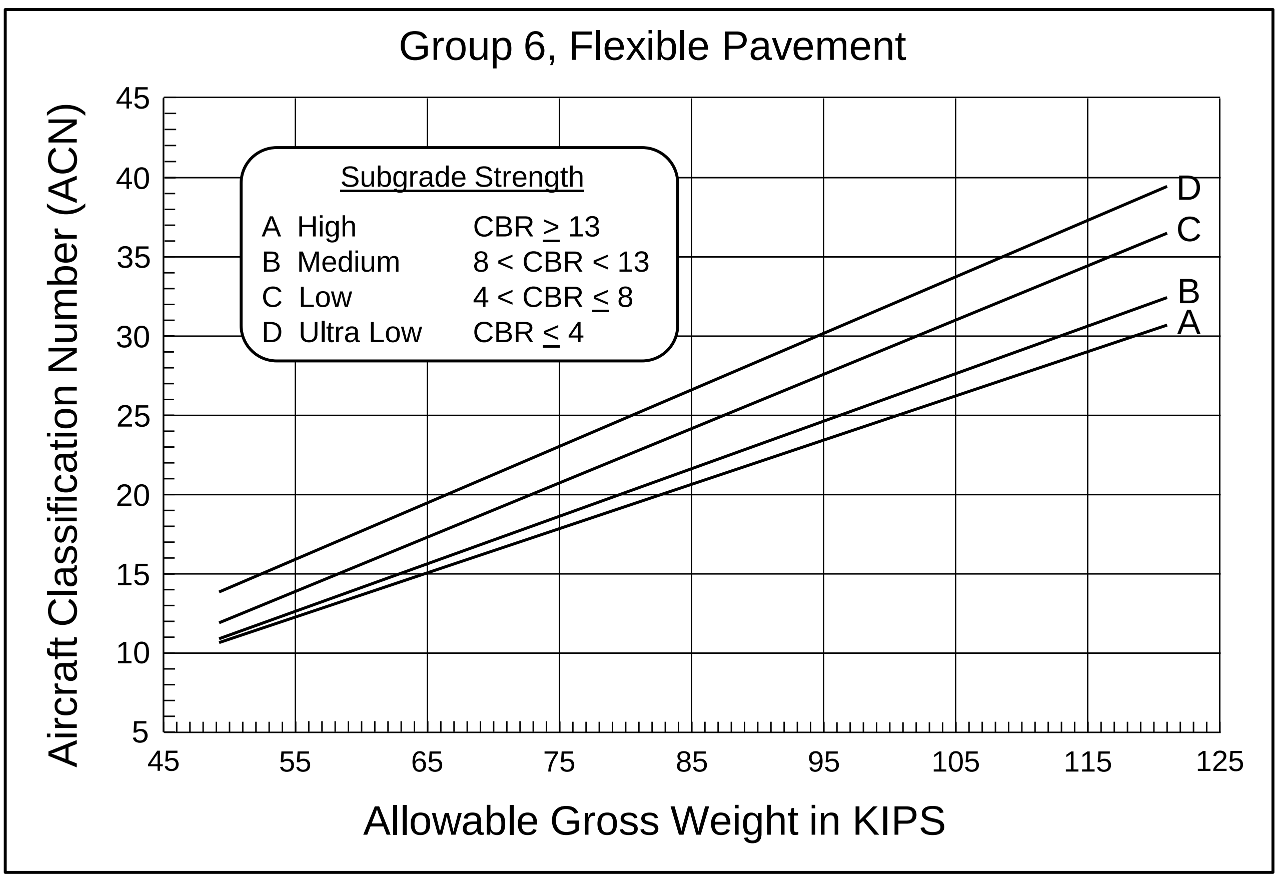


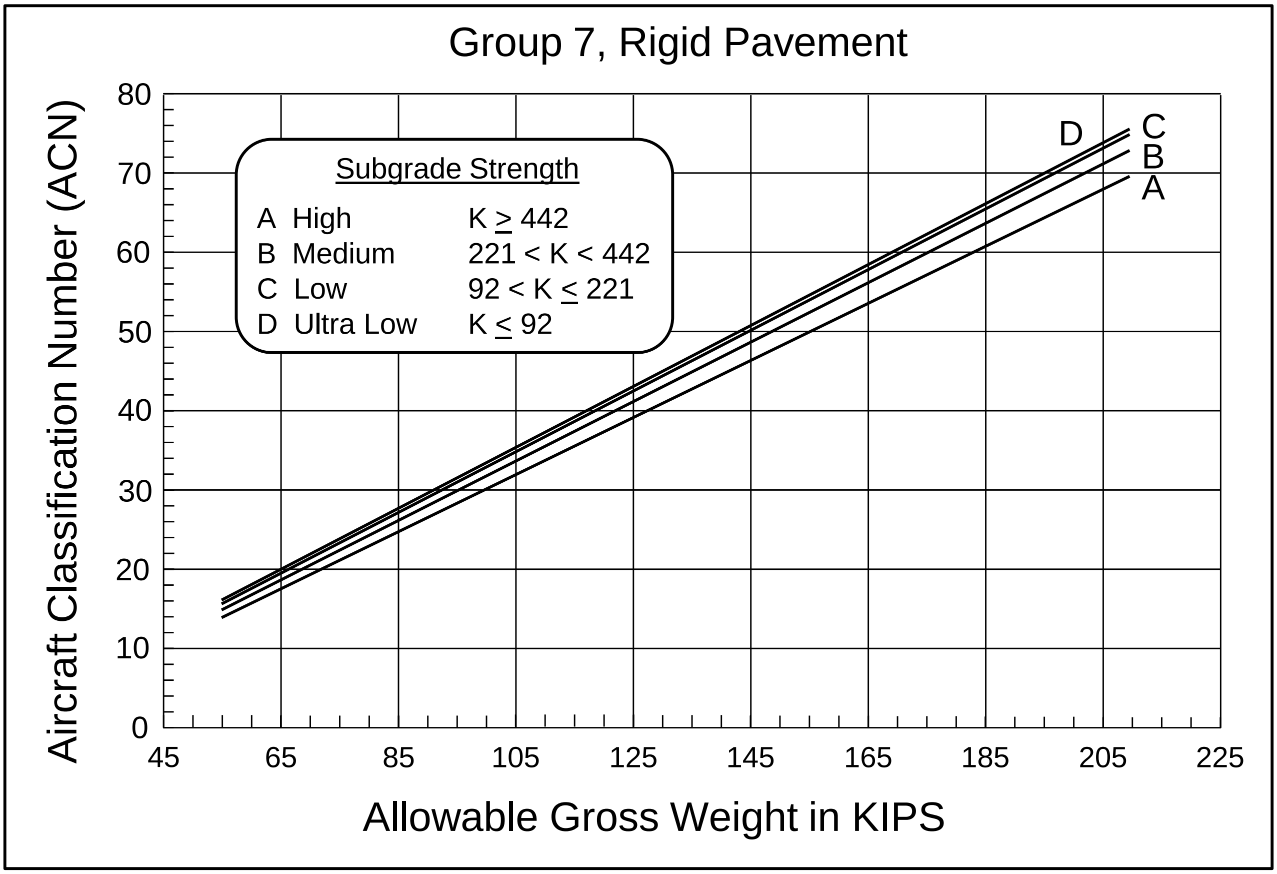
Group 5, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
D
C
B
A
35
30
25
20
15
10
5
35
45
55
65
75
85
95
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS

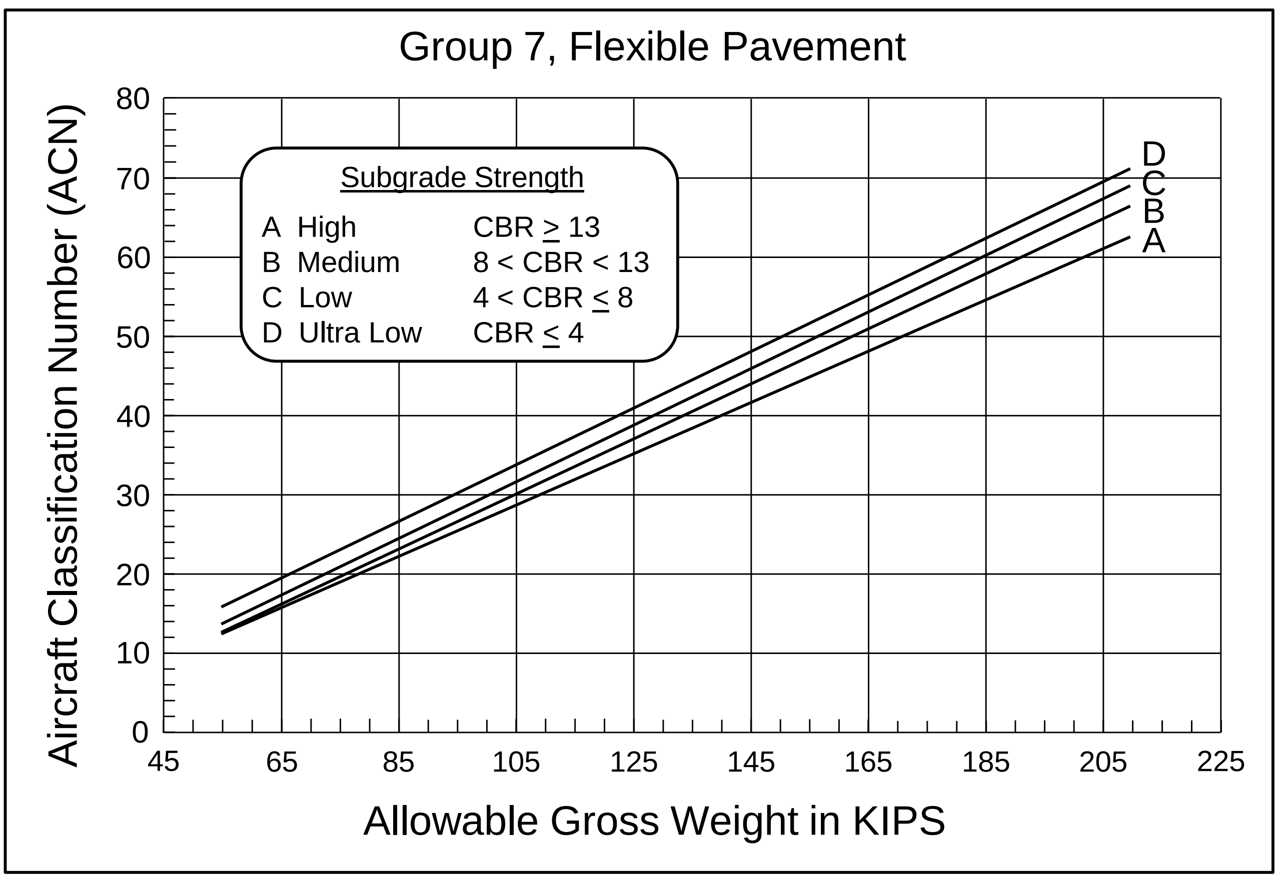


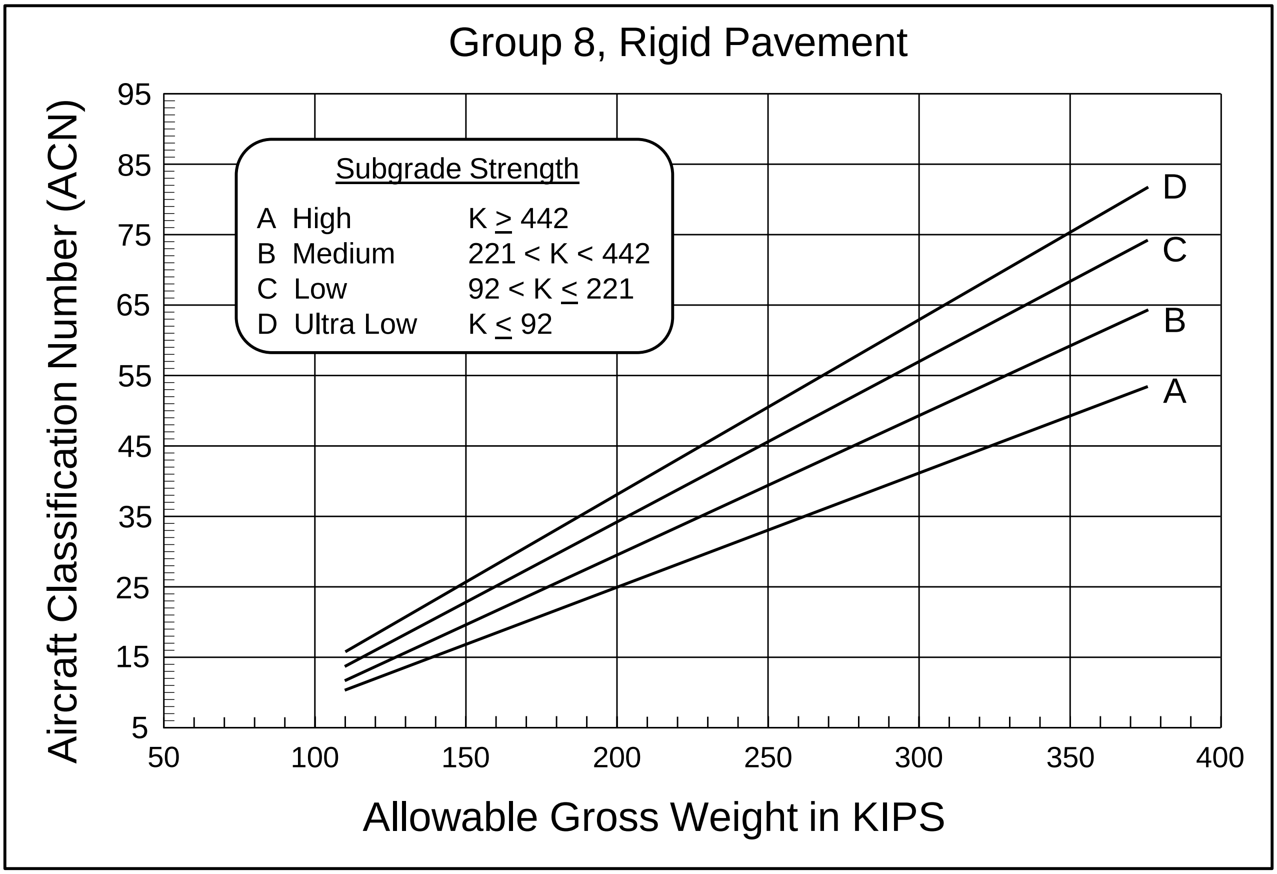
Group 5, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
D
C
B
A
35
30
25
20
15
10
5
35
45
55
65
75
85
95
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS

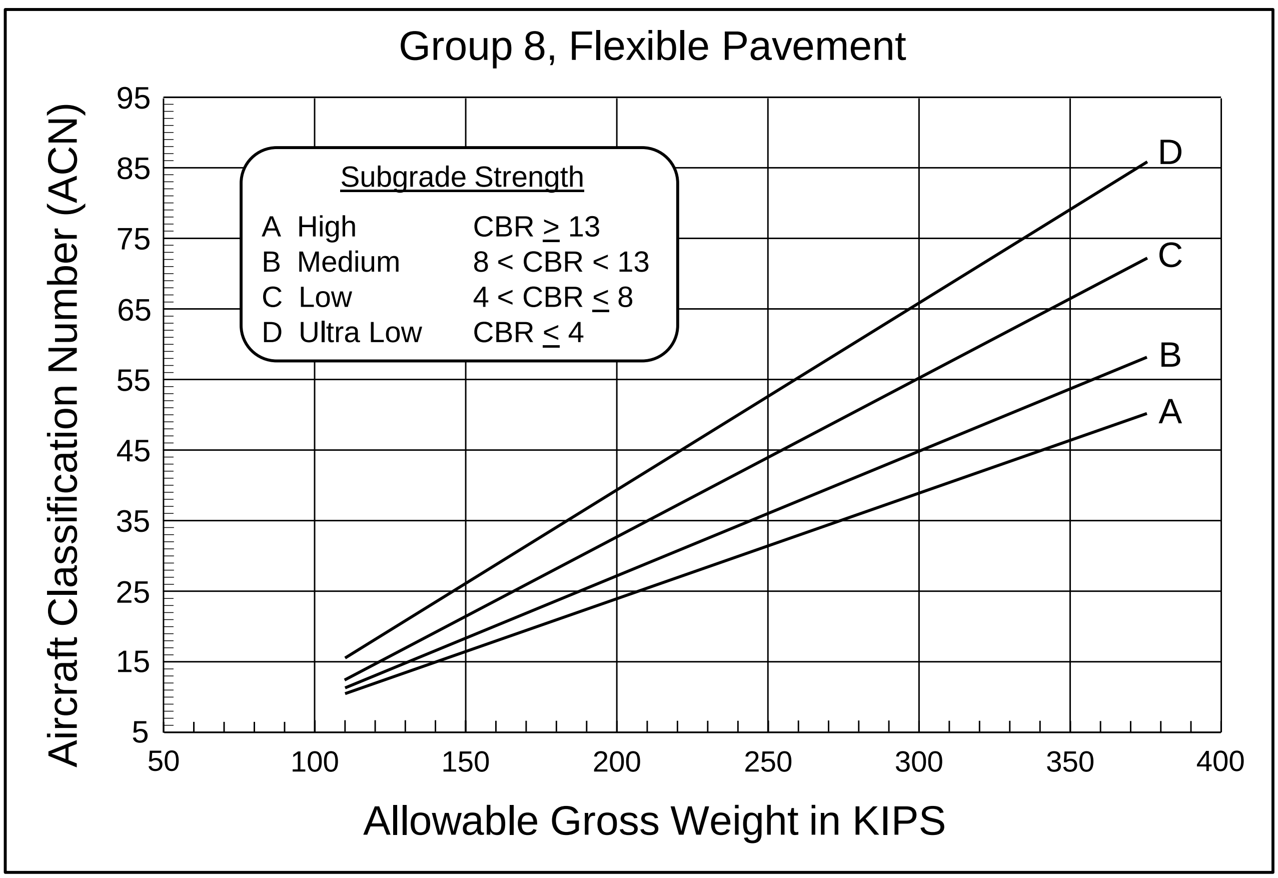

Group 6, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
D C B A
Aircraft Classification Number (ACN)
5 10 15 20 25 30 35 40 45
45 55 65 75 85 95 105 115 125
Allowable Gross Weight in KIPS

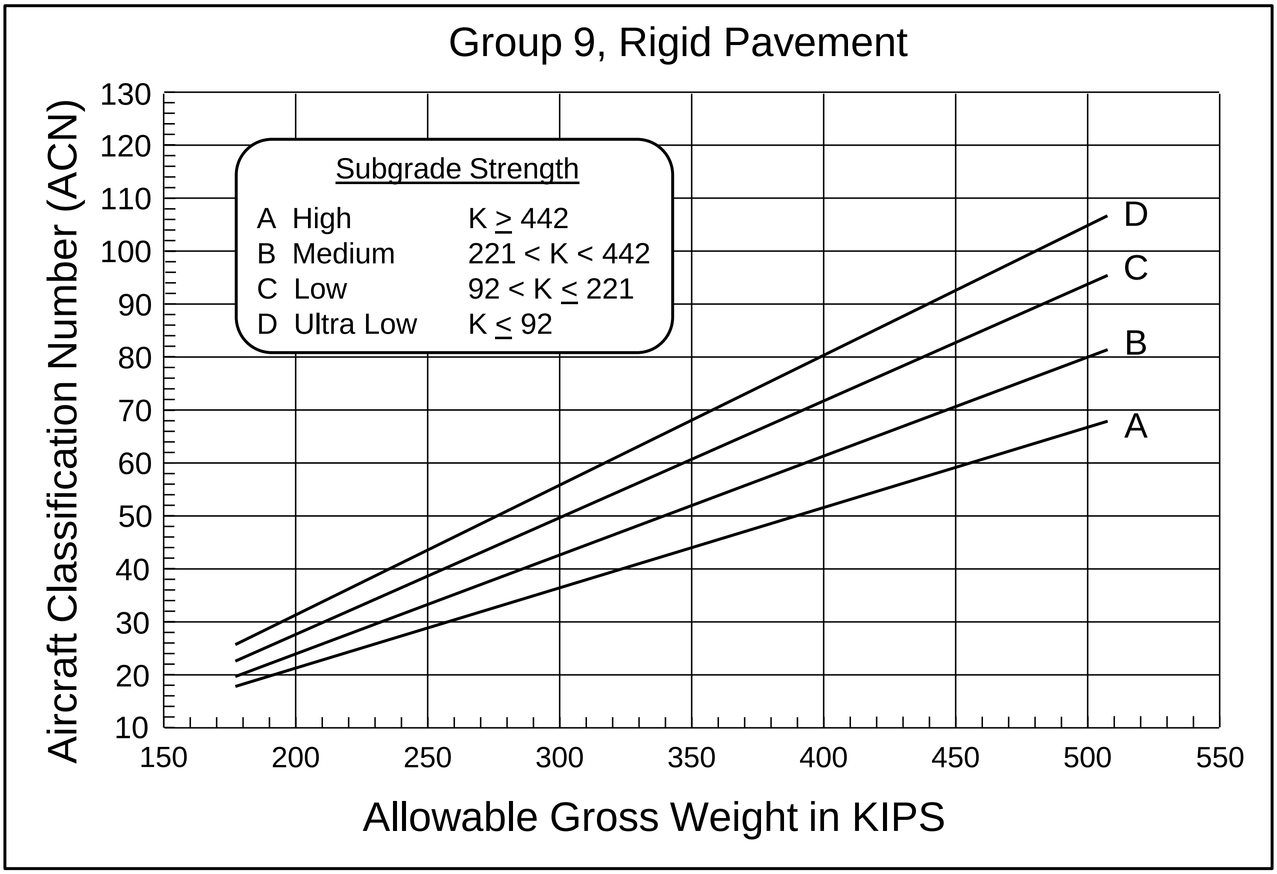

Group 6, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
D C B A
Aircraft Classification Number (ACN)
5 10 15 20 25 30 35 40 45
45 55 65 75 85 95 105 115 125
Allowable Gross Weight in KIPS

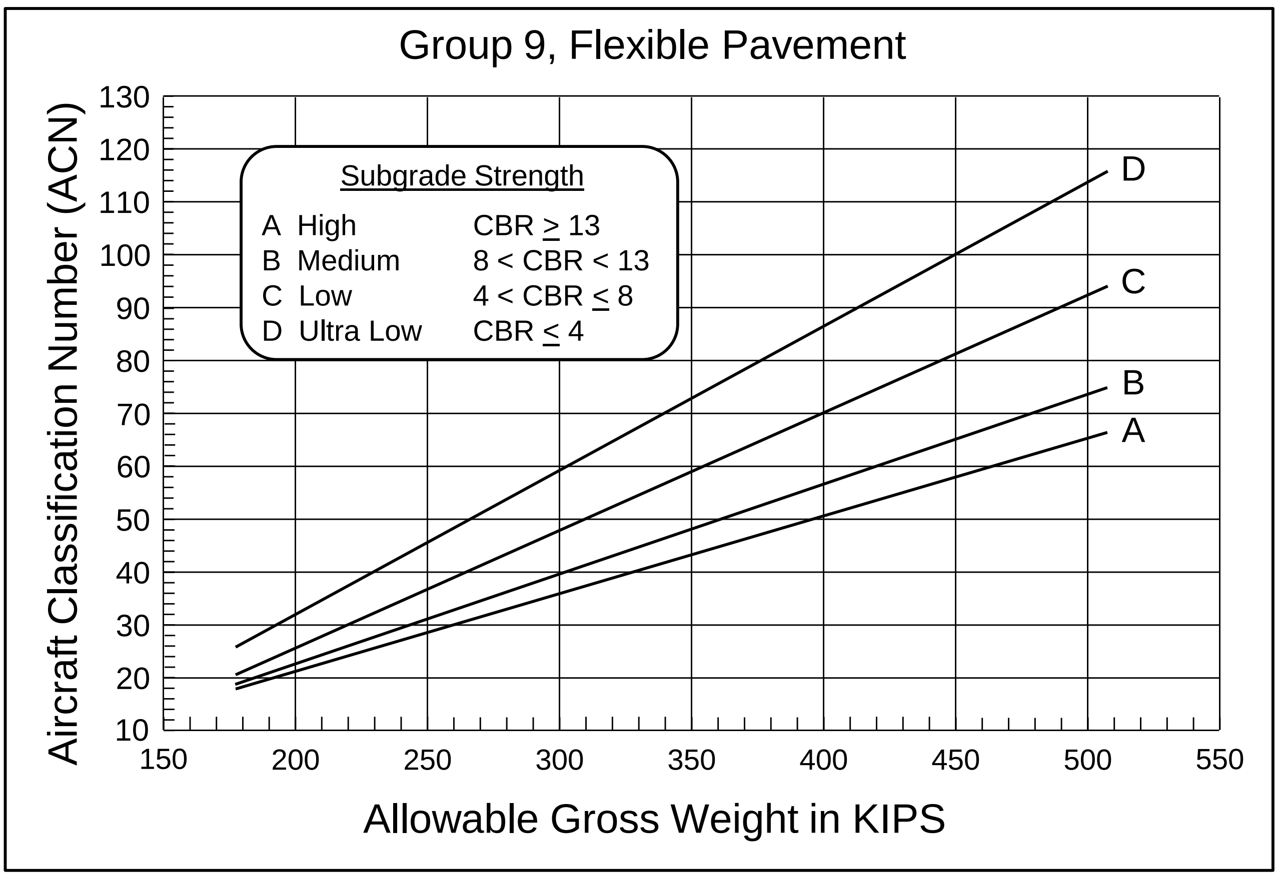

Group 7, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
0
10
20
30
40
50
60
70
80
45
65
85
105
125
145
165
185
205
225
A
B
C
D

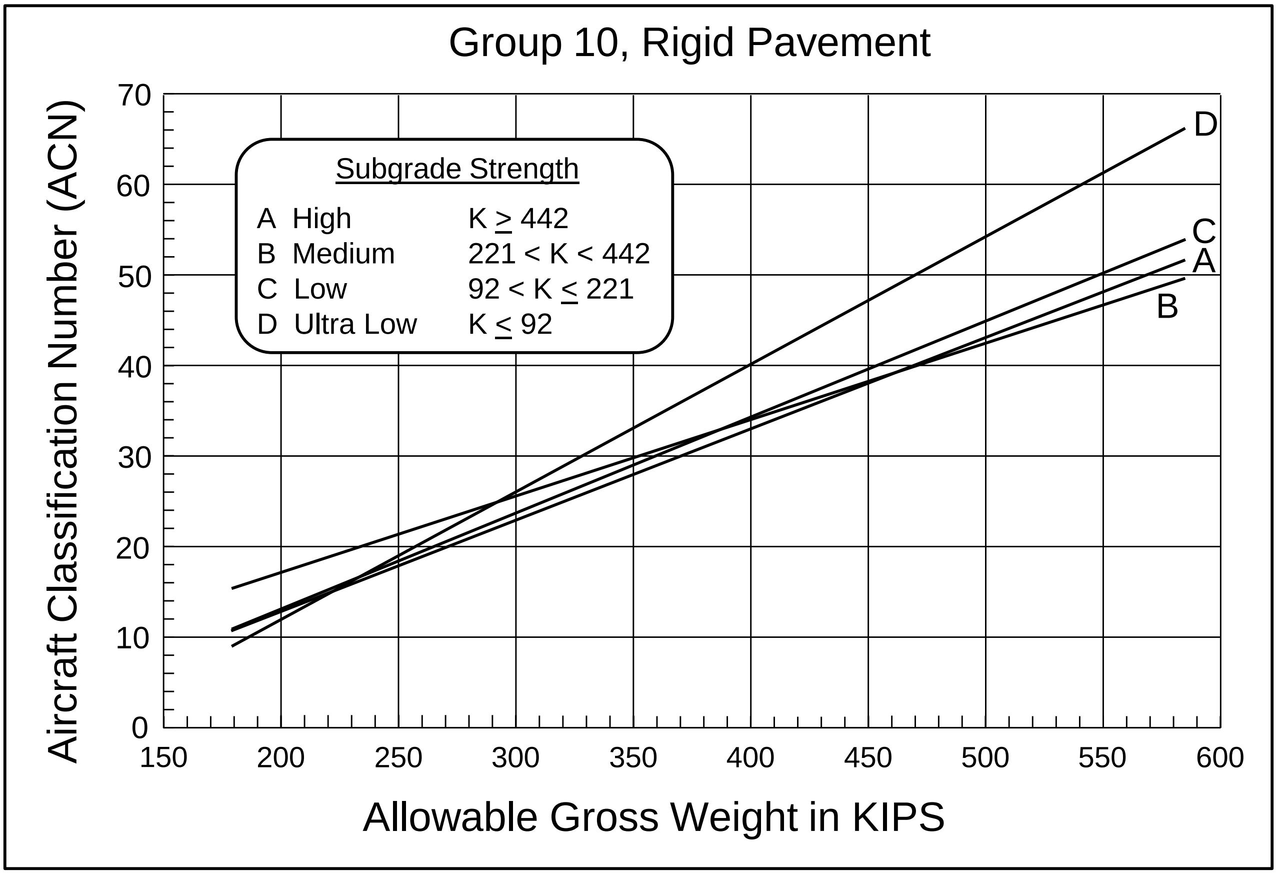

Group 7, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
0
10
20
30
40
50
60
70
80
45
65
85
105
125
145
165
185
205
225
A
B
C
D

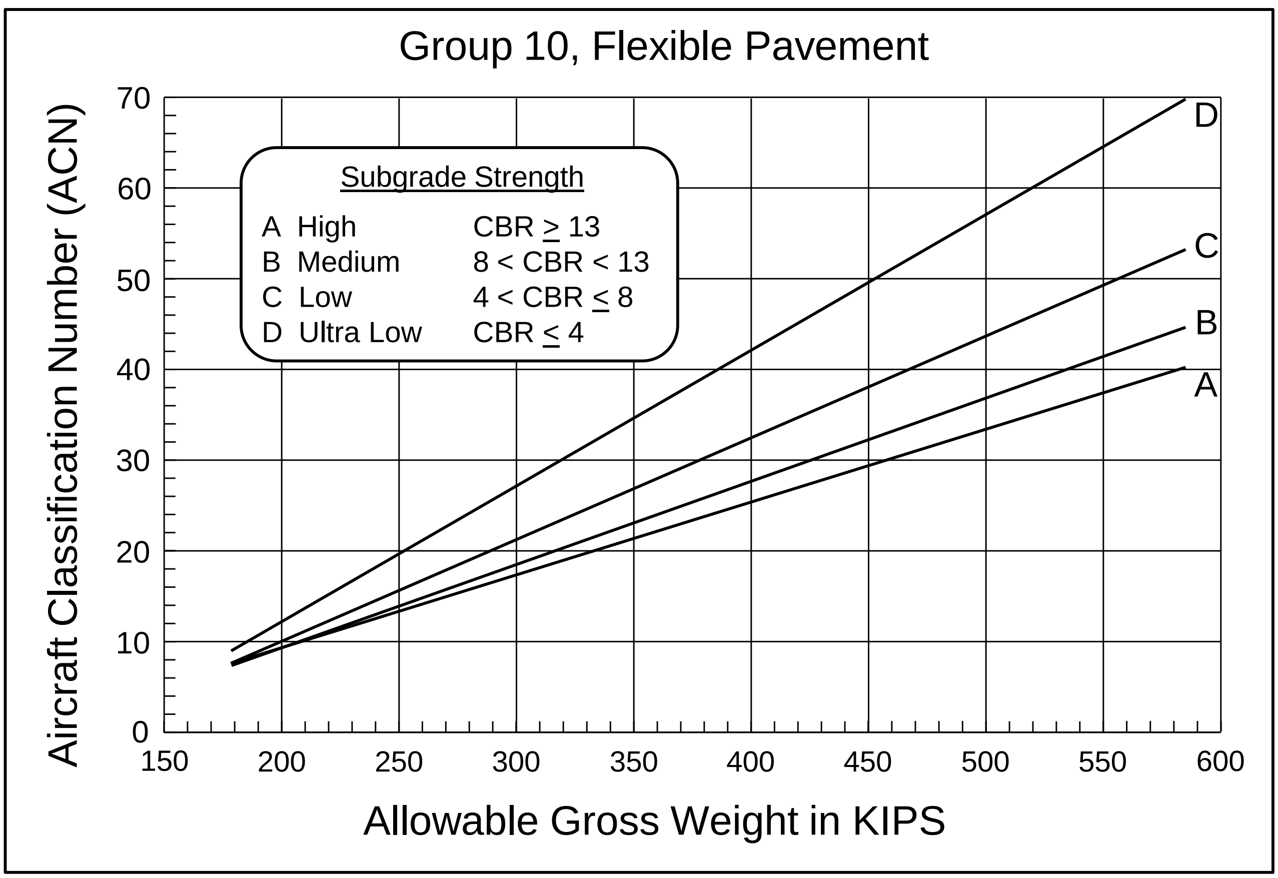

Group 8, Rigid Pavement
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
A
B
C
D
95
85
75
65
55
45
35
25
15
5
50
100
150
200
250
300
350
400

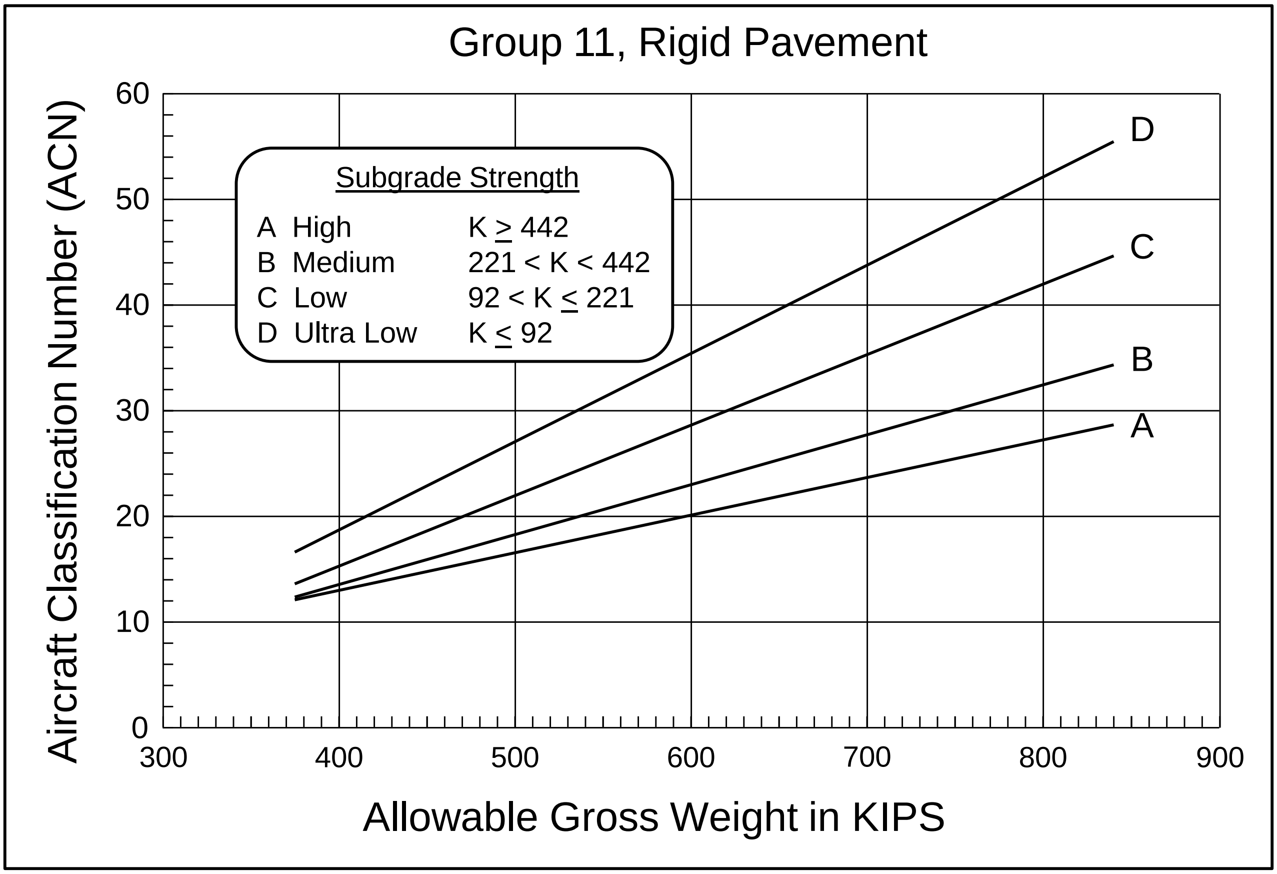

Group 8, Flexible Pavement
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
A
B
C
D
95
85
75
65
55
45
35
25
15
5
50
100
150
200
250
300
350
400


Group 9, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
A
B
C
D
10 20 30 40 50 60 70 80 90 100 110 120 130
150 200 250 300 350 400 450 500 550

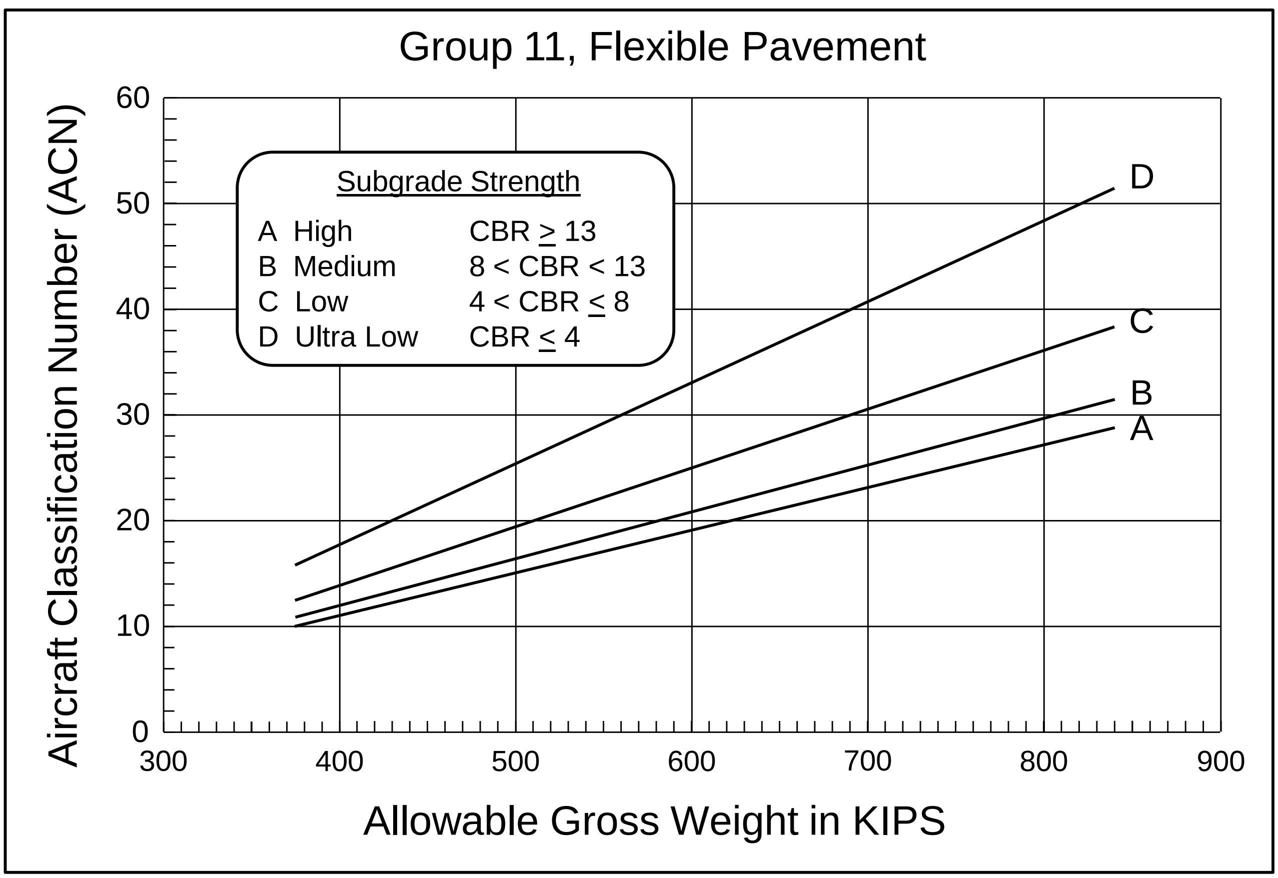

Group 9, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
A
B
C
D
10 20 30 40 50 60 70 80 90 100 110 120 130
150 200 250 300 350 400 450 500 550


Group 10, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
0
10
20
30
40
50
60
70
150
200
250
300
350
400
450
500
550
600
D
C
A
B

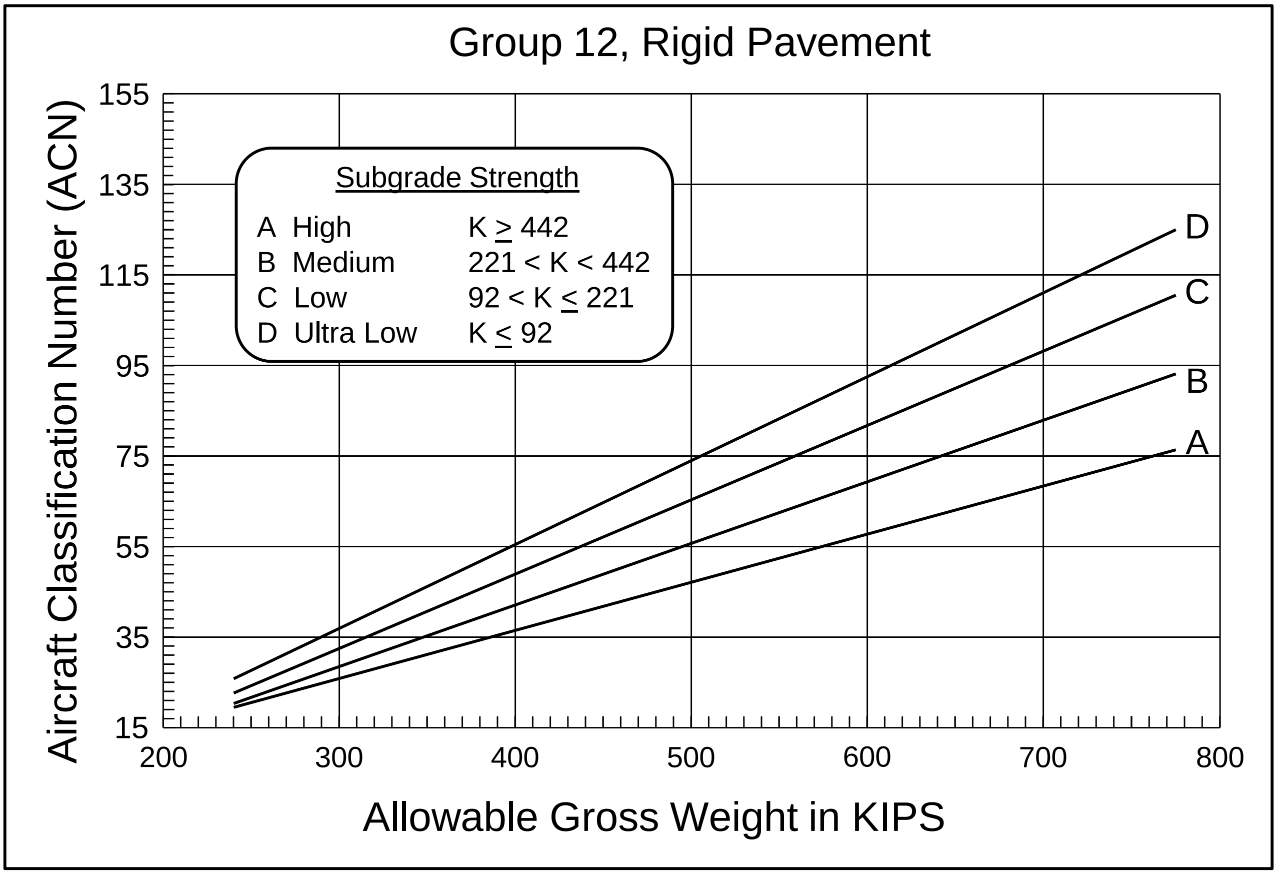

Group 10, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
0
10
20
30
40
50
60
70
150
200
250
300
350
400
450
500
550
600
D
C
B
A

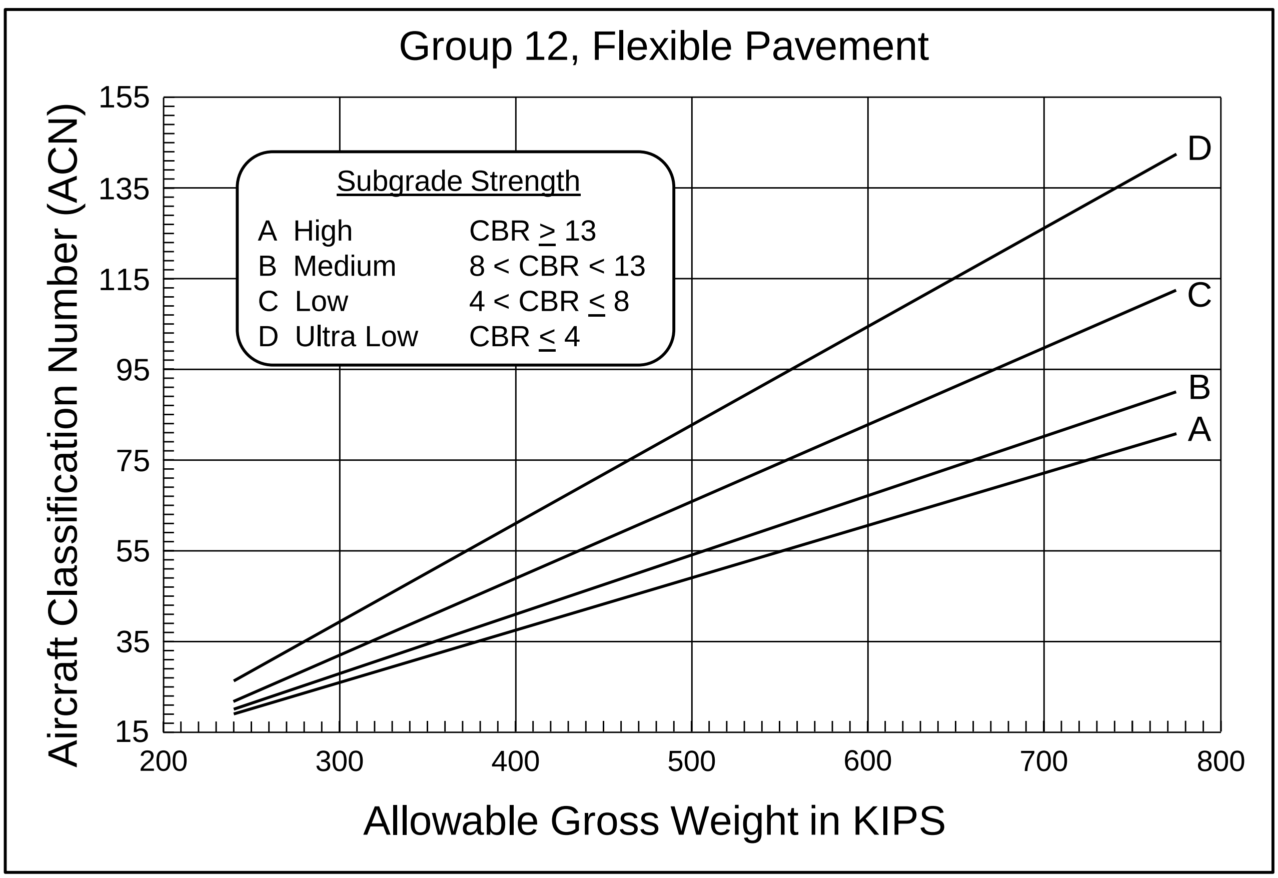


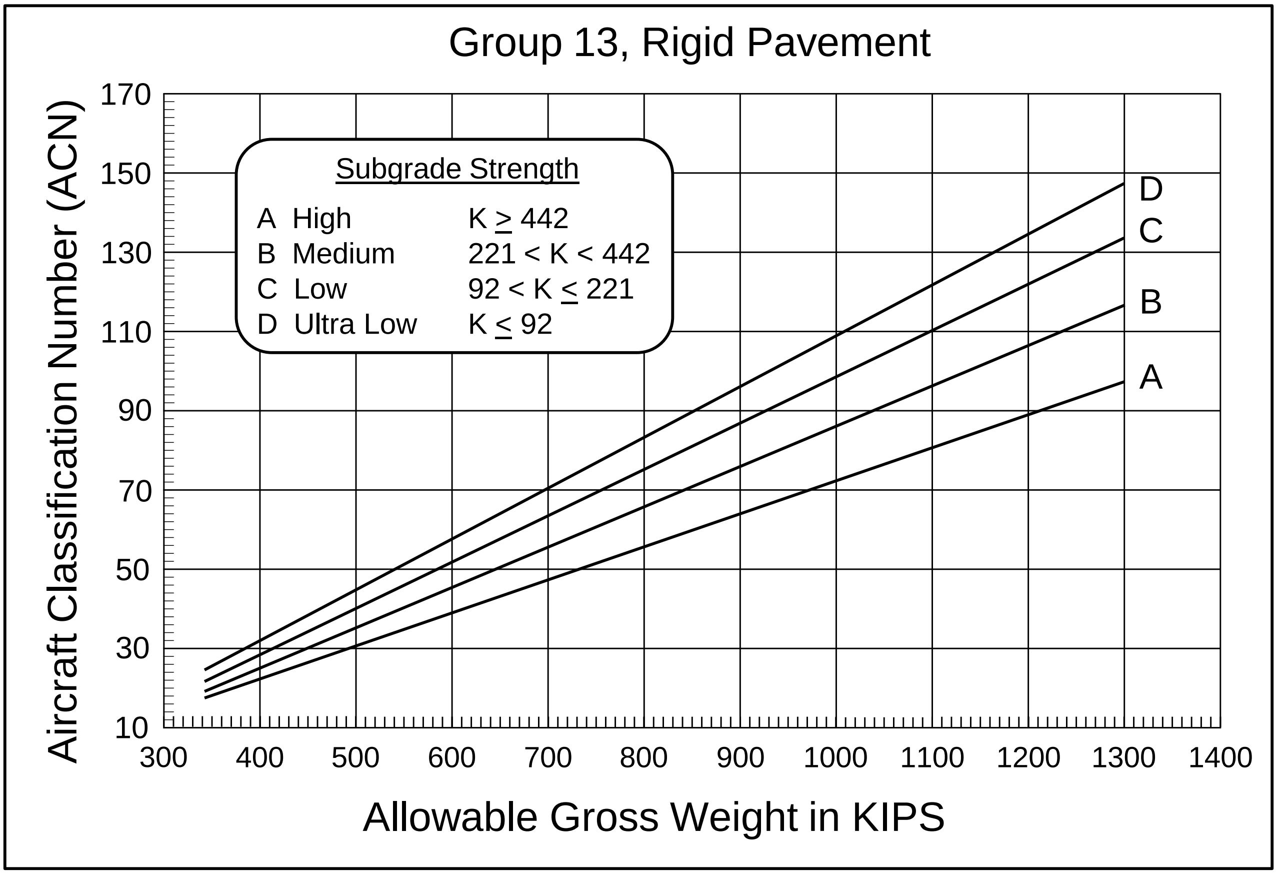
Group 11, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
D
C
B
A
Aircraft Classification Number (ACN)
0
10
20
30
40
50
60
300
400
500
600
700
800
900
Allowable Gross Weight in KIPS

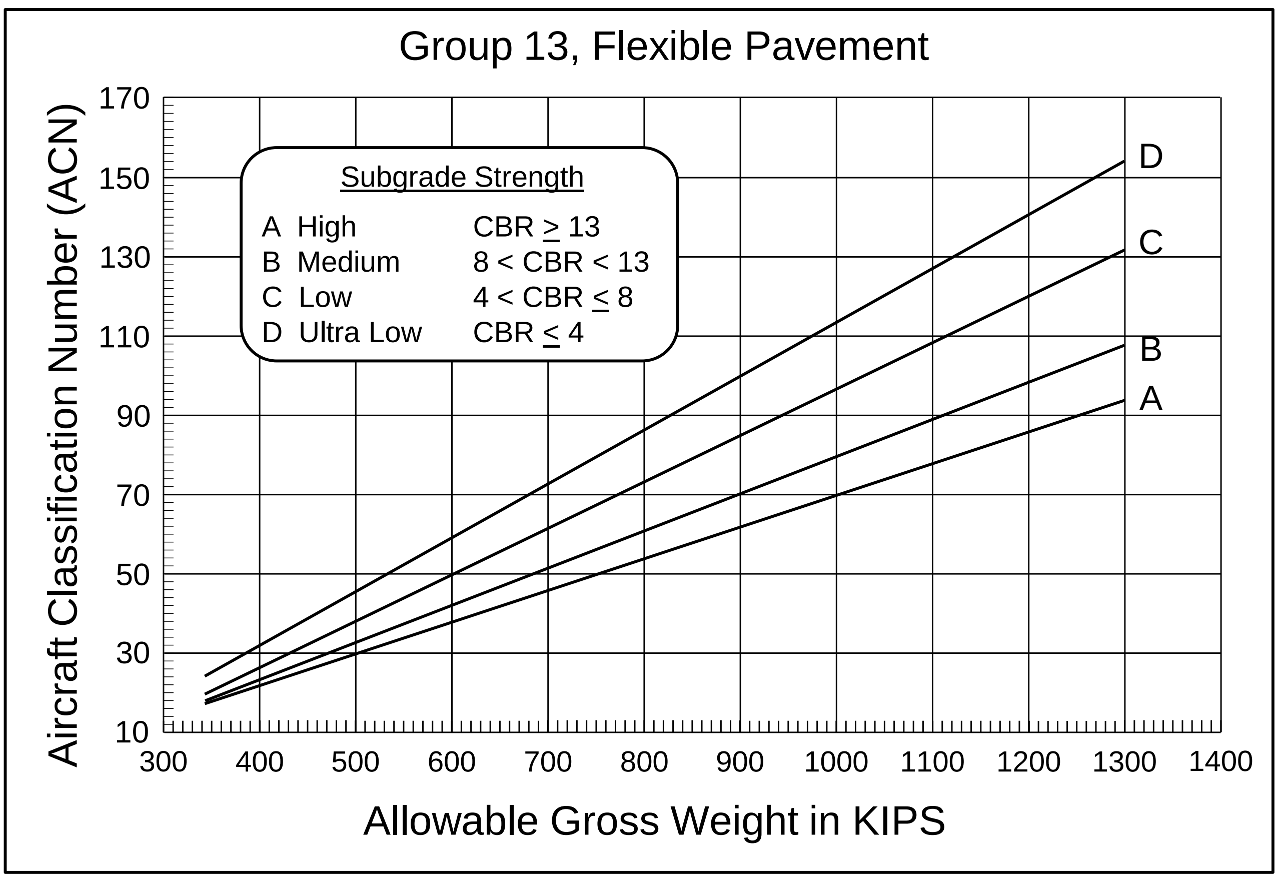


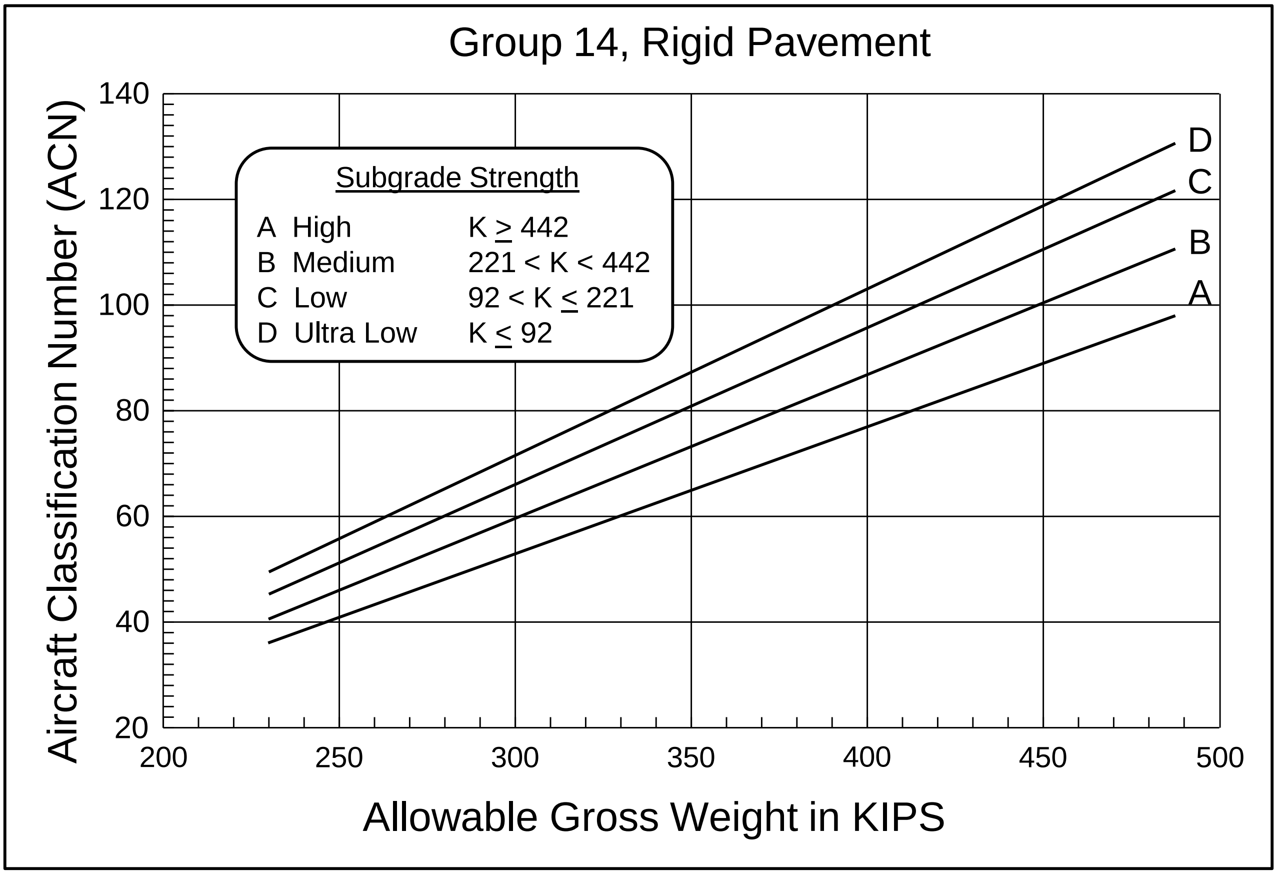
Group 11, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
D
C
B
A
Aircraft Classification Number (ACN)
0
10
20
30
40
50
60
300
400
500
600
700
800
900
Allowable Gross Weight in KIPS

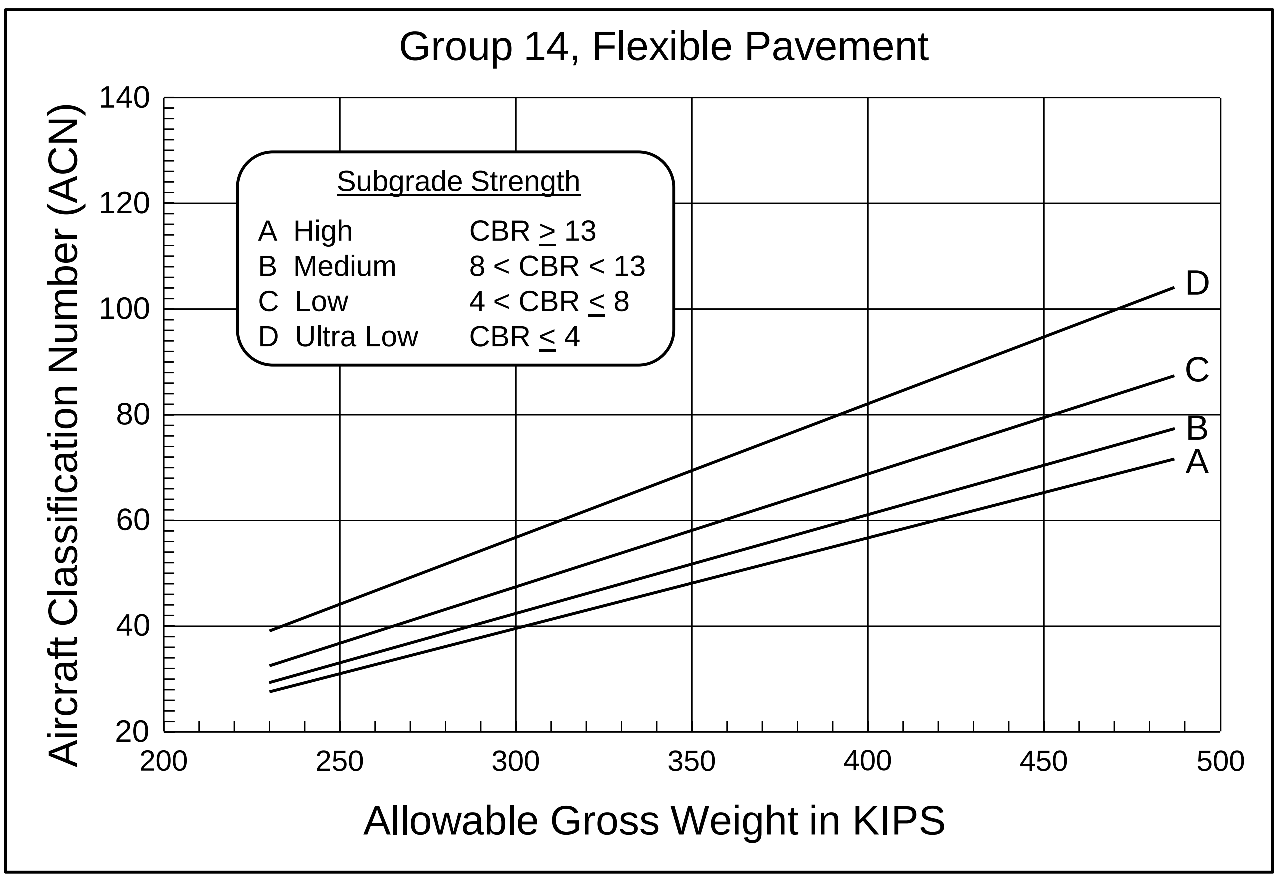

Group 12, Rigid Pavement
Aircraft Classification Number (ACN)
155
135
115
95
75
55
35
15
200
300
400
500
600
700
800
Allowable Gross Weight in KIPS
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
D
C
B
A


Group 12, Flexible Pavement
Aircraft Classification Number (ACN)
155
135
115
95
75
55
35
15
200
300
400
500
600
700
800
Allowable Gross Weight in KIPS
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
D
C
B
A


Group 13, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
170
150
130
110
90
70
50
30
10
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
D
C
B
A


Group 13, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
Aircraft Classification Number (ACN)
Allowable Gross Weight in KIPS
170
150
130
110
90
70
50
30
10
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
D
C
B
A


Group 14, Rigid Pavement
Subgrade Strength
A High K ≥ 442
B Medium 221 < K < 442
C Low 92 < K ≤ 221
D Ultra Low K ≤ 92
Aircraft Classification Number (ACN)
140
120
100
80
60
40
20
200
250
300
350
400
450
500
Allowable Gross Weight in KIPS
D
C
B
A


Group 14, Flexible Pavement
Subgrade Strength
A High CBR ≥ 13
B Medium 8 < CBR < 13
C Low 4 < CBR ≤ 8
D Ultra Low CBR ≤ 4
Aircraft Classification Number (ACN)
140
120
100
80
60
40
20
200
250
300
350
400
450
500
Allowable Gross Weight in KIPS
D
C
B
A

Related Data: Aircraft Group Index

Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Included Aircraft	C-12 C-21 C-23 C-38A C-41A HH-60 RC-26 RQ-4-Bk 10 T-1* T-6 T-37 UH-1H	A-10 AT-38 F-15* F-16 F-22 F-35 F-117 RQ-4-Bk 20+ T-38	CV-580* MH-53 MV-22 CV-22	C-130* C-27J C-295 CN-235	C-20* C-37	B-717* C-9 DC-9 T-43	A-320 A-321 B-727 B-737 C-22 C-40 MD-81 MD-82 MD 83 MD 87 MD 90 P-3*	A-300 A-310 B-2A B-707 B-720 B-757 C-32A* DC-8 E-3 E-8C KC-135 VC-137	A-330 B-1 B-767 DC-10-10 L-1011 MD-10 B-767 -400ER* KC-46A	C-17* IL-76	C-5*	A-340 B-777 DC-10-30 DC-10-40 KC-10 MD-11*	A-380 AN-124 B-747 E-4 VC-25 B-747 -400*	B-52*

Note: * Denotes Controlling Landing Gear Configuration in Group

Pass Intensity Levels (in Passes)

Level	1	2	3	4	5	6	7	8	9	10	11	12	13	14
I	300,000			50,000										15,000
II	50,000			15,000										3,000
III	15,000			3,000										500
IV	3,000			500										100

Gross Weight Ranges for Aircraft Groups (in KIPs)

Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Lowest Gross Weight	4	8	23	22	39	49	55	110	177	178	374	240	342	230
Highest Gross Weight	27	84	61	175	91	121	210	376	507	585	840	775	1,301	488

Appendix I

Friction Test Results

Slope Measurements

Surface Texture Measurements

Average Friction Values

Grip Tester Friction Plots

Estimation of Rubber Deposits

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MAXWELL AFB, AL
RUNWAY 15/33
SLOPE MEASUREMENTS

33

1.2	1.3	0.4	0.2	1.1	1.1	**8000**
1.5	1.5	1.1	0.7	1.2	0.5	
0.8	0.9	0.7	0.6	0.8	0.5	**7000**
0.8	1.1	0.4	0.6	1.4	0.1	
0.9	0.9	1.5	0.5	0.9	0.7	**6000**
1.1	1.3	0.8	0.1	0.2	0.5	
0.6	0.8	0.8	-0.4	-0.4	-0.2	**5000**
0.4	0.4	0.3	0.4	0.3	0.3	
0.8	1.1	1.1	0.6	0.4	0.5	**4000**
0.7	1.3	1.1	0.6	0.5	0.3	
1.5	1.4	0.7	1.2	0.5	0.9	**3000**
1.1	1.9	0.8	-0.4	0.9	0.6	
1.1	1.1	1.4	1.2	0.9	0.6	**2000**
1.1	1.5	0.5	1.1	0.7	0.2	
1.1	1.3	1.4	0.9	1.1	1.0	**1000**
1.0	1.1	1.0	1.0	1.1	0.7	
0.7	1.1	1.0	0.7	0.6	0.9	**0**

15

0.0	**8000**
-0.1	
-0.2	**7000**
-0.1	
0.0	**6000**
-0.1	
0.1	**5000**
-0.2	
-0.2	**4000**
-0.2	
-0.5	**3000**
-0.1	
0.1	**2000**
0.5	
0.5	**1000**
0.4	
0.4	**0**

TRANSVERSE SLOPE

1. ALL TRANSVERSE SLOPE MEASUREMENTS WERE TAKEN WITHIN 30 ft (6.1 m) OF THE CENTERLINE.
2. POSITIVE SLOPE VALUES INDICATE DRAINAGE AWAY FROM CENTERLINE.
3. NEGATIVE SLOPE VALUES INDICATE DRAINAGE TOWARD CENTERLINE.

LONGITUDINAL SLOPE

1. ALL LONGITUDINAL SLOPE MEASUREMENTS WERE TAKEN AT CENTERLINE.
2. POSITIVE SLOPE VALUES INDICATE DRAINAGE TOWARD THE 33 END OF THE RUNWAY.
3. NEGATIVE SLOPE VALUES INDICATE DRAINAGE TOWARD THE 15 END OF THE RUNWAY.

MAXWELL AFB, AL

RUNWAY 15/33

SURFACE TEXTURE MEASUREMENTS

LOCATION FT FROM PRIMARY END	PAVEMENT TYPE	SURFACE RUBBER	OUTFLOW METER TIME (SECONDS)	MEAN TEXTURE DEPTH (MILLIMETERS)	AVERAGE OUTFLOW METER TIME (SECONDS)
0	UNGROOVED PCC	NONE	-	-	19.34
500	UNGROOVED PCC	VERY LIGHT	22.99	0.771	
1000	UNGROOVED PCC	LIGHT TO MEDIUM	-	-	
1500	UNGROOVED PCC	LIGHT TO MEDIUM	15.68	0.835	
2000	UNGROOVED PCC	LIGHT TO MEDIUM	-	-	
2500	UNGROOVED AC	LIGHT TO MEDIUM	6.70	1.101	6.70
3000	UNGROOVED AC	VERY LIGHT	2.21	2.047	2.21
3500	UNGROOVED AC	NONE	-	-	
4000	UNGROOVED AC	NONE	-	-	
4500	UNGROOVED AC	NONE	-	-	
5000	UNGROOVED AC	NONE	-	-	
5500	UNGROOVED AC	NONE	-	-	
6000	UNGROOVED AC	VERY LIGHT	2.22	2.039	
6500	UNGROOVED PCC	VERY LIGHT	25.30	0.759	25.30
7000	UNGROOVED AC	VERY LIGHT	4.43	1.339	5.20
7500	UNGROOVED AC	VERY LIGHT	5.97	1.158	
8000	UNGROOVED AC	NONE	16.65	0.823	16.65

Dynamic Hydroplaning Potential	Low	Poor	Fair	Good	High
Outflow Meter Time (sec.)	<2	2 to 5	5 to 20	20 to 80	>80

MAXWELL AFB, AL
RUNWAY 15/33
Average Friction Values

Distance			40 MPH				60 MPH			
			Left of Center	Right of Center	Average (500 ft)	Average (1/3)	Left of Center	Right of Center	Average (500 ft)	Average (1/3)
0	-	500	0.73	0.81	0.77	0.76	0.52	0.58	0.55	0.60
500	-	1000	0.70	0.76	0.73		0.58	0.73	0.66	
1000	-	1500	0.74	0.77	0.76		0.62	0.70	0.66	
1500	-	2000	0.78	0.82	0.80		0.61	0.49	0.55	
2000	-	2500	0.79	0.90	0.85	0.84	0.84	0.78	0.81	0.81
2500	-	3000	0.87	0.89	0.88		0.90	0.77	0.84	
3000	-	3500	0.87	0.87	0.87		0.86	0.77	0.82	
3500	-	4000	0.85	0.86	0.86		0.84	0.79	0.82	
4000	-	4500	0.84	0.85	0.85		0.83	0.79	0.81	
4500	-	5000	0.81	0.84	0.83		0.80	0.81	0.81	
5000	-	5500	0.80	0.80	0.80		0.80	0.83	0.82	
5500	-	6000	0.81	0.84	0.83		0.82	0.80	0.81	
6000	-	6500	0.73	0.73	0.73	0.75	0.63	0.62	0.63	0.65
6500	-	7000	0.72	0.73	0.73		0.74	0.55	0.65	
7000	-	7500	0.75	0.80	0.78		0.77	0.56	0.67	
7500	-	8000	0.75	0.78	0.77		0.75	0.55	0.65	

Notes:
1. DISTANCES ARE MEASURED FROM THE PRIMARY END 15.
2. YELLOW SHADING DENOTES AVERAGE FRICTION VALUES BETWEEN MINIMUM PLANNING LEVEL AND MINIMUM ACTION LEVEL.
3. RED SHADING DENOTES AVERAGE FRICTION VALUES LESS THAN THE MINIMUM ACTION LEVEL.

A. Friction deterioration below the Maintenance Planning Level for 500 Feet.
When the average Friction value is: > .43 and < .53 at 40 mph AND > .24 and < .36 at 60 mph for a distance of 500 ft, AND adjacent 500 ft segments are: > .53 at 40 mph AND > .36 at 60 mph **No corrective action is required.** These readings indicate that the pavement friction is deteriorating but the situation is not within an unacceptable overall condition. The area in question should be monitored closely by conducting friction surveys to establish the rate and the extent of friction deterioration.

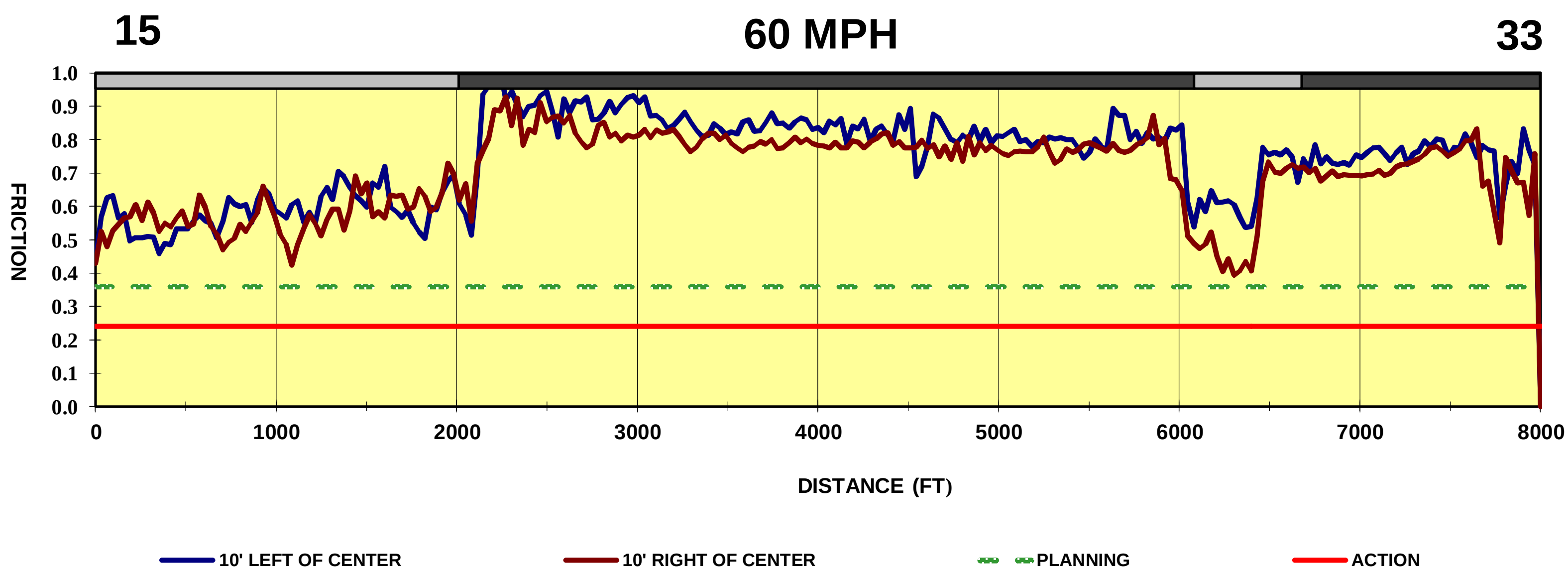
B. Friction deterioration below the Maintenance Planning Friction Level for 1000 Feet.
When the average Friction value is: < .53 at 40 mph AND <.36 at 60 mph for a distance of 1000 feet or more, conduct an extensive evaluation into the cause(s) and extent of the friction deterioration and **take appropriate corrective action.**

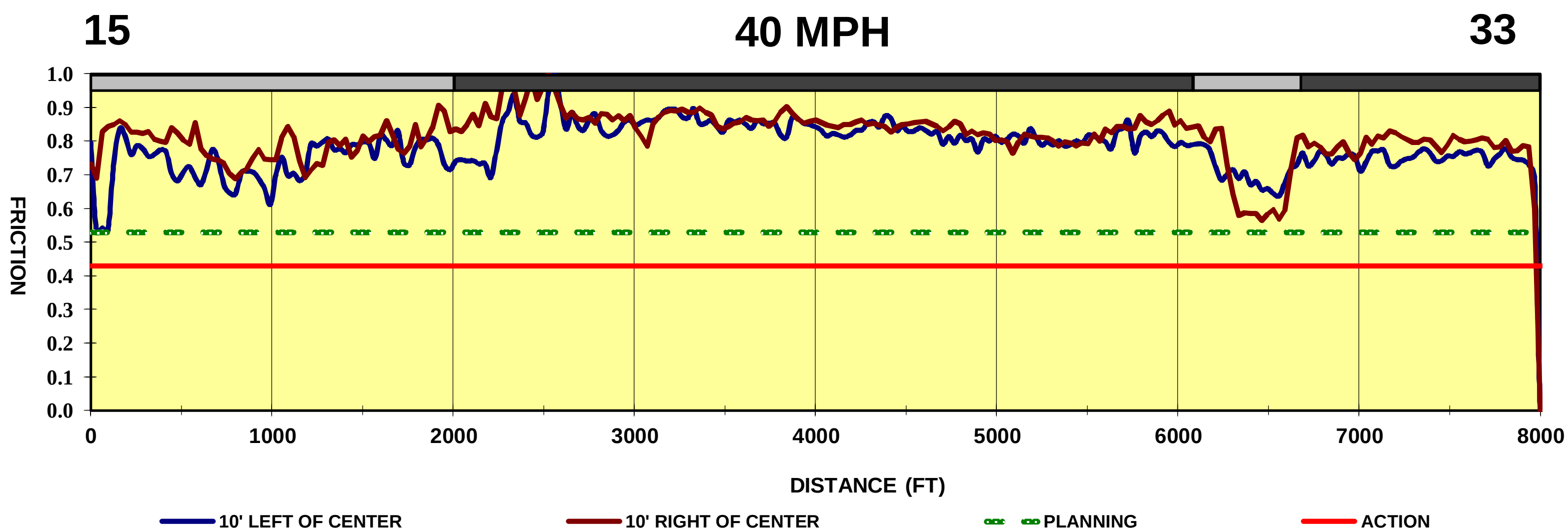
C. Friction deterioration below Minimum Friction Level.
When the average Friction value is: < .43 at 40 mph AND < .24 at 60 mph for a distance of 500 feet, AND the adjacent 500 ft segments are: < .53 at 40 mph AND < .36 at 60 mph **Corrective action should be taken immediately after determining the cause(s) of the friction deterioration**. The overall condition of the entire runway pavement surface should be evaluated with respect to the deficient area before undertaking corrective measures.

D. New Design/Construction Friction Level.
For newly constructed runway surfaces that are either saw-cut grooved or have a porous friction course overlay, the average Friction value of the wet runway pavement surface for each 500 ft segment should be: > .74 at 40 mph AND > .64 at 60 mph.

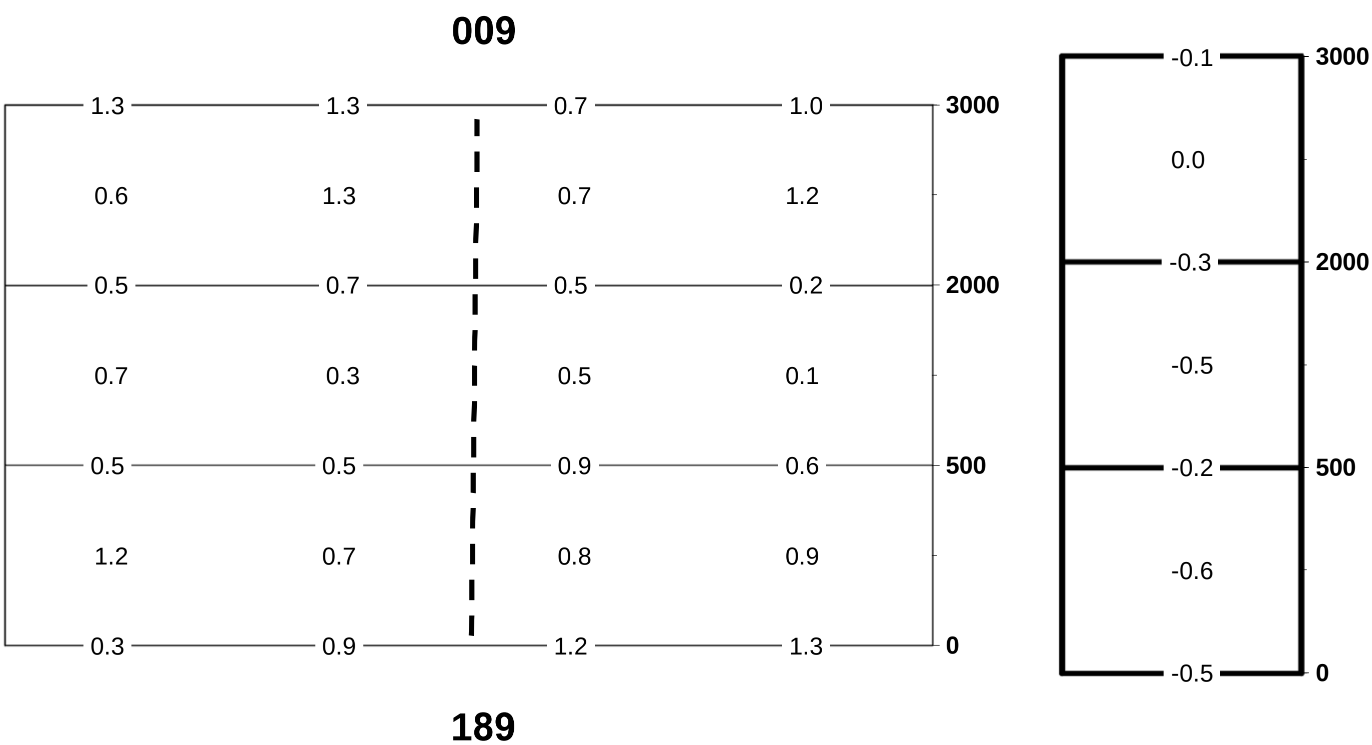
ALL measurements are on wet pavement surface conditions as per FAA AC150/5320-12C.

MAXWELL AFB, AL
RUNWAY 15/33
SELF-WETTING GRIP TESTER FRICTION PLOTS





MAXWELL AFB, AL
ASSAULT LANDING STRIP 009/189
SLOPE MEASUREMENTS



TRANSVERSE SLOPE

1. ALL TRANSVERSE SLOPE MEASUREMENTS WERE TAKEN WITHIN 20 ft (6.1 m) OF THE CENTERLINE.
2. POSITIVE SLOPE VALUES INDICATE DRAINAGE AWAY FROM CENTERLINE.
3. NEGATIVE SLOPE VALUES INDICATE DRAINAGE TOWARD CENTERLINE.

LONGITUDINAL SLOPE

1. ALL LONGITUDINAL SLOPE MEASUREMENTS WERE TAKEN AT CENTERLINE.
2. POSITIVE SLOPE VALUES INDICATE DRAINAGE TOWARD THE 189 END OF THE RUNWAY.
3. NEGATIVE SLOPE VALUES INDICATE DRAINAGE TOWARD THE 009 END OF THE RUNWAY.

MAXWELL AFB, AL
ASSAULT LANDING STRIP 009/189
SURFACE TEXTURE MEASUREMENTS

LOCATION FT FROM PRIMARY END	PAVEMENT TYPE	SURFACE RUBBER	OUTFLOW METER TIME (SECONDS)	MEAN TEXTURE DEPTH (MILLIMETERS)	AVERAGE OUTFLOW METER TIME (SECONDS)
0	UNGROOVED PCC	NONE	13.46	0.867	6.46
250	UNGROOVED PCC	LIGHT TO MEDIUM	6.37	1.125	
500	UNGROOVED PCC	MEDIUM	2.66	1.807	
1000	UNGROOVED PCC	NONE	3.33	1.571	
2000	UNGROOVED AC	NONE	10.38	0.936	9.78
2500	UNGROOVED AC	VERY LIGHT	9.17	0.976	
3000	UNGROOVED AC	NONE	-	-	

Dynamic Hydroplaning Potential	Low	Poor	Fair	Good	High
Outflow Meter Time (sec.)	<2	2 to 5	5 to 20	20 to 80	>80

MAXWELL AFB, AL
ASSAULT LANDING STRIP 009/189
Average Friction Values

Distance			40 MPH				60 MPH			
			Left of Center	Right of Center	Average (500 ft)	Average (1/3)	Left of Center	Right of Center	Average (500 ft)	Average (1/3)
0	-	500	0.72	0.72	0.72	0.74	0.58	0.58	0.58	0.63
500	-	1000	0.73	0.77	0.75		0.70	0.66	0.68	
1000	-	1500	0.69	0.73	0.71	0.81	0.64	0.64	0.64	0.72
1500	-	2000	0.83	0.89	0.86		0.81	0.78	0.80	
2000	-	2500	0.83	0.88	0.86		0.78	0.77	0.78	
2500	-	3000	0.81	0.84	0.83		0.72	0.73	0.73	

Notes:
1. DISTANCES ARE MEASURED FROM THE PRIMARY END 189.
2. YELLOW SHADING DENOTES AVERAGE FRICTION VALUES BETWEEN MINIMUM PLANNING LEVEL AND MINIMUM ACTION LEVEL.
3. RED SHADING DENOTES AVERAGE FRICTION VALUES LESS THAN THE MINIMUM ACTION LEVEL.

A. Friction deterioration below the Maintenance Planning Level for 500 Feet.
When the average Friction value is: > .43 and < .53 at 40 mph AND > .24 and < .36 at 60 mph for a distance of 500 ft, AND adjacent 500 ft segments are: > .53 at 40 mph AND > .36 at 60 mph **No corrective action is required.** These readings indicate that the pavement friction is deteriorating but the situation is not within an unacceptable overall condition. The area in question should be monitored closely by conducting friction surveys to establish the rate and the extent of friction deterioration.

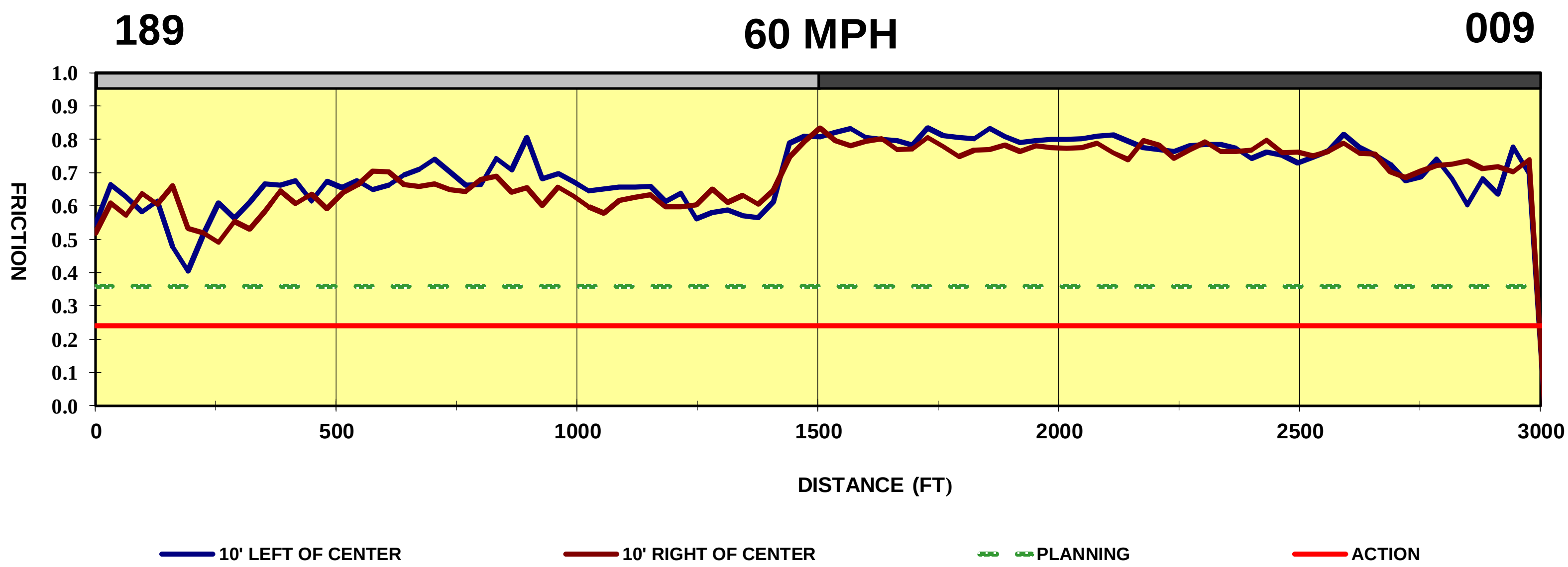
B. Friction deterioration below the Maintenance Planning Friction Level for 1000 Feet.
When the average Friction value is: < .53 at 40 mph AND <.36 at 60 mph for a distance of 1000 feet or more, conduct an extensive evaluation into the cause(s) and extent of the friction deterioration and **take appropriate corrective action.**

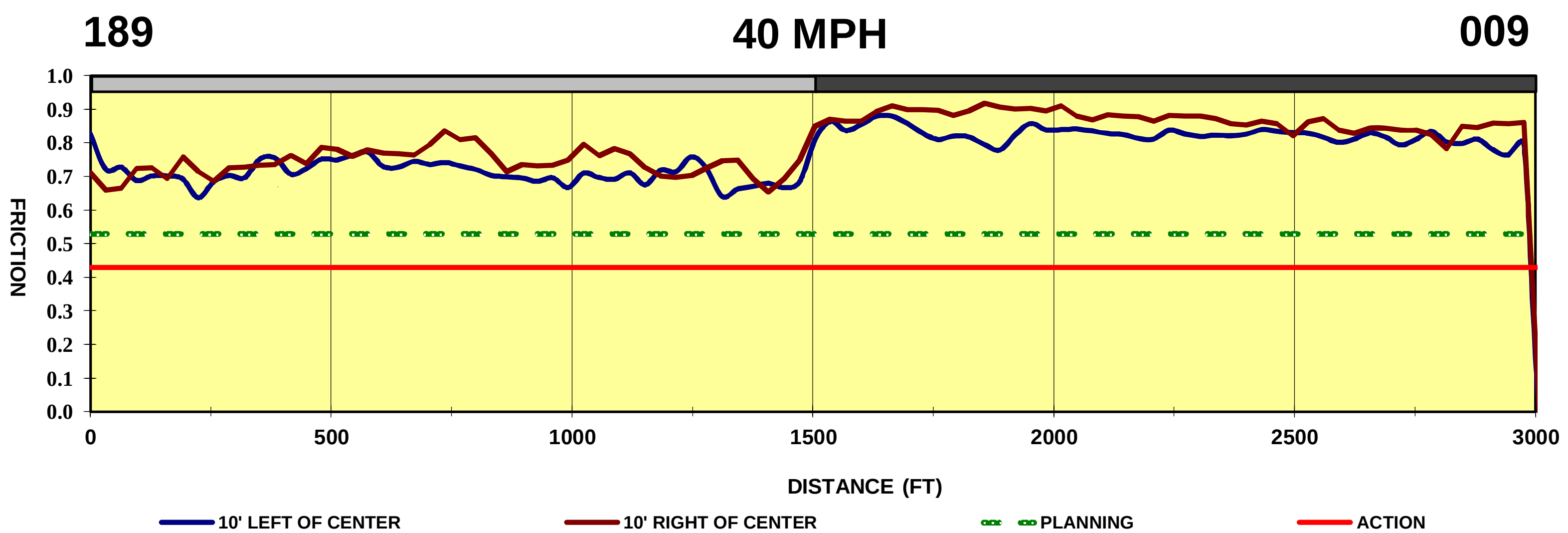
C. Friction deterioration below Minimum Friction Level.
When the average Friction value is: < .43 at 40 mph AND < .24 at 60 mph for a distance of 500 feet, AND the adjacent 500 ft segments are: < .53 at 40 mph AND < .36 at 60 mph **Corrective action should be taken immediately after determining the cause(s) of the friction deterioration**. The overall condition of the entire runway pavement surface should be evaluated with respect to the deficient area before undertaking corrective measures.

D. New Design/Construction Friction Level.
For newly constructed runway surfaces that are either saw-cut grooved or have a porous friction course overlay, the average Friction value of the wet runway pavement surface for each 500 ft segment should be: > .74 at 40 mph AND > .64 at 60 mph.

ALL measurements are on wet pavement surface conditions as per FAA AC150/5320-12C.

MAXWELL AFB, AL
ASSAULT LANDING STRIP 009/189
SELF-WETTING GRIP TESTER FRICTION PLOTS





Estimation of Rubber Deposits

Classification of rubber deposit accumulation	Estimated percentage of rubber covering pavement texture in touchdown zone of runway	Description of rubber covering pavement texture in touchdown zone of runway as observed by evaluator
Very Light	Less than 5%	Intermittent tire tracks; 95% of surface texture exposed.
Light	6 – 20%	Individual tire tracks begin to overlap; 80 – 94% surface texture exposed.
Light to Medium	21 – 40%	Central 6m traffic area covered; 60 – 79% surface texture exposed.
Medium	41 – 60%	Central 12m traffic area covered; 40 – 59% surface texture exposed.
Medium to Heavy	61 – 80%	Central 15-foot traffic area covered; 30 – 69% of rubber vulcanized and bonded to pavement surface; 20 – 39% surface texture exposed.
Heavy	81 – 95%	70 – 95% of rubber vulcanized and bonded to pavement surface; will be difficult to remove; rubber has glossy or sheen look; 5 – 19% surface texture exposed.
Very Heavy	96 – 100%	Rubber completely vulcanized and bonded to surface; will be very difficult to remove; rubber has striations and glossy or sheen look; 0 – 4% surface texture exposed.

Note.- With respect to rubber accumulation, there are other factors to be considered by the airport operator; the type and age of the pavement, annual conditions, time of year, number of wide-body aeroplanes that operate on the runways, and length of runways. Accordingly, the recommended level of action may vary according to conditions encountered at the airport. This table is modified from Airport Services Manual Part 2, Pavement Surface Conditions, Appendix 2, Doc 9137-AN/898.

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Appendix J

Engineering Assessment Summary Table

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Engineering Assessment - Maxwell AFB

Branch Name	Section ID	PCI Index	PCI Rating	FOD Potential	FOD Rating	Structural Index	Structural Rating	Friction Rating	EA Rating
APRONS									
Aero Club Apron	A10C	14	Poor	67	Poor	0.40	Good	N/A	Unsatisfactory
North Apron	A01B	85	Good	18	Good	1.06	Good	N/A	Adequate
	A04B	84	Good	0	Good	1.03	Good	N/A	Adequate
	A08B	99	Good	0	Good	0.67	Good	N/A	Adequate
	T02B	84	Good	3	Good	1.00	Good	N/A	Adequate
	T15B	81	Good	16	Good	0.45	Good	N/A	Adequate
	T30B	97	Good	0	Good	0.43	Good	N/A	Adequate
	T31B	96	Good	0	Good	0.64	Good	N/A	Adequate
West Apron	A05B	88	Good	6	Good	0.93	Good	N/A	Adequate
	T07B	89	Good	0	Good	0.97	Good	N/A	Adequate
	T29B	90	Good	0	Good	0.48	Good	N/A	Adequate
Maintenance Hangar Access Apron	A11C1	81	Good	17	Good	0.49	Good	N/A	Adequate
	A11C2	77	Good	23	Good	0.49	Good	N/A	Adequate
	A11C3	90	Good	20	Good	0.49	Good	N/A	Adequate
Engine Run-up Apron	A07B	92	Good	11	Good	0.83	Good	N/A	Adequate
	T14B	100	Good	0	Good	0.68	Good	N/A	Adequate
OVERRUNS									
Overrun Runway 15/33	O03C	100	Good	0	Good	0.18	Good	N/A	Adequate
	O04C	100	Good	0	Good	0.31	Good	N/A	Adequate
	O05C	94	Good	4	Good	0.52	Good	N/A	Adequate
	O06C	94	Good	4	Good	1.80	Poor	N/A	Unsatisfactory
Overrun Runway 009/189	O01C	43	Poor	N/A	Poor	27.00	Poor	N/A	Unsatisfactory
	O02C	98	Good	0	Good	1.50	Poor	N/A	Unsatisfactory
	O08C	100	Good	0	Good	1.33	Fair	N/A	Degraded
RUNWAYS									
Runway 15/33	R03A	88	Good	0	Good	1.03	Good	Good	Adequate
	R04C1	58	Fair	31	Good	0.33	Good	Good	Degraded
	R04C2	93	Good	5	Good	0.36	Good	Good	Adequate
	R05A1	61	Fair	28	Good	0.18	Good	Good	Degraded
	R05A2	94	Good	4	Good	0.18	Good	Good	Adequate
	R06A1	96	Good	0	Good	0.52	Good	Good	Adequate
	R06A2	96	Good	0	Good	0.52	Good	Good	Adequate
	R07A	97	Good	0	Good	0.48	Good	Good	Adequate
	R08C1	93	Good	0	Good	0.39	Good	Good	Adequate
	R08C2	98	Good	0	Good	0.43	Good	Good	Adequate
	R09C	60	Fair	29	Good	0.21	Good	Good	Degraded
	R10C	80	Good	13	Good	0.18	Good	Good	Adequate
	R12A	100	Good	4	Good	0.18	Good	Good	Adequate
	R13C	83	Good	5	Good	0.21	Good	Good	Adequate
Runway 009/189 (Assault Strip)	R01A	88	Good	15	Good	0.61	Good	Good	Adequate
	R02A	83	Good	12	Good	0.22	Good	Good	Adequate
TAXIWAYS									
Runway 15/33 Turn Around	T11A	95	Good	0	Good	0.48	Good	N/A	Adequate
Runway 009/189 Turn Around	T06A	60	Fair	29	Good	0.18	Good	N/A	Degraded
Taxiway A	T04A	100	Good	0	Good	0.40	Good	N/A	Adequate

Engineering Assessment - Maxwell AFB

Branch Name	Section ID	PCI		FOD		Structural		Friction Rating	EA Rating
		Index	Rating	Potential	Rating	Index	Rating		
Taxiway A (North)	T16A	93	Good	10	Good	0.37	Good	N/A	Adequate
	T18A1	80	Good	24	Good	0.64	Good	N/A	Adequate
	T18A2	91	Good	14	Good	0.62	Good	N/A	Adequate
	T34A	81	Good	14	Good	0.76	Good	N/A	Adequate
Taxiway A (Northwest)	T03A	89	Good	1	Good	0.60	Good	N/A	Adequate
	T13A	100	Good	0	Good	0.37	Good	N/A	Adequate
	T17A	86	Good	0	Good	1.00	Good	N/A	Adequate
	T19A	100	Good	0	Good	**1.45**	**Poor**	N/A	**Unsatisfactory**
	T33A	92	Good	0	Good	0.95	Good	N/A	Adequate
Taxiway B	T05C	100	Good	0	Good	0.23	Good	N/A	Adequate
	T12C	100	Good	0	Good	0.30	Good	N/A	Adequate
Taxiway C	T08A	100	Good	0	Good	0.33	Good	N/A	Adequate
Taxiway D	T32B	100	Good	0	Good	0.23	Good	N/A	Adequate
Taxiway E	T09A	100	Good	0	Good	0.19	Good	N/A	Adequate

Appendix K

Probability of Failure Data

Recommended Project List

Transportation AMP Airfield Pavement Repair Requirement Prioritization	
Overview	PCI Surveys have been conducted on a cyclical basis for most Air Force airfields for the past 20 years. Historically PCI reports and PAVER databases with PCI data were provided to the bases, but were not rolled up into a central database. The intent was for the bases to use the requirements data and recommendations in the report to develop a pavement asset management plan for the base, with the report data serving as justification for funding and prioritization. While we will continue to use PCI report data for this purpose, we are taking a more structured approach using the probability of failure and consequence of failure to help prioritize requirements not only for each base, but across the enterprise.
Limitations	PCI reports are a snapshot in time and do not capture all of the work that was done in the time since the report was published. In addition, there may be gaps in the data due to oversight or lack of information at the time of the survey. For example, many bases do not have RPUID or Facility Number information for their pavement sections; the linear segmentation initiative will resolve this data gap. Another limitation of this rollup-spreadsheet relates to the cost data. The costs shown are the costs to maintain and repair only the pavement itself and do not capture associated costs like rubber removal, airfield marking, drainage repairs associated with reconstruction, etc.
Explanations for the Probability of Failure Data Fields	
Pavement Surface Type	This column is included for informational purposes. It contains the Pavement surface type data (AC, APC, PCC) listed in the current PAVER database. This field is used when assigning families to determine deterioration rates and is used when determining the FOD Potential Rating (listed in the Engineering Assessment Table) in the latest PCI report that will be used to populate column V, FOD Potential Rating.
Pavement Rank	Validate the Pavement Rank according rules in AFI 32-1041: The cell currently contains the Pavement Rank data (P - primary, S - secondary, T - tertiary, and U - unused) listed in the current PAVER database. This field is determine based on a number of factors outlined in AFI-32-1041. It is used to determine the Critical PCI listed in column O and is used in this spreadsheet in column AE to adjust the MDI as part of computing the CoF. While there are ongoing efforts to develop a more robust definition of these ranks based on more objective data such as number and/or type of aircraft operation as well as efforts looking at changes to the MDI, use the current definitions in AFI 32-1041. In general, we expect Ranks would be reduced rather than increased and we ask bases enter U for any pavements that are currently unused. Note any changes in the remarks column.
Critical PCI	This column is included for informational purposes. It contains the Critical PCI for the section based on the pavement Rank. Per AFI 32-1041 Critical PCI is currently defined as 70 for primary pavement and 55 for secondary and tertiary pavements. The critical PCI serves as a trigger for various M&R requirements. In general M&R done above critical PCI is more cost effective than that done below critical PCI. Note that these critical values are currently established as a matter of policy. Efforts are underway to develop a more concise critical PCI process that will identify an economic breakpoint for each section.
PCI	This column is included for informational purposes. It contains the pavement condition for the section as of the last PCI Inspection date. It is determined based on the full PCI process outlined in ASTM 54340 and AFI-32-1041.
PCI Deterioration Rate	This column is included for informational purposes. It contains the pavement deterioration rate for the pavement familiy that the section belongs to e.g. Primary Asphalt Pavement. The deterioration rate was determined as of the last PCI Inspection date.
POF	The following formula is used to determine the PoF: PoF - 100 - 2015 Adjusted PCI value as outlined in Business rules.
Adjusted Mission Dependency Index (MDI)	The following formula is used to determine the Adjusted MDI: IF (Pavement Rank = P THEN AdjMDI = MDI) ELSE IF (Pavement Rank = S THEN AdjMDI = MDI * 0.9) ELSE IF (Pavement Rank = T THEN AdjMDI = MDI * 0.7) ELSE (Put check in cell). The Adjusted MDI will be used to compute CoF as outlined in AFCEC Business Rules.
2015 PCI Before M&R	The PCI values reported in this column are based on the (PCI at the time of the last inspection) - (the number of years since the last inspection * the deterioration rate for the section).
2016 to 2021 PCI Before M&R	The PCI reported in columns for 2016 to 2017 are based on the (PCI at the time of the last inspection) - (the number of years since the last inspection * the deterioration rate for the section). While we show funding for specific fiscal fiscal years, it may be more cost effective for the base to combine requirements from two or more fiscal years into a single project.

Probability of Failure Table - Maxwell AFB

Branch Name	Section Name	Pavement Surface Type	Pavement Rank	Critical PCI	PCI Deter Rate	POF	Adjusted MDI	2015 PCI Before M&R	2016 PCI Before M&R	2017 PCI Before M&R	2018 PCI Before M&R	2019 PCI Before M&R
Aprons												
Aero Club Apron	A10C	PCC	T	55	1.08	93	67	14	13	99	98	97
North Apron	A01B	PCC	S	55	1.08	18	86	85	84	83	82	81
	A04B	PCC	P	70	0.40	17	95	84	84	83	83	82
	A08B	PCC	P	70	0.40	1	95	99	99	98	98	97
	T02B	PCC	P	70	0.40	16	95	84	84	83	83	82
	T15B	PCC	P	70	0.40	21	95	81	81	80	80	79
	T30B	PCC	P	70	0.40	3	95	97	97	96	96	95
	T31B	PCC	P	70	0.40	4	95	96	96	95	95	94
West Apron	A05B	PCC	P	70	0.40	13	95	88	88	87	87	86
	T07B	PCC	P	70	0.40	11	95	89	89	88	88	87
	T29B	PCC	P	70	0.40	10	95	90	90	89	89	88
Maintenance Hangar Access Apron	A11C1	PCC	S	55	1.08	21	86	81	99	98	97	96
	A11C2	PCC	S	55	1.08	25	86	77	99	98	97	96
	A11C3	PCC	S	55	1.08	12	86	90	89	88	87	86
Engine Run-up Apron	A07B	PCC	S	55	1.08	9	86	92	91	90	89	88
	T14B	AC	S	55	1.35	0	86	100	98	97	97	97
Overruns												
Overrun Runway 15/33	O03C	AC	T	55	1.35	0	63	100	98	97	97	97
	O04C	AC	T	55	1.35	0	63	100	98	97	97	97
	O05C	AC	T	55	1.35	6	63	98	96	97	97	97
	O06C	AC	T	55	1.35	16	63	98	96	97	97	97
Overrun Runway 009/189	O01C	DBST	T	55	1.35	N/A	63	100	99	97	96	97
	O02C	AC	T	55	1.35	12	63	98	96	97	97	99
	O08C	AC	T	55	1.35	7	63	100	98	97	97	97
Runways												
Runway 15/33	R03A	PCC	P	70	0.40	13	99	88	88	87	87	86
	R04C1	AAC	P	70	0.86	45	99	58	99	98	97	97
	R04C2	AAC	P	70	0.86	8	99	93	92	91	90	91
	R05A1	AAC	P	70	0.86	42	99	61	99	98	97	97
	R05A2	AAC	P	70	0.86	6	99	94	93	92	91	92
	R06A1	PCC	P	70	0.40	4	99	96	96	95	95	94
	R06A2	PCC	P	70	0.40	4	99	96	96	95	95	94
	R07A	PCC	P	70	0.40	3	99	97	97	96	96	95

Probability of Failure Table - Maxwell AFB

Branch Name	Section Name	Pavement Surface Type	Pavement Rank	Critical PCI	PCI Deter Rate	POF	Adjusted MDI	2015 PCI Before M&R	2016 PCI Before M&R	2017 PCI Before M&R	2018 PCI Before M&R	2019 PCI Before M&R
Runway 15/33	R08C1	PCC	P	70	0.40	7	99	93	93	92	92	91
	R08C2	PCC	P	70	0.40	2	99	98	98	97	97	96
	R09C	AAC	P	70	0.86	43	99	60	99	98	97	97
	R10C	AAC	P	70	0.86	21	99	80	79	78	78	79
	R12A	AAC	P	70	0.86	0	99	83	82	82	81	80
	R13C	AC	P	70	0.86	18	99	82	79	77	74	71
Runway 009/189 (Assault Strip)	R01A	PCC	P	70	0.40	14	99	88	88	87	87	86
	R02A	AAC	P	70	0.86	18	99	83	84	83	82	82
Taxiways												
Runway 15/33 Turn Around	T11A	PCC	P	70	0.40	5	95	95	95	94	94	93
Runway 009/189 Turn Around	T06A	AC	P	70	0.86	43	95	60	99	98	97	97
Taxiway A	T04A	AAC	P	70	0.86	0	95	100	99	98	97	96
Taxiway A North	T16A	PCC	P	70	0.40	8	95	93	93	92	92	91
	T18A1	PCC	P	70	0.40	22	95	80	80	79	79	78
	T18A2	PCC	P	70	0.40	10	95	91	91	90	90	89
	T34A	PCC	P	70	0.40	20	95	81	81	80	80	79
Taxiway A Northwest	T03A	PCC	P	70	0.40	11	95	89	89	88	88	87
	T13A	AAC	P	70	0.86	0	95	100	99	98	97	96
	T17A	PCC	P	70	0.40	14	95	86	86	85	85	84
	T19A	AC	P	70	0.86	9	95	100	99	98	97	96
	T33A	PCC	P	70	0.40	8	95	92	92	91	91	90
Taxiway B	T05C	AC	S	55	0.86	0	86	100	98	97	97	97
	T12C	AC	S	55	1.35	0	86	100	98	97	97	97
Taxiway C	T08A	AC	P	70	0.86	0	95	100	99	98	97	96
Taxiway D	T32B	AC	S	55	1.35	0	86	100	98	97	97	97
Taxiway E	T09A	AC	P	70	0.86	0	95	100	99	98	97	96

Sections recommended for Major Maintenance and Repair (M&R) in the next five (5) years include:

Section	Branch	Recommended M&R
R04C1, R05A1, and R09C	Runway 15/33	Seal Cracks and Mill and Overlay
T06A	Runway 009/189 Turn Around	Seal Cracks and Mill and Overlay
T19A	Taxiway A Northwest	Complete Reconstruction

All of the above recommendations will be implemented through the below projects already programmed or the list of additional projects to program. Note the lists are in no particular priority order.

Programmed Projects:

Replace Sections Runway 15/33 – FY16, PNQS157400, PNQS167500, PNQS172593 ($17M)

Replace Concrete Pads North End Assault Zone – FY16, PNQS137001 ($40K)

Additional Projects to Program:

Repair Runway 009/189 Turn Around: Includes crack sealing and mill and overlay of section T06A. New overlay should correct slope deficiencies in this area that contribute to ponding issues; as such, adjoining shoulder sections may need to be repaired concurrently.

Reconstruct T19A: Includes demolition of existing composite pavement structure, preparation and placement of adequate base course material, and placement of AC pavement of greater thickness in order to reduce current AGL restrictions for installation-assigned aircraft.

Demolish Closed Airfield Pavements: Includes selective demolition (removal) of closed asphalt pavements adjacent to current operational surfaces throughout the airfield. Removing these pavement sections will prevent excess moisture from infiltrating into the subgrade and becoming trapped in the pavement structure. Additionally, removal of these non-operational pavements will allow for re-grading efforts along Runway 15/33 and Assault Runway 009/189 that should mitigate current ponding issues on shoulder pavements.

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